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MATHTATAG: A MOBILE APPLICATION FOR ONLINE ASSESSMENT AND PERFORMANCE MONITORING WITH FILIPINO TEXT-TO-SPEECH FOR GRADE 1 IN MATHEMATICS

A Capstone Project

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In Partial Fulfillment of the Requirements for the Degree

Bachelor of Science in Information Technology

by

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CERTIFICATION

This capstone, ***MATHTATAG: A MOBILE APPLICATION FOR ONLINE ASSESSMENT AND PERFORMANCE MONITORING WITH FILIPINO TEXT-TO-SPEECH FOR GRADE 1 IN MATHEMATICS*** prepared and submitted by Cedick L. Artiaga, Engelbert R. Morales, Ken P. Cas, and Lloyd Jernell C. Loteriña in partial fulfilment of the requirements for the degree, Bachelor of Science in Information Technology has been examined and recommended for Oral Examination:

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CERTIFICATION OF ORIGINALITY

This is to certify that the Capstone Project entitled *MATHTATAG: A MOBILE APPLICATION FOR ONLINE ASSESSMENT AND PERFORMANCE MONITORING WITH FILIPINO TEXT-TO-SPEECH FOR GRADE 1 IN MATHEMATICS* for the degree Bachelor of Science in Information Technology at the Polytechnic University of the Philippines embodies the result of original and scholarly work carried out by the undersigned. This capstone project does not contain words or ideas taken from published sources or written works that have been accepted as basis for the award of a degree from any other higher education institution, except where proper referencing and acknowledgement were made.

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ABSTRACT

Title	:	MATHtatag: A Mobile Application for Online Assessment and Performance Monitoring with Filipino Text-to-Speech for Grade 1 in Mathematics
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This paper describes the design, development, and evaluation of MATHtatag—a mobile-based assessment and performance monitoring app for Grade 1 Mathematics based on the MATATAG Curriculum. The project primarily sought to address the difficulty of learners to read and comprehend written instructions, little opportunity for formative assessments, and late feedback and monitoring to the teachers and the parents. Some of the features are interactive math activities, Filipino text-to-speech, automated scoring, and real-time performance dashboards.

A pretest–posttest was deployed among forty-five (45) Grade 1 learners from a public elementary school in Gumaca, Quezon. Descriptive statistics are presented to show the learning gains, while the acceptability of the system based on the ISO/IEC 25010 software quality model among the teacher and the parent respondents. The performance of the learners improved significantly from 12.64 (63.22% Mean Percentage Score) in the pretest to 19.18 (95.89% Mean Percentage Score) in the posttest, with an average gain of 6.54. All learners

have improved. The general mean based on ISO/IEC 25010 evaluation of the system among teacher and parent respondents is 4.63, indicating a high agreement to system quality and usability.

The results suggested that mobile-based formative assessment with Filipino audio support can boost learner engagement and mathematics performance while aiding teachers and parents in progress monitoring. The study signifies the potential benefit of digital learning tools as a supplementary support to strengthen further the early mathematics education in public schools.

Keywords: Polytechnic University of the Philippines, mobile learning, Grade 1 Mathematics, formative assessment, educational technology, performance monitoring, MATATAG Curriculum.

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Chapter 1
INTRODUCTION

PROJECT CONTEXT

Studies from years past suggested that many Filipino students were not doing well in international exams. Based on the 2018 results of the Program for International Student Assessment (PISA), more than half learners from the Philippines received scores that did not reach the minimum proficiency level in Mathematics. This means many Filipino learners experienced difficulties to answer basic Maths (Lapinid, 2022). In the latest 2022 PISA report, the Philippines is still behind and languishes in math, reading and science which showed that the Philippines was ranked 78/78 in 2018, and 77/81 in 2022, the lowest possible ranking (Philstar, 2023; Congressional Policy and Budget Research Department [CPBRD], 2024). Such results revealed a tremendous challenge as it showed that so many Filipino children could not even reach the most basic level of learning in mathematics

Research also said that children struggled in Math because of different reasons like weak understanding of concepts, problems in how lessons were taught, and sometimes anxiety that made them afraid of math (Mangarin & Caballes, 2024). Because of these problems, many learners cannot fully learn math in school.

On the other hand, using mobile learning had been studied by many researchers. A meta-analysis with 22 studies reported that mobile learning produced a positive influence on math learning and helped improve the scores of learners (Güler et al., 2022). Other research findings also showed that the use of mobile learning provided benefits in different school settings because it allowed children to learn with the help of digital tools.

But even with these good results, many apps in the Philippines were still limited. Some were only in English, which was hard for Grade 1 pupils who were just starting to read. In the Philippines, children in Kindergarten to Grade 3 were taught in their own first language (mother tongue) so they could understand lessons better before shifting to Filipino or English (DepEd, 2016). Some did not include features where parents and teachers can easily check the progress of the child. Others lacked audio or voice to guide children, which is very important, especially in the early years.

Thus, there was a need for a mobile app that could conduct online math assessments, track the performance of the learner in real-time, and use Filipino text-to-speech so that the child could listen and make more sense of the instructions. Enter MATHtatag. The project was targeted for Grade 1 mathematics, considering this is the level where the child needs basic foundation in math, and that if they're guided well at this point, it would hold them in good stead as they progress to higher years.

TECHNICAL BACKGROUND

Equipment/Hardware

The system was intended to be used mainly on mobile devices, as these are more available to learners/parents and teachers. Mobile phones were also more practical both at home and at the classroom for young learners. The computer is only needed during development. A reliable internet connection is also a must if the real-time features are to be supported. The equipment consists of the following:

Mobile Devices (Smartphones/Tablets) – The main devices of student/parents/teachers and

admins to access the MATHtatag application. The system is for Android mobile devices as these are found more in households and easier for the young learners to use. Android phones and tablets also more compatible and accessible. Thus, it can be used for classroom and home learning.

Laptops/Computers – Were just needed by the developers for designing, code, and and checking the system's functions during development. These are not necessary by the end-users as the teachers and administrators will access everything in the mobile app.

Internet Connection – A reliable one is needed so that the app can connect to database, upload and download, and real-time monitoring support.

Software

A combination of development tools/services was required to build and operate and maintain the features of the system. Among these are the principal frameworks for development, backend service for handling database and additional libraries making the app functionally and user-friendly. Some of these are:

React Native with Expo – Served as the main development framework used in building the mobile app. These allow the system to run smoothly on different devices like Android.

Firebase – An easier backend service for the system. These handle data storage, live real-time database, user login and media file storage (images and audio files) by the application.

Visual Studio Code – The coding tool or IDE used by the developer.

ElevenLabs API – Provides the Filipino Text-to-Speech feature that would convert the written questions/instructions to spoken words so that the Grade 1 student and its parent can better understand it.

Gemini API – Used for generating texts in the Parent Dashboard especially summaries and feedback.

Peopleware/Manpower

The progress and implementation of MATHtatag comprised different members with separate tasks in the development and deployment of the app. The student developers were the app developers who programmed the mobile application, integrated features with Firebase, and ensured that everything worked well, and the group was involved in all the design, programming, testing and documenting of the entirety of the system. Each member was assigned an individual role but each is not solely responsible for every aspect to build the whole system. Roles were assigned to - Ken P. Cas– Lead Developer/System Architect, who handled the development of core features and modules integration; Engelbert R. Morales and Cedick L. Artiaga – Front-End/UI Designer, who made the child-friendly user interface of the app simple for Grade 1 learners; Ken P. Cas and Cedick L. Artiaga – Back-End Developer/Database Handler, who handles the Firebase Realtime Database and securing of the exercises, results, and analytics data into it; Lloyd Jernell C. Loteriña and Engelbert R. Morales – Quality Assurance Tester, who tests the features of the app and reports errors; and Lloyd Jernell C. Loteriña and Engelbert R. Morales – Documentation and Researcher, who wrote the document deliverable and researched if lessons that were to be integrated into the

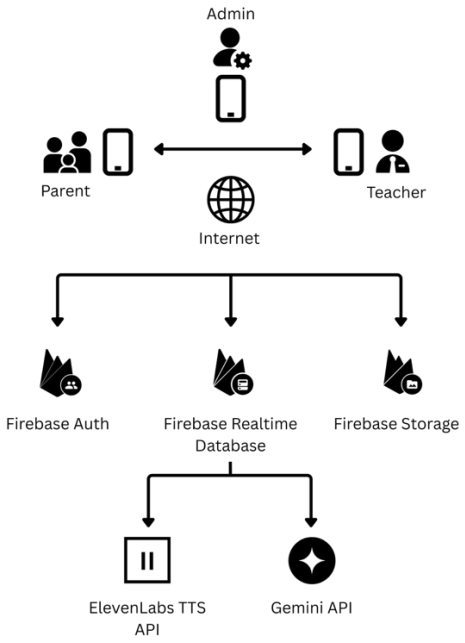
app met the curriculum of Mathematics Grade 1 and its learning outcomes.

To make sure that the system indeed aligns to the correct teaching pedagogy the group also met with a consultant Grade 1 teacher, who checked and verified that the information, exercises, and assessment items included in the app are relevant and true to the Grade 1 Mathematics curriculum.

In terms of the actual users who will access the app, the system will primarily be the students but also use of the app by their respective teachers and parents, teachers, involved in writing assignments and monitoring student performance, their parents tracking their child's progress and availing of the analysis, and ultimately the students who will answer the exercises on the app and learn through interactive assessments additionally the assistance of Filipino text to speech. The system is specifically intended for Grade 1 pupils who still do not read and learn to an extent by themselves but are facilitated and assisted by their respective teacher in school and their parents at home whenever they use the app. Also, part of the manpower is school administrator and application administrator in charge of maintaining, users, certificates, reports with the technical information. The group truly worked together with each person involved with their own task contributing to hopefully make a successful working product.

Network Infrastructure/Architecture

Figure 1. Network Infrastructure/Architecture



MATHtatag's networking uses rather simplistic architecture that follows the client–server model where all stakeholders use the mobile application. MATHtatag's parents, students, teachers, and administrators each have their own version of the app but all access the internet using HTTPS. Once connected, these requests are sent and interpreted by Firebase services. Firebase Authentication lets users login (or sign up), and handle role access ensuring each user only has access to the features they have been allotted. Firebase Realtime Database stores and updates the application's exercises, scores, announcements and report in real time (user A scores and user B reviewing their performance and high scores), and Firebase Storage allows it to store and serve uploaded media files such images and audio. For the analysis of parent questions, interface annotations, and summaries helpful for parents to figure out how to assist their child's performance, we used ElevenLabs API to generate Filipino

text to speech to allow our young learners to hear instructions and rhetorical questions in their own language, and Gemini API generates the part of the text that helps parents know what they need to do next to assist their child.

Storage, Backup and Recovery Procedure

MATHtatag's data storage relies on Firebase services, providing a secure and dependable cloud-based solution. The core data for student scores, exercises, announcements, teacher accounts, and parent details were stored in the Firebase Realtime Database, and the media files like images and audio that were created by text-to-speech along with the posted screenshots are all in Firebase Cloud Storage. These services handle the organization of data for us and make sure that we can readily access our information whenever we need to. Backup is included in Firebase's inbuilt system. Our data is continually synchronized and protected in the cloud. Even if a user's device is lost, destroyed or is just disconnected, we do not lose these records. Users may always view their records again when they sign in with their account. For recovery, Firebase gives redundancy and automatic stored data restoration. Reducing the risk of data loss, this allows the app to have record consistency and keep records secure for our teachers, parents and administrators at all times.

Security Procedures

Protecting MATHtatag's data is very important as it contains sensitive information pertaining to learners, parents and teachers. To protect this data, we use Firebase Authentication to handle logins for parents, teachers and administrators, each role has its own permissions. Teachers are able to create and check exercises, parents can monitor and review

their child's progress while administrators can manage accounts and system maintenance. Learners themselves do not have 'accounts' as per se but rather their profile and performance data is accessed through the Parent Dashboard and Teacher Dashboard. To further protect their identity, each learner is assigned a unique system generated ID used to store and retrieve their records, meaning personal details are not exposed when handling or analyzing data.

All communication between the mobile app and Firebase is done over secure HTTPS, which ensures that scores, reports and logins are encrypted during transfer. Passwords are protected using Firebase's automatic encryption so our developers can't see them.

The system also enforces rules on access to various uploaded files like images or audio in Firebase Cloud Storage too, so that only authorized users may view or even download them. While dedicated administrators and designed system administrators scan for suspicious account and activities on that platform, they also need to respond to tech reports by reporting possible misuse.

On the development side, GitHub is a great version control system they use in storing source code, and with this afforded easier for the team who worked on it to "track changes, save an older version if mistakenly edited or changed" and to can make along fixes too to keep it stable and safe from vulnerabilities.

Policies and Procedures

MATHtatag was developed with policies and procedures to ensure system reliability, ease of use, and user safety. The project utilized an Agile Development approach to develop the system in small chunks or phases. Using sprints allowed features to be planned,

developed, tested and refined regularly. Rather than having to wait until surface rise up the end of the project. It allows the inspectors to act and adapt quickly if there are issues, or improvements are identified.

During the development, regular code reviews and debugging were performed to ensure that any errors were caught early and fixed before deployment. Different levels of testing were applied, like if login works properly, if exercises can be created and answered, and if the data is logged in the database.

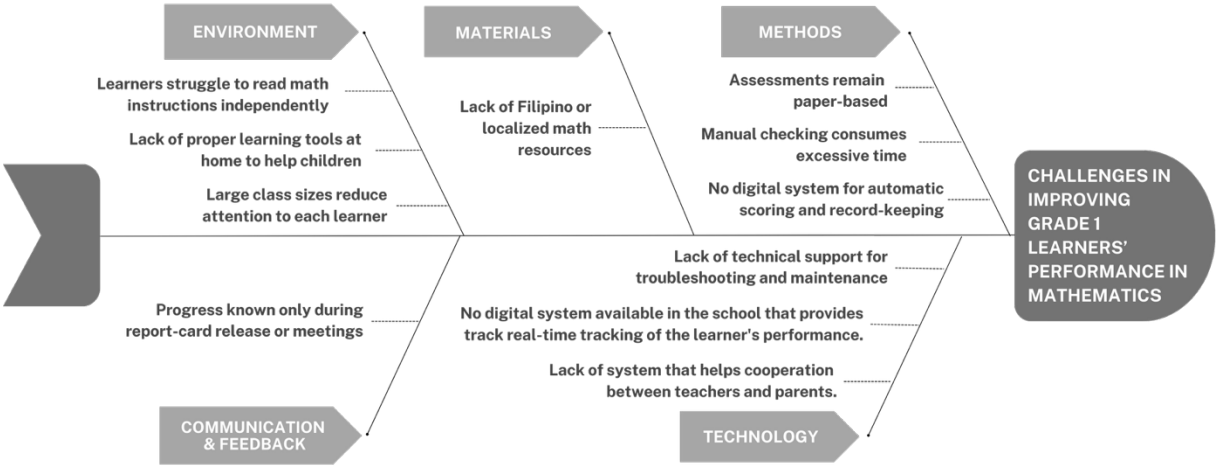
To operate the system, policies pertaining to data handling and access are observed. Parents, teachers, and administrators are the only role that have permission to access learner records from the developer and learners are dispatched by their parents and teachers and child do not have accounts. Uploaded files are safely stored in Firebase and the generated audio are with access restrictions to prevent from misuse.

In the case of some issues rising up, the app's reporting feature allow the administrators received feedback and acted on issues. Regular updates and maintenance were planned to keep the app secure and compatible with the latest devices. Overall, these guidelines and operating procedures ensure that MATHtatag's development, implementation, and use were organized, safe, and effective for all users.

PROBLEM ANALYSIS

Fishbone Diagram

Figure 2. Fishbone Diagram



Environment

Many learners can't read instructions in mathematics on their own, especially when they do not have enough learning tools at home or in large classes classrooms where 1 teacher handles about 25 to 30 learners. Although that's an acceptable pupil–teacher ratio for early grades cited in DepEd's guidelines, research from the Philippines indicates that classes this size can still limit how much of individualized support teachers are able to give (Department of Education, 2023; Susulan, et al., 2022). In one Filipino study students' reading levels were very low, showing 7% mastery at one school, and this illustrates the ways reading difficulty limits comprehension of instruction (Caraig & Quimbo, 2022). Other Filipino research into home-based learning showed that when teachers supported students at home with math activities students' performance improved (Refugio, Maligro & Zamoras, 2024). Although we found less research on home/school size effects more generally we did find broader

international studies that confirm large classes limit personalized instruction and engagement (Antoniou, Alghamdi & Kawai, 2024). Take together these provide evidence that reading-instruction comprehension issues, insufficient at home, lack of resources, and class sizes weaken the learning environment in the Philippine context in ways justifying tools like Filipino TTS and home-accessible learning systems.

Materials

A big part of the challenge was the lack of localized or Filipino-language math resource, leaving students to receive materials that don't fit their linguistic or cultural needs. In Filipino studies of strategies for teaching reading (not necessarily math materials) - When they have a strategy that they can comprehend more easily in their language, they understand (the Interactive Reading-Aloud Strategy in the Philippines had a remarkable impact e.g. Requilme, 2024). While we don't have specific Philippine evidence localized math resources, the international meta-analyses of TTS and assistive tech support that when students have instructional materials that match their context, their comprehension improves (Wood et. al, 2018). This gave me confidence to do work on developing Filipino-language, context relevant math materials in the system.

Methods

Because teachers moving through their regular assessment tasks still adhere to traditional assessment practices of paper and pencil assessment testing, manual checking, and leaving no automatic record keeping behind for another teacher, where they can - of course - cause delay in learning and adding more task to educators. Within the Philippine setting, a recent study noted that many classroom teachers still use traditional assessment

tools even in online or blended learning, and that they found it difficult to conduct authentic assessment, with some of them having blended their learning also using some digital apps for easy evaluation (Agtarap, Januto, Aglibot & Toquero, 2024). International research shows computer-based assessments with automatic scoring significantly reduce teacher workload and speed feedback (Morris, 2023). So, for the Philippine context project's shift to automatic scoring and digital record-keeping is relevant.

Communication and Feedback

When parents and students receive feedback only on report-card day or in sporadic meetings, windows for intervention close quickly. There is limited research on parent-teacher digital feedback in-the-moment in the Philippines but international studies show that tech-mediated communications between home and school has helped strengthen engagement and learning outcomes (See et al., 2020). Moreover, Filipino-based reading-intervention studies point to the power of feedback (specifically continuous) as a cog or gear within the process of comprehension improvement machine (Sampang, 2023). So, real time progress monitoring charts and dashboards parents can access were a no brainer in the Philippine setting.

Technology

Technological barriers in schools like insufficient technical support, and infrastructure, together with the absence of digital tracking systems, are also detractors to innovation. Work in the Philippines around readability and language resources for local languages (Cebuano readability model), did show that a lack of infrastructure resulted in not being able to use any tools, or the adaption of these tools, and the need for infrastructure that is responsive to the local context (Reyes et al., 2022). Internationally studies have also shown that without a strong infrastructure, training, and support, ICT integration fails (Msafiri & Kangwa., 2024). So the

lack of a technical support system and a real-time tracking system, is a very real barrier for a Philippine school educator, and supported the need for the digital system that the project proposes.

Problem and Solution Statement

Problem Statement

Mathematics remains one among the most difficult subjects to grasp for Filipino learners, particularly in the formative early grades as basic skills are developed. Most of the Grade 1 pupils are unable to follow other people's math instructions because of their inability to read, poor home learning support, and extremely large class size which prevents individualized approach to the pupils (Caraig & Quimbo, 2022). Among local learners, only a low percentage is able in achieve a satisfactory level of reading. It has been found that "home based intervention with teachers' support improves their performance" (Refugio, Maligro, & Zamoras, 2024). Clearly, home based learning is beneficial for the learners, but they have limited that kind of support. For young learners, who "learn better in their mother tongue" (Raquilme, 2024), there is a "lack of localized math instructional materials in Filipino" (ibidem). Assessment designs that are merely paper based, checked manually, and result in delayed feedback are also problematic". Most schools lack the capacity to automate scoring and provide real time monitoring, technical and other technology support" (Agtarap et al., 2024; Reyes et al., 2022; Msafiri & Kangwa, 2024). These issues deny prompt instruction, adequate feedback and genuine learner engagement in mathematics.

Solution Statement

In response to these learning difficulties, the researchers created MATHtatag. This app introduces a needed localized, interactive, and data-driven platform for teachers, parents, and learners to improve math learning outcomes. After all, teachers remain the creators of exercises and assessments, and MATHtatag merely provides a way for teachers to upload and manage these activities digitally for easier distribution and real-time tracking. The app also assisted learners who struggled with reading comprehension of math concepts via Filipino Text-to-Speech (TTS) to read the questions and instructions out loud so that the parent and child will know how to understand and answer activity. It also championed home-based learning, allowing parents a convenient way to guidance their child's practice, even without printed materials.

As a further response to these concerns, MATHtatag offered learning activities for Filipino language math about MATATAG Curriculum content to ensure inclusivity and comprehension. MATHtatag automated the scoring and recording of learner responses to free-up time for teachers and immediate feedback for the learners, as well as help educators identify learning gaps to remediate them early. A performance dashboard allows teachers and parents to view student progress, spot weak areas, and intervene early in the student's learning before they fall behind. Lightweight and user-friendly, the application was designed to function even in low-connectivity and with minimal technical support. MATHtatag straddles environment, materials, methods, communication, and technology toward a more inclusive, efficient, yet responsive way of supporting Grade 1 Mathematics learning in Philippine classrooms.

Problem – Requirements Matrix

Table 1.
Problem - Requirements Matrix of MATHtatag

Problem	System Requirement
Learners have trouble understanding math questions because of weak reading skills and big class sizes.	• Include a Filipino Text-to-Speech TTS that reads the questions and directions aloud. • Create a simple, colorful interface that young learners can easily navigate. • Allow teachers to create activities that learners can answer both in school and at home.
There are few math materials written in Filipino that match the Grade 1 curriculum.	• Create math activities in Filipino based on the MATATAG Curriculum. • Teachers add or modify questions so it suits their lessons. • All materials can be saved in one secure source for easy access.
Checking paper-based tests takes too much time and delays feedback.	• Use automatic scoring so results appear right after the learner answers. • Show results through real-time dashboards for teachers and parents. • Allow teachers to download reports in Excel format for records.
Parents only learn about their child's progress during report-card release.	• Provide a Parent Dashboard that shows the child's scores and progress any time. • Send notifications and updates from the teacher through the app. • Include short feedback or tips for parents on how to help their child.
Schools lack digital systems and technical help for tracking performance.	• Use Firebase Realtime Database to save and sync all data automatically. • Add an Admin Dashboard to manage users and fix reported issues. • Give a System Administrator control over updates, data maintenance, and troubleshooting.

	• Make sure the app works even with slow internet connections.

PURPOSE AND DESCRIPTION

Purpose

This project aimed to produce MATHtatag: A Mobile Application for Online Assessment and Performance Monitoring with Filipino Text-to-Speech for Grade 1 in Mathematics. The system was developed to support MATATAG Curriculum, primarily by reinforcing the foundation of numeracy in early grades. It aimed to provide a digital platform that makes assessment easier, provides continuous monitoring of the performance of the learners, and offers language support via Filipino text-to-speech to make learning accessible and personalized for Grade 1 pupils and to foster better collaboration among the teachers, parents, and the learners by giving them a common tool that connects school and home in guiding the child's line of progress.

Description

Mathematics, enabling them to answer assessments that go through the official curriculum as well as reads out the tasks and instructions to the children using our trusted Filipino text-to-speech reader, relaying the tasks even to those learners who do not possess reading skills yet.

Teachers are allowed to create exercises within the system, assigned them, then readily view the results via dashboards and analytics listed in real time, which helps them keep

track of the more demanding lessons to adjust what performing lessons are as needed. Parents also gain access to these dashboards too, allowing them to follow their child throughout, and provide backing at home should the need arise. School admins can track overall use of the system & manage the accounts to ensure proper usage and take the lead in overseeing execution.

With the assessment results, the performance monitoring, and a Filipino function, MATHtatag is basically a platform that enhances how mathematics is taught and learnt and catered for Grade 1 learner and so much more. The connection of the school and the home is strengthened, while timely feedback for learners can be also provided as a way to strengthen their foundation in mathematics in line with MATATAG Curriculum.

SPECIFIC OBJECTIVES

To design, develop and evaluate a mobile application that improves Grade 1 learners' performance in Mathematics by giving them accessible assessment tools, learning supportive features such as Filipino text to Speech, and effective monitoring systems that further strengthen collaboration between teachers, parents and learners.

1. To develop a mobile application that enables teachers to create and manage mathematics assessments aligned with the MATATAG Curriculum for Grade 1 learners.

2. To integrate a Filipino text-to-speech feature that helps learners understand questions and instructions more clearly.

3. To provide real-time analytics and progress dashboards that allow teachers and parents to monitor learner performance effectively.

4. To promote stronger collaboration between teachers, parents, and learners in supporting mathematics learning at the Grade 1 level.

5. To evaluate the acceptability of the developed mobile application using the ISO 25010 standards

SCOPE AND LIMITATIONS

The present study discusses MATHtatag, a mobile-based tool designed for online assessment and performance monitoring in the Maths subjects in Grade 1 Mathematics of MATATAG Curriculum with Filipino text-to-speech. The online assessment will be created and sent by the teacher and the learners will answer through their mobile phones that are text-to-audio supported for better understanding of the text for learners.

This study took place at Camohaguin Elementary School, Gumaca, Quezon where data was collected from 45 Grade 1 learners (15 from each class) through purposive sampling. Teachers collected data of learners with an android device and mobile internet connection to allow them to continuously use the system. As trending in the country aligned with national ICT trends, most Filipino households keep mobile phone as the primary device (PSA 2023), hence it was used on a system designed for Grade 1 numeracy skills.

MATHtatag provides teachers and parents access to real-time analytics to monitor learners' performance, while school administrators manage the users and system usage. It was deployed solely on Android as it is affordable for Filipino Families.

The project scope is constrained to the peer-reviewed Grade 1 Mathematics based on MATATAG Curriculum, not covering other Grade levels nor subject matters. Filipino text-to-speech is limited for reading questions and instructions, not for full-fledged conversations and translations. We deploy this system as a means of digital learning support supplement to printed modules and classroom time, not as a full-fledged substitute.

The app relies on connectivity for features such as analytics, account management, and automated reporting, is not instantly accessible or available at all times if at all. It does give assessment results and progress monitoring, but the learning modules and lessons are not automatically generated by the system. A downside is deployment was limited to learners with available devices and mobile data, which may not represent all Grade 1 learners from device-limited and low-connectivity households.

Chapter 2
REVIEW OF LITERATURE AND SYSTEMS

RELATED LITERATURE

The MATATAG Curriculum prioritizes the building blocks of mathematical proficiency in the early grades, and sees mathematics as a means of seeing patterns and solving everyday problems. To support learners' development of numeracy, content standards reflect n/a, number and algebra, measurement and geometry, and data and probability (Department of Education [DepEd], 2023). Learners in Grade 1 are expected to engage with simple graphs, conceptualize basic operations, and identify patterns using concrete and pictorial representations to develop the skills necessary for making informed decisions. The curriculum does describe what knowledge and performance standards we can expect of students at this level, but does not provide guidance on how child-friendly digital technologies, gamified assessments, localized Filipino, and parent-facing analytics will be integrated in support of this early-grade mathematics delivery. Seamless, technology-driven support tools are especially ripe for the relevance of a system like MATHtatag, which operationalizes MATATAG-aligned competencies within a mobile, approachable, and age-appropriate learning platform.

The theoretical underpinnings make the case for interactive and multimodal learning to continue. The Constructivist Learning Theory purports that children are knowledge builders who assimilate new information to past experiences and learn most effectively through contextual, hands-on, and socially supported actions (Kurt, 2021). Exegetic i.e., "reading of visual texts, numerical relationships", is crucial for early mathematics learning but not often translated into digital learning tools in localized formats for Grade 1 Filipino learners. The fillips

of constructivist principles do not protean itself into “localized formats” in tools made for Grade 1 Filipino learners, and rarely embed such helps as Filipino text-to-speech (TTS) of instructions that could help students understand the task. Engagement Theory likewise stresses the importance of interactive, meaningful tasks that leverage technology in their implementation, operating on the principles of Relate, Create, and Donate (Malik, 2021). Most applications that are inspired by these principles are tailored for older students and high-caliber project-based learning. MATHtatag not only hews to this principles of constructing a deep learning environment, but it also fuses the two theories when it operationalizes principles of constructivist theories and engagement approach as it implements engaging design principles through interactive exercises, real-time feedback, and TTS-mediated supports for more active and continued engagement of Grade 1 learners.

National development priorities offer another rationale for digital math learning tools for the early grades. The Philippine Development Plan 2023–2028 describes the need for immediate actions aimed at improve access to and quality within basic education, reform educational assessment system, and strengthen the digitalization of the education sector, especially to recover loss in learning exacerbated by the pandemic (National Economic and Development Authority, 2023). At the same time the PDP accentuates priority areas like foundational skills and digitalization, however it does not specify the types of child-friendly assessment platforms for children, particularly those intended for use in the early grades, that could help achieve these goals. The lack of awareness of one or two platforms outside the classroom that could serve children in such a way makes MATHtatag even more important. Additionally there are ICT statistics from the Philippine Statistics Authority which reveal that “Mobile phone and mobile Internet continued to be the most accessible technologies among

Filipino households" (Philippine Statistics Authority, 2023). The implication of these statistics is that mobile-first, in the home access to learning, may be feasible for children in the Philippines. However, among things to be explored, are the types of technologies that are the most widely available and how they could provide mathematics assessments for early grades and dashboards for teachers.

The expanding research base regarding the benefits linked to gamification and game-supported learning provides ample justification for the inclusion of such an interactive digital approach in mathematics instruction. According to Zainuddin et al. (2020), their review showed that game-based features like points, rankings, and achievement badges increase motivation and engagement in ways that leads to subsequently increased achievement "when closely integrated within solid theories of learning". They also note that vast majority of gamified interventions are short-term, non-theoretical, and directed toward higher education, with few such gamified assessments either setting or application for early grades, particularly in Philippine classrooms or lower tech avenues. Li et al. (2023) further show a large positive effect for learning as a result of gamification; they also note that gamified interventions most often lack discipline-specific and long-term implementation for early mathematics. This supports the "design-based research methodology" of continuing gamified assessments with MATHtatag, integrating with Grade 1 competencies.

Buray and Alferez (2025) found positive results with use of Digital Game-Based Learning framework, able to improve numeracy with interactive activities ranging from counting games to engaging story and topic based activities, with Grade 1 students, that contained data showing that their findings can translate to improved math skills. Threats to validity include small sample size as this study used face to face only data, without mobile or TTS supported

learning pathways. Similar benefits of a structured interactive and play-based, to improve logical reasoning, early numeracy, motivation, and problem solving. Fatima et al. (2021), Warmansyah et al. (2023), and Adipat et al. (2021), all used similar interventions, showing their benefits as well, though only in traditional/hands on, studies, and the results do not offer an extension of work towards mobile, TTS supported, analytics informed environments such as MATHtatag, giving it added importance, then, as the first of its class in this growing field.

This was also found very often in the literature wherein logical reasoning is an important predictor of achievement in mathematics and is influenced by prior knowledge, teaching and learning conditions, and effective learning environment, per Bello (2024). Likewise, the digital tools in the form of digital resources explored in the study of Espinoza-Cedeño and colleagues (2024) show how “interactive technology-based materials can improve logical thinking and autonomous learning” - though only on older students and do not provide interventions for early grades. The study of Mondragon et al. (2023) also reported that digital playful materials improve kindergarten learners’ reading and counting abilities. These all, however, do not align directly with the structure and demands of Grade 1 mathematics as part of the MATATAG Curriculum. MATHtatag, on the other hand, creates digital, structured activities building on competency-based learning outcomes intended to support early-grade learners in honing logical reasoning and numeracy.

Research also underlines the significant value of parent involvement and school–family partnerships for early learning. Joyce Epstein’s School-Family-Community Partnership Model identifies six major forms of family and community participation (including communication and home-based learning support), which promote student success (Vance, 2024). Similarly, Bartolome and Mamat (2020) find that structured home–school engagement helps children’s

whole development and is a significant factor in learning to read, though gaps remain in areas regarding inconsistent implementation and the lack of national policy guidance. Studies of digital home–school communication like that of Pöntinen et al. (2023) show that e-portfolios and digital platforms showing children's work improve transparency of learning processes for parents and promote parental involvement, though varying degrees of accessibility and digital competence can limit effectiveness just as the personalization of folders to unique individuals can be an obstacle. The Annie E. Casey Foundation (2025) reaches similar conclusions when it emphasizes that early and sustained parent involvement leads to better academic, and socio-emotional, development for children. However, across all of these studies, we see little evidence of tools designed to be child-friendly, subject-specific and aimed at ensuring both children and their parents grow together in specific subjects; MATHtatag attempts to fill these gaps through the embedding of parent dashboards and real-time progress monitoring that help families understand student learning and how best to support it, thus operationalizing parent involvement in new ways the existing research has yet to explore.

Alongside parent-facing, teacher-facing learning analytics for tracking progress and provision of additional guidance are also available. Van Leeuwen (2022) notes how good dashboards reflect real-time learner data on comprehensible metrics that guide timely action. However, in most cases, such analytics tools are formulated for older students and large-scale e-learning systems. Early-grade applications are less common, and those that exist often do not borrow from foundational competencies frameworks or take into consideration the facts-on-the-ground lived experiences of Filipino classrooms. This only further communicates the relevance of MATHtatag's minimal teacher dashboard, tailoring performance data, pretest–posttest growth, and activity completion data to Grade 1 learners.

Another important aspect involves reading barriers of young students accessing mathematics tasks; studies of text-to-speech technology by Wei (2024), Raffoul and Jaber (2024), and Wood et al. (2018), show that TTS promotes decoding, fluency, comprehension, and engagement of struggling readers. Although illuminating of the effectiveness of TTS as a pedagogical tool, all of these studies work with western older learners, with outcomes beyond just reading, with a focus mostly beyond early-grade maths, and none looking at Filipino-language TTS or even just TTS in child-friendly digital maths assessments. But MATHtatag uses Filipino TTS to enable emergent readers to more easily access MATHs by providing them with the language they need to follow procedure so that their focus can be on the mathematics, rather than decoding text.

Finally, studies indicating young learners' needs tell us what sort of intervention is required. Aviles (2022) found that Grade 1 pupils in Quezon City continue to struggle with literacy, numeracy, communication, and social participation without the support of highly manualized, classroom-based interventions. Gnawali (2023) talks about linguistic barriers, socioeconomic factors, and further variation in practices complicating young learners' mathematics struggles, but leaves the door open to technological assistive. Finally, large scale assessments referred to in the World Banks' Public Expenditure Review show that after projects like K-12, the share of primary school pupils lacking foundational skills in math remains high. (World Bank, 2020). All these studies point to something missing: plentiful, low-cost, locally relevant, child-friendly platforms accessible on basic phones or low spec computers, with active learning supports for learners on their early math journey.

All in all, our literature paints a clear picture of the need for an easily accessible, curriculum-based, gamified, analytics-powered digital platform for our early-grade Filipino

learners. While some research highlight promise of constructivist learning, gamification, digital tools, parental involvement, learning analytics, TTS, none of them integrate all of that into one system created for Grade 1 mathematics, with MATATAG Curriculum as the backdrop. The gaps we identified: for example, limited early-grade digital interventions, the inadequacy of localized materials, a lack of a mobile-first, parent-supported assessment tool, lead on to the development of MATHtatag. National and research-aligned pedagogically sound digital activities, integrated with widely available mobile technology and focused on foundational numeracy, fulfilling the inexpensive local needs of learners and parents while striving to improve home–school-support systems and early-grade mathematics learning in the Philippine setting.

RELATED METHODOLOGIES IN RELATED STUDIES

Methodologies in educational technology research continue to stress the potential evaluation designs, validated instruments, and analysis provisions for clearly demonstrating the effectiveness of learning innovations. A common method for studies in the same domain is a reliance on quasi-experimental designs, pretest–posttest comparisons to assess the learning afforded by digital access to records. Wang et al. (2020) implemented two quasi-experimental studies that obtained comparisons between students experiencing an AI-based adaptive learning system and a control experience in which students received teacher-led instruction. Detailed random assignment, expert-reviewed item validation, and statistical analyses (ANCOVA, effect size computation) show how such controlled comparisons and standardized assessments can demonstrate the advantages of technology-supported learning. The short length of implementation and less wide reach of topics mean that their

lessons are limited, however, warranting longer-term and more integrated exposures. Our present study benefits from their work procedurally and derives from their strengths a validated pretest–posttest design extended here to Grade 1 Mathematics aligned to MATATAG Curriculum.

In addition to such experimental work, literature also points to the need for timely attention to responsible and fair AI use within educational settings. Chinta et al. (2024) carried out an extensive systematic review of literature where using a rigorous systematic set of inclusion and exclusion criteria in top-tier research databases, the authors screened, evaluated, and synthesized studies to catalogue fairness metrics, bias mitigation procedures, and ethical frameworks in educational AI. While their work is essentially a roadmap for a methodology for assessing fairness in AI, they do not look at implementation at the algorithm, for designing fair child-centric learning tools or assessment systems for early-grade learners. This implies that fairness is perhaps useful but there exist barriers to implementation in terms of “rendering accountability” by translating such principles into adequate methods for the young humans it was conceived to assist. This study indirectly informs the one at hand by pointing to such a gap to be filled along such lines which MATHtatag attempts to do through features like Filipino text-to-speech integration and developmentally appropriate assessment paths.

The value of such gaming pedagogy in the classroom is epitomized by Al Ali et al. (2024) who “examined the effects of gamified teaching on developing the creative reading skills of young learners by conducting a quasi-experimental study”. They validated their pretest–posttest instruments, had their lesson plans reviewed by experts, certified that teachers were adequately trained, and used t-test and ANOVA statistics to examine between-group differences. They found value in gamified digital interventions, but confined the study to

reading, which does not quite align itself to the early-grade mathematics curricula. They also do not touch analytics dashboards nor a mobile-based environment for assessment. This paper aims to address that gap, taking a similar validated pretest–posttest methodology and extending that into a mobile gamified mathematics environment for Grade 1 learners, enhanced with real-time analytics and a Filipino TTS.

Research drawing on design and development also supports child-focused applications. Özbek and Torppa (2025) used a method somewhat similar to that in the present study, employing co-design storms with experts, parents, and teachers iteratively refine designs, and then used controlled pretest–posttest trials of their early literacy application intended for children who may experience reading-related difficulties. Their mixture of qualitative design “refinements” with quantitative “validation” signifies how iterative development improves usability and learning responsiveness, although, notably, they did not venture into curriculum-aligned mathematics and real-time feedback and monitoring dashboards. We follow their outline of using iterative, developmental cycles, but blend that into learning grade 1 mathematics materials with potentially gamified activities, TTS support, and real-time performance monitoring through teacher–parent dashboards.

More sophisticated quantitative methods have been touched on in work on more general learning questions; for instance, Mim et al. (2025) used statistical analyses and explainable machine learning methods to explore predictors of hard early learning, with careful preprocessing and feature selections for their data, SMOTE for the class imbalance, and multiple classifiers evaluated through cross-validation. Their use of SHAP and LIME to interpret model outputs also indicates how machine learning can lead to better understanding of factors that influence learning outcomes. However, their study was based entirely on

secondary data, and it neither designed nor evaluated any instructional tasks so it still left a methodological gap applying that knowledge to study curricular-aligned learning tasks. We follow their systematic data management to some extent, while extending the method toward evaluating a real-classroom intervention we conducted with the MATHtatag app.

Beyond performance-focused approaches, methodologies for calculating a system's value offer important information on how to assess quality of educational software. Manglapuz and Lacatan (2019) conducted a descriptive research design using ISO 25010:2011 Software Quality Model, asking experts to evaluate using a Likert-scale instrument and performing analysis using weighted mean. This gives consideration of various system aspects such as function and reliability, yet lacks measurement of a resulting student performance or accomplishment of curricular learning standards. This is addressed in the present study by integrating ISO 25010 with learner performance including pretest-posttest and monitoring insights from analytics.

Studies in user acceptability research are also valuable for methodology considerations in mobile learning development. Guillen and Nisperos (2024) adopted methodical mapping and expert validation approaches for the creation of an accessible User Acceptability Testing instrument for mobile game-supported learning. Through their work, "we discuss how the well-known models of user acceptance: TAM and UTAUT; and evaluation standards like ISO 9241 and ISO 25010, are used to identify and validate constructs" which provides methodological help regarding user acceptability evaluation. As their tool is still focused on acceptability rather than learning outcomes, further research is needed that combine user experience evaluation with measurable improvement of it through actual performance, a twin-process that this study mimics by combining UAT inspired validation with research on learning gains.

Overall, the methods embraced in this body of background literature suggest a general trend: successful educational technology research will implement strong experimental designs with validated instruments, cyclic development models, structured fairness considerations, and thorough software evaluation frameworks. Very few studies, however, synthesize these methodological strengths into a single intervention focused on early-grade mathematics, mobile-first learning, or localized contexts like the Philippines. Drawing on these methodological lessons, the current research uses validated pretest–posttest assessment, ISO 25010 evaluation, analytics-supported monitoring, and participatory design as a basis for carefully evaluating MATHtatag and its effectiveness as a mobile application for Grade 1 students.

RELATED SYSTEMS

Existing digital learning systems show how mobile, gamified and interactive technologies can enhance children's learning experiences, but they also highlight significant gaps in curriculum alignment, early-grade mathematics coverage, localized language support, and analytics-driven monitoring that the proposed MATHtatag application seeks to address. Many related systems demonstrate the efficacy of gamification in enhancing learner engagement and helping improve cognitive development of young children. Gallera (2023), for example, developed an interactive gamified mobile application including points, badges, levels and rewards that improved preschoolers' performance in puzzles, memory tasks and quizzes. Gallera reported high acceptability rating with promising outcomes for early cognitive development, however, his system was focused on general cognitive and recreational activities and not curriculum-based mathematics. Gallera's system does not include Filipino text to

speech, nor does it include teacher and parent analytics dashboard which are more aligned with MATHtatag needs.

Systems that are also rooted in design-based research show promise through mobile learning environments for early learners. Laranjeiro (2021) outlines the Aprender XXI project, a collection of game-based apps intentionally tied to preschool curricular guidelines that evaluated their efficacy through authentic classroom engagement. While these exploratory and multimodal activities come closer to what is asked of teachers in providing an engaging mobile learning experience, they lack the adaptive but low-stakes assessment, dashboards with learning analytics that explore language-specific elements about the Filipino learner in unknown terms, etc. The “games” also miss the target by focusing on preschool-level content, not NGSSS-mapped or Grade 1 mathematics, while MATHtatag precisely identifies with MATATAG competencies, infusing real-time feedback, Filipino TTS, performance dashboards, etc.

Likewise, the Learning4Kids project presented by Niklas et al. (2020) investigates how app-based interventions may support vocabulary and math skills at home while encouraging parents to facilitate the activities and track usage. Its RCT design demonstrates some methodological rigor, however, the system is designed specifically for kindergarten students in Germany, and relies on parents for math instruction as opposed to linking to a national mathematics curriculum or providing shared analytics to both parents and teachers. This stands somewhat in contrast to the goals of MATHtatag, which seeks to build bridges between home and school learning and assesses the child’s performance within the MATATAG Curriculum.

More sophisticated systems demonstrate AI and student modeling's potential to customize students' learning experiences, some through Bayesian networks (Logic-Muse from Nkambou et al., 2023) and some through deep knowledge tracing. Logic-Muse is specifically targeted to older students learning propositional logic, though, and misses a child-friendly mobile interface and a parent–teacher dashboard. Even while more sophisticated, these systems are not aligned with our goals in MATHtatag and explain why we use child-appropriate adaptive features and Filipino-language scaffolds in the Grade 1 mathematics learners.

Systems that prioritize interactive human–computer interaction design also succeed in leveraging a multimedia-rich platform. The system developed by Li et al (2025), which incorporates HCI principles, utilized gamified activities, non-gamified activities, AR/VR, content segmentation across subjects, instructional content with multiple types (problem sets, theory based quizzes, teaching modules) in a single course, with the outcome from tests showing significant learning gain and high satisfaction rates from users. Again, the system is broader and multi subject based, rather than focused on a specific mathematics curriculum; as such it is less suited to early grade learners who require structured curriculum-aligned progression. Lack of Filipino TTS and role-based analytics brought together also reduces the local applicability of the model in the context of Philippine basic education.

Other systems articulate the importance of online teaching of mathematics through accessible platforms. Ulum (2023) showed that Grade 4 students learning through Khan Academy scored better than colleagues at Turkey's national EBA platform, showing that interactive exercises and logging progress can lead to greater achievement. But these systems are web-based, designed for older students, and convey little localized language support or analytics dashboards for parents and teachers of younger children. What's resoundingly clear

is that online tools can create learning gains, but must be adapted to early-grade development and national curricular structures. MATHtatag does just that.

Gamified quiz platforms such as Wayground (formerly Quizizz) show how digital assessments can encourage engagement, provide instant feedback, and enable independent learning (Smith Budhai, 2025). While very widely used, such general-purpose tools do not map activities to the MATATAG Curriculum, do not come with Filipino TTS or early-grade learners progress dashboards, point to the need for systems that provide a specific curriculum's worth of activities, local language support, and analytics-based monitoring.

Studies of family-focused gamified learning also turn up promising opportunities. Cha-Ching! Money Adventures, which Tally et al. (2022) found to result in statistically significant gains in financial literacy among Filipino children aged 7–10 with strong engagement, has interesting narrative-based modules and features for home engagement demonstrating potential for how gamified mobile tools can initiate parent–child conversations and deepen learning at home, but is not a mathematics teaching system, does not tackle Grade 1 competencies or teacher-facing analytics, gaps that MATHtatag aims to fill with integrated home–school communication and support, curriculum-based content, and dashboards for different role types.

Even works such as Math World by Maitem et al. (Math World, 2012) show the promise of gamification and virtual environment designs to motivate young learners in mathematics. Although using its 3D interface and actually matching Grade 2 textbooks, Math World is still a desktop-based, self-report measure that does not support mobile delivery, TTS, or analytics

dashboards for parents and teachers, a glaring signal of the need for mobile-first, data-informed, curriculum-specific systems that cater to early-grade learners.

Overall, MATHtatag is informed by systems that demonstrate strength in mobile, gamified, and interactive learning tools, but consistently reveal gaps in curriculum alignment, adaptability for early-grades learners, Filipino-language English language learning, and analytics-driven communication between home and school. These inform the development of a localized, mobile, MATATAG-aligned Grade 1 mathematics platform with gamified assessments, Filipino TTS, and a role-based dashboard for teachers and parents, that addresses educational gaps created by these other systems.

SYNTHESIS OF THE REVIEWED LITERATURE, STUDIES, AND SYSTEMS

Taken together, the reviewed studies read as a whole emphasizes an awareness of the need for numeracy foundations among early-graders and in the MATATAG Curriculum, but also gaps in the availability of child-friendly, developmentally appropriate, technology-mediated assessment experiences for learners. The works examined have some common threads including the need for constructivist and engagement-learning environments, the applicability of gamified and digital teaching approaches, parent and teacher involvement, the importance of accessible analytics and progress monitoring, and the impact of reading problems particularly among Grade 1 learners. These align roughly with the priority of moving towards improved assessment practices as contained in the Philippine Development Plan 2023-2028; priority for mobile first assessments as illustrated by the penetration stats for ICT access by the Philippine Statistics Authority. But despite this policy direction, few interventions as

identified in the literature translated these into specific curriculum-aligned digital tools for early mathematics, which points towards the niche for the MATHtatag application".

An obvious thread that connected local and foreign works alike was the desire to build multimodal, interactive learning experiences, especially for very young children who learn by engaging in concrete things often presented visually within meaningful contexts. Again, theory and values in local frameworks such as Constructivist Learning Theory (Kurt, 2021) and the Engagement theory (Malik, 2021) both value interactive and meaningfully situated tasks yet I found little in the literature on systems which aim to turn this theory into an actual localized digital platform catering to Filipino Grade 1 learners. Relatedly, the nationwide campaign to recover pandemic learning losses back to right (National Economic and Development Authority, 2023) hints at the need also for tools accessible by various means under local conditions yet there is little in the literature which folds theory into curriculum needs and forward into a practical digital delivery that is nimble enough for local contexts. This apparent lack of "translation" from theory to national priorities and down to early grade level practice is a unique space that we occupy with MATHtatag which I have mentioned earlier is a mobile, MATATAG-aligned, multimodal learning experience cross enriched with Filipino text-to-speech.

Across studies on gamification and digital learning, there is consensus that interactivity and gamified methods increase motivation, engagement and performance; systematic reviews by Zainuddin et al. (2020), as well as the meta-analytic work of Li et al. (2023) support this effect across grade levels, while local studies such as Buray and Alferez (2025) show gains in numeracy among Grade 1 learners as well. Fatima et al.'s (2021) play-based studies, is followed by Warmansyah et al (2023) and Adipat et al (2021), who also find improvements in logical reasoning as well as problem solving. Many among these interventions are also short-

term, do not completely align to a national curriculum, or for now, do not involve digital, or, maths learning - while literature on game-based/digital strategies suggest great potential, they also remind that these are often not accompanied by analytics in real-time, by the voice of the parent, by TTS capabilities delivering voice overs for emergent readers, or are embedded into early grade mathematics. You can begin to see why MATHtatag's design would be more holistic than most.

Another recurring note is that logical ability and logical reasoning training promotes success in mathematics. Several studies, including Bello (2024), Espinoza-Cedeño et al. (2024), and Mondragon et al. (2023), show how much direct effect such logical thinking has on learners' school performance, and how important it is to support learners' development of logical skills with digital tools. Most of these tools are for older students, not designed for focused on literacy or cognitive skills, or not aligned to structured curriculum standards, suggesting a gap in early-grade, mobile-delivered, competency-based mathematics learning that MATHtatag begins to address with its MATATAG-competencies-aligned structured activities.

Parental involvement is also a key aspect, as stressed in Joyce Epstein's model (Vance, 2024) and also evident in local research such as Bartolome and Mamat (2020). Digital platforms such as those described by Pöntinen et al (2023) illustrate the potential of home–school partnership with technology, but generally do not have personalization and subject focus, or provide parents insights that are actionable. A report from the Annie E. Casey Foundation (2025) also emphasizes how consistent family engagement can contribute to improved learning. Yet common digital techniques to boost learning isn't kid-safe, as far as subject-focused analytics or parent dashboards to help early mathematics goings-on and

progress. Enter an important role offered through MATHtatag, filling this void of parent dashboards and real-time insights to empower families to help kids in Grade 1 really shine in math.

Teacher-facing analytics are also a prominent pattern. Van Leeuwen (2022) mentions the key role of dashboards as useful in providing teachers with the proper data for intervention on-the-fly and preplanned informed instruction. Most existing visual dashboards support use by teachers of secondary and older students, though few described in the literature affordances for adaptations for very young children or can easily integrate with the curriculum targeting students in early grades. The literature seems to consistently go back to the point that teachers need simple, actionable, but developmentally appropriate analytics, not high-volume flow graphs and numbers that loom text-heavy and complicated. MATHtatag meets this need also by providing teachers with essential small bites of progress indicators contextualized in MATATAG competencies.

Reading barriers are threads that course through the literature. Text-to-speech technology comes up repeatedly as an important support for learners who struggle in reading. Wei (2024), Raffoul and Jaber (2024), Wood et al. (2018) show across studies that, TTS does have effects that significantly help students “decode, read fluently, and comprehend, among students with reading difficulties (Sleeman et al., 2022)” with no negative performance impacts on others. But these studies focus upon writers’ center age group conditions, on reading or literacy tasks rather than those in mathematics, and do not speak specifically about Filipino-language contexts, leaving a gap in our early-grade mathematics instruction. MATHtatag fills this gap with application using Filipino text-to-speech, running alongside each learner as emergent readers to better follow instructions and focus on thinking mathematically.

The learning crisis is laid out in studies such this by Aviles (2022), Gnawali (2023) and the World Bank (2020): So pushes the need for specialized tools for accessible support, proper delivery and principle intervention - kids are left behind not only by syllabus, and vocabulary, but also coaches, cash and class. Yet most tools are manual, verbally delivered and not web-based. So the lessons here advocate for mobile delivery for universal learners, MATHtatag is designed as such, and helps users along their learning journey.

The review by others is on some of these gaps; other mobile apps (Gallera, 2023; Laranjeiro, 2021; Niklas et al., 2020) hint at the potential of gamified and mobile environments but not in terms of curriculum scope, analytics, or early grade math or Filipino-language scaffolding. More advanced systems such as Logic-Muse (Nkambou and colleagues, 2023) highlight how AI may be applied for educational purposes but address older students and whose interfaces are not developmentally fit. HCI-based systems (Li et al., 2025) as well as Khan Academy (Ulum, 2023) as an online platform show interactivity or progression tracking some of the power of these modalities but, again, do not serve Grade 1 Filipino learners and do not use TTS or parent–teacher dashboards. Even successful family-oriented tools like Cha-Ching! Money Adventures (Tally and colleagues, 2022) hint at how mobile devices can support gamified learning but do not operate within mathematics, and are not aligned to MATATAG competencies. The comparisons illuminate consistency: lack of early grade focus, lack of Filipino support, lack of parents analytics, and critical curriculum alignment gaps for MATHtatag.

Across the review across studies/systems led to a synthesis showing that although many of the interventions support gamification and digital learning, parental involvement, analytics, and TTS - no tool reviewed combined all of these into one cohesive, developmentally

appropriate, mobile-first, curriculum aligned application for Grade 1 mathematics. This is a clear gap in research and practice that MATHtatag addresses. By ensuring the pedagogy aligns with MATATAG standards, embedding Filipino TTS, facilitating real time analytics for teachers and parents, and providing gamified exercises accessible on very common mobile devices, MATHtatag extends theory and practice into early-grade mathematics instruction.

Chapter 3
METHODOLOGY

REQUIREMENTS ANALYSIS

MATHtatag is an application designed to enhance the way that learners aged 0-8 experience mathematics through fun and interactive exercises as opposed to more traditional means (i.e., paper and pencil). The app has been developed to provide learners with an easy and engaging method of learning mathematics while offering teachers the ability to assess their students digitally and request instant feedback about their students' progress, thus eliminating the delay associated with checking homework and the limited ability of the school to communicate with parents.

There are four primary users of the app: teachers, parents, learners, and administrators. Teachers can create exercises and review results; parents can follow their children's progress throughout the year; learners can participate in directed learning activities using a text-to-speech (TTS) option; and administrators manage the users of the app. The app also has a software developer who provides maintenance, updates, and support for any technical issues experienced by users of MATHtatag.

In summary, MATHtatag provided teachers, learners, and parents with one simple platform to utilize for supporting the teaching, learning, and assessment of mathematics across one unified system that helped promote the mastery of the Grade 1 mathematics curriculum, while also enhancing the level of engagement of learners.

Requirements – Features Matrix

A Requirements-Features Matrix was created to record and structure the needs of the system, allowing each to be traceable and verifiable by a unique feature. This also became a point of reference to validate that the functionalities had been developed to meet the requirements of all users. The following table displays the primary requirements and features related to the MATHtatag application.

Table 2.
Features Matrix of MATHtatag

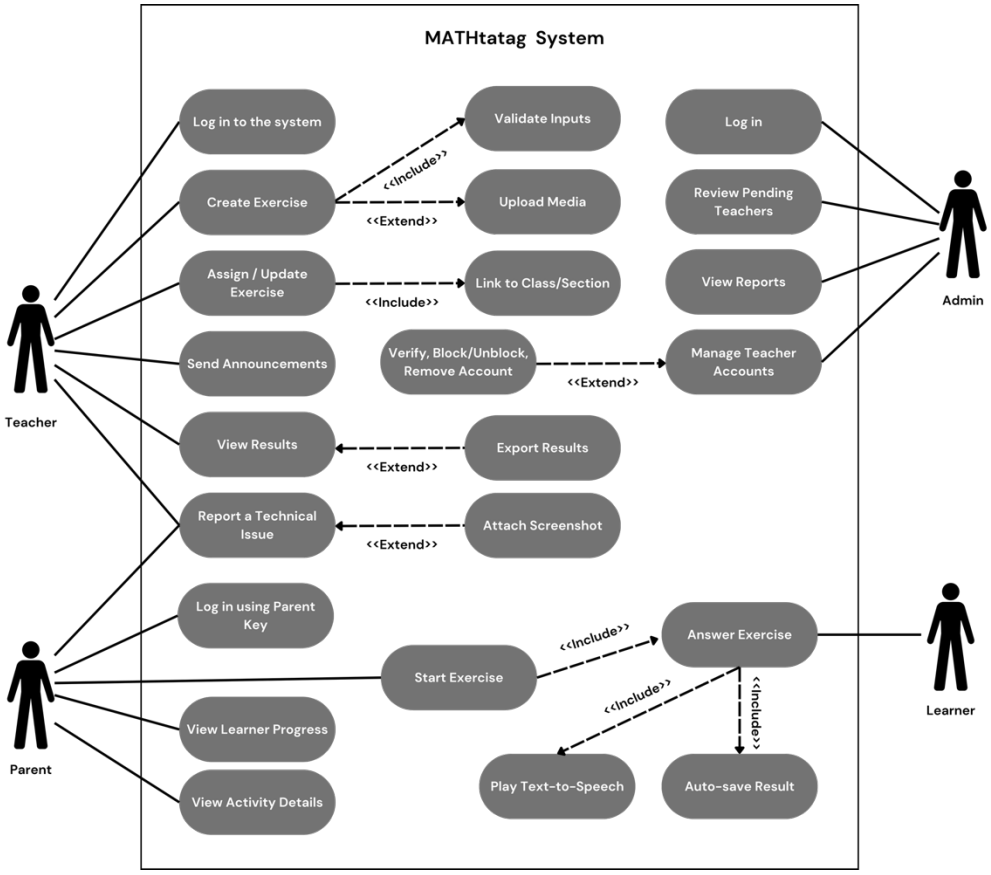
Feature	Description	Importance	Notes
Teacher Dashboard	Displays the list of learners, their performance results, average scores, and completion rates. Provides options to export reports and monitor results per quarter.	High	Helps teachers track learner performance efficiently and provide timely feedback.
Exercise Creation and Management	Allows teachers to create, edit, and assign math exercises aligned with the MATATAG Curriculum. Supports uploading of images and audio for activity instructions.	High	Simplifies activity preparation and promotes localized, curriculum-based assessments.
Interactive Activity Interface	Enables learners to answer digital exercises such as matching, multiple choice, and re-order type questions. Provides instant scoring and feedback.	High	Keeps learners engaged and promotes self-paced learning with immediate correction.
Filipino Text-to-Speech (TTS)	Reads questions and instructions aloud to assist learners who are still developing reading skills.	High	Enhances understanding, inclusivity, and accessibility for young learners.

Parent Dashboard	Shows the learner's scores, activity summaries, and progress by quarter. Allows parents to monitor and support learning at home.	High	Strengthens parent involvement and encourages home-based learning support.
Admin Management Panel	Enables administrators to verify teacher accounts, manage users, and review system usage reports.	High	Ensures proper access control and smooth management of user activities.
System Developer Panel	Allows the developer to handle maintenance, update notifications, technical reports, and management of TTS API keys.	Medium	Keeps the application updated, stable, and secure during deployment.
Firebase Authentication	Provides protected account access with role-specific permissions for teachers, parents, as well as administrators.	High	Protects user credentials and prevents unauthorized access.
Firebase Realtime Database	Stores, updates, and synchronizes user data, results, and activities across all devices in real time.	High	Ensures data accuracy and reliability throughout the system.
Technical Report Module	Allows users to submit reports regarding errors or technical issues with screenshots.	Medium	Helps in quick identification and resolution of system problems.
Maintenance and Update Notification	Displays update alerts and maintenance messages for users during app startup.	Medium	Keeps users informed of new versions and ensures app reliability.
Child-Friendly Interface	Uses simple layout, bright colors, large buttons, and visual icons suitable for Grade 1 learners.	High	Improves usability, comprehension, and learner engagement.

Export and Printing Module	Allows teachers to export class results and performance data into Excel or printable reports.	Medium	Supports documentation and school reporting needs.
Data Privacy and Security Compliance	Follows data protection measures under the Philippine Data Privacy Act (RA 10173).	High	Safeguards sensitive information and builds user trust.

Use Case Diagram

Figure 3. Use Case Diagram



The use-case diagram shows the different ways that each group of users (Teachers, Parents, Learners, and Administrators) can interact with MATHtatag. A Teacher uses MATHtatag to manage Classroom tasks such as creating exercises for their students,

assigning activities to students, sending announcements to students and/or parents, and, finally, reviewing the results of each student in their class. Parents can use MATHtatag to check their child's progress and see what specific activities were completed. Learners interact with MATHtatag by beginning and answering exercises and using the supported features like Text to Speech and auto-save; and, lastly, the Administrator oversees Teacher Verification and manages accounts for all of the users in MATHtatag. Together, the above mentioned groups of users and their interactions with the MATHtatag platform demonstrate how they all work together in an integrated manner to support seamless learning, tracking, and management of MATHtatag.

Use Case Report

Table 3.
Use Case Report

Use Case	Description	Actors	Preconditions	Postconditions	Main Success Scenario
Teacher Creates an Exercise	Allows teachers to create and publish interactive activities for learners.	Teacher	1. Teacher is logged in. 2. Internet connection is active.	New exercise is saved and linked to a class or section.	1. Teacher logs in. 2. Opens “Create Exercise” form. 3. Inputs title, instructions, and items. 4. Adds image or Filipino TTS audio. 5. Validates and saves the activity. 6. Publishes it for the class.

Teacher Sends Announcement	Allows teachers to send announcements to all learners or specific classes.	Teacher	1. Teacher account is verified. 2. Internet connection is active.	Announcement is displayed on parent and learner dashboards.	1. Teacher logs in. 2. Opens “Announcements” tab. 3. Types a message and selects a class. 4. Sends announcement. 5. System updates Firebase node and shows message in dashboards.
Teacher Views and Exports Results	Enables teachers to view and analyze class performance with the option to export to Excel.	Teacher	1. Teacher account is active. 2. Students have completed exercises.	Teacher successfully exports results as an Excel file.	1. Teacher logs in. 2. Opens “Results/Analytics” page. 3. Selects a class and quarter. 4. System displays results. 5. Teacher taps “Export to Excel”. 6. File is generated and saved or shared.
Learner Answers an Activity	Enables learners to answer assigned exercises with Filipino TTS and instant feedback.	Learner	1. Learner’s account is linked to parent key. 2. Exercise is assigned and active.	Answers are saved with computed score and attempts.	1. Learner opens app via Parent Key login. 2. Selects assigned activity. 3. Listens to TTS instructions. 4. Submits answers. 5. System checks and gives feedback. 6. Results auto-save in Firebase.

Parent Views Learner Progress	Lets parents track learner performance per quarter and per activity.	Parent	1. Parent key is valid. 2. Learner has recorded results.	Updated summary of progress is displayed.	1. Parent logs in using key. 2. Opens "Progress" page. 3. Selects a quarter. 4. System shows activity scores and completion data.
User Reports a Technical Issue	Allows users to submit an issue report with optional screenshots.	Teacher / Parent / Admin / Developer	1. User is logged in. 2. Issue encountered during app use.	Report is recorded and assigned to developer for review.	1. User opens "Report Issue". 2. Describes problem. 3. Optionally uploads a screenshot. 4. Submits report. 5. Developer receives and reviews the report.
Admin Manages Teacher Accounts	Admin verifies, blocks, or removes teacher accounts and handles reports.	Admin	1. Admin is logged in. 2. Teacher accounts exist in pending list.	Updated teacher status is saved in the database.	1. Admin logs in. 2. Opens teacher accounts list. 3. Selects pending account. 4. Verifies or blocks teacher. 5. System updates account status.
System Developer Applies Updates and Maintenance	Developer manages app maintenance, version updates, and TTS API keys.	Developer	1. Developer credentials are verified. 2. Firebase connection is active.	App is updated or maintenance mode is toggled.	1. Developer logs in to panel. 2. Reviews reports and system logs. 3. Updates API keys or deploys fixes. 4. Activates maintenance mode if needed. 5. Saves updates and notifies users.

User Logs In / Out	Authenticates and ends a user session.	All actors	1. User credentials or Parent Key are valid.	User is directed to their dashboard; session stored or ended.	1. User opens app. 2. Inputs credentials or key. 3. System validates login. 4. Redirects to assigned dashboard. 5. On logout, user session clears.

DESIGN SPECIFICATIONS

The MATHtatag system was designed to provide an interactive and accessible platform for Grade 1 mathematics learning. It presented questions and instructions in a more interactive, simple, and accessible manner via online assessment, real-time scoring, and Filipino text-to-speech for guiding the learners to ask questions. The app becomes the 'glue' of teachers, parents, learners, and administrators into one mobile system and allows for interactive and seamless collaboration toward progress.

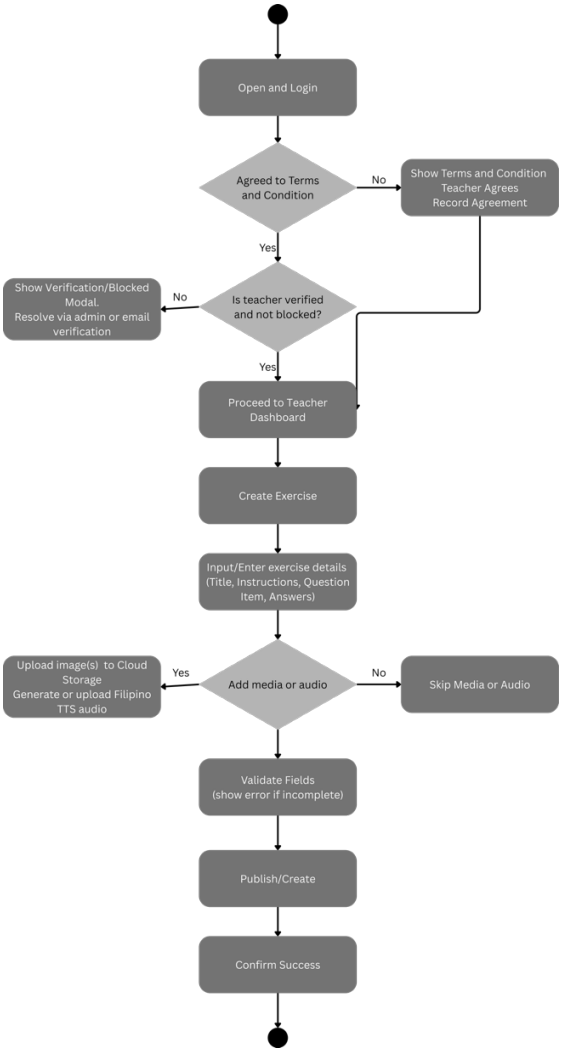
MATHtatag app was built using React Native with Expo for cross-platform support. It's optimized for use on Android devices, uses Firebase Authentication for sign-in, the real-time simulator stores information using the Firebase Realtime Database, and stores media using Firebase Cloud Storage. Updates are delivered via Expo Updates to keep the system current and stable.

Everyone has their own module – Create and assign exercises for the teacher, answering activities for the learner with voice guidance. The parents can view their child's progress and, finally, the administrators can manage accounts and reports. The system

developer updates, and the design follows a simple memorable child-friendly interface with large buttons, and attractive images for an easy touch and go experience.

Activity Diagram

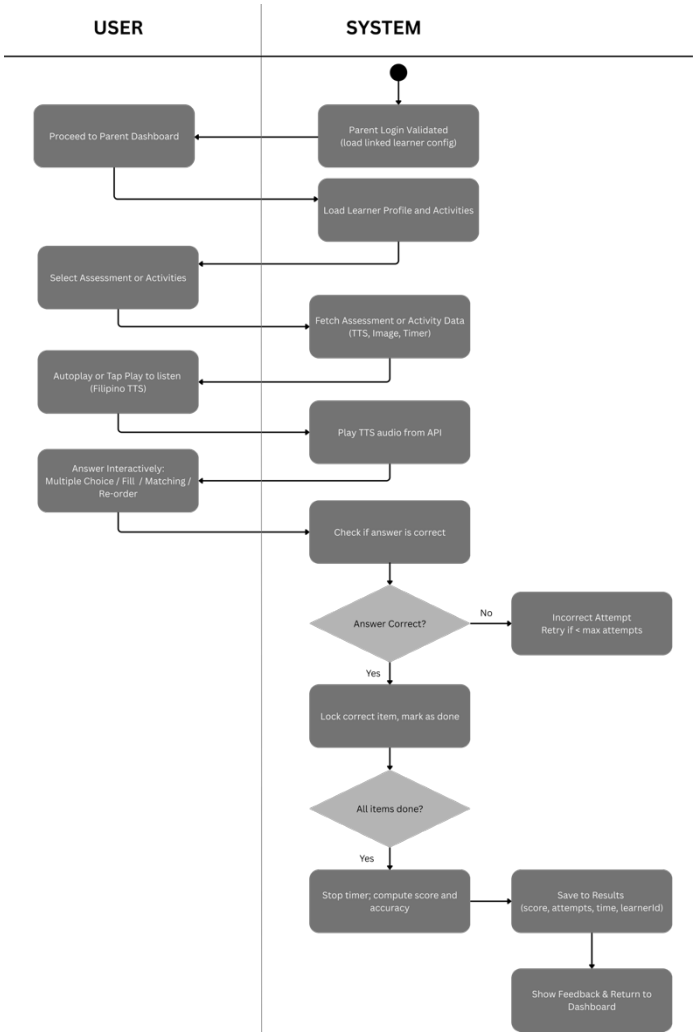
Figure 4. Teacher Creates an Exercise



This flowchart shows step by step how would a teacher publish and create an exercise on the system, starting from the login to the application and accepting the terms and conditions.

The agreement is noted by the system and checks if the teacher's account is verified and is not blocked, if its verification is needed, the teacher would be requested do so by the admin or email. When confirmed the teacher would be taken to the dashboard and select "Create Exercise". The teacher then enters the exercise information including the title, the instructions, the questions and the answers. The system provides for image upload or TTS (Filipino Text-to-Speech audio). If Media is included it is uploaded to the cloud storage, the system checks all information and if a field is not filled, it displays an error message. When validated the teacher Uploads the exercise, the system stores it into the database and notify that creation is successful.

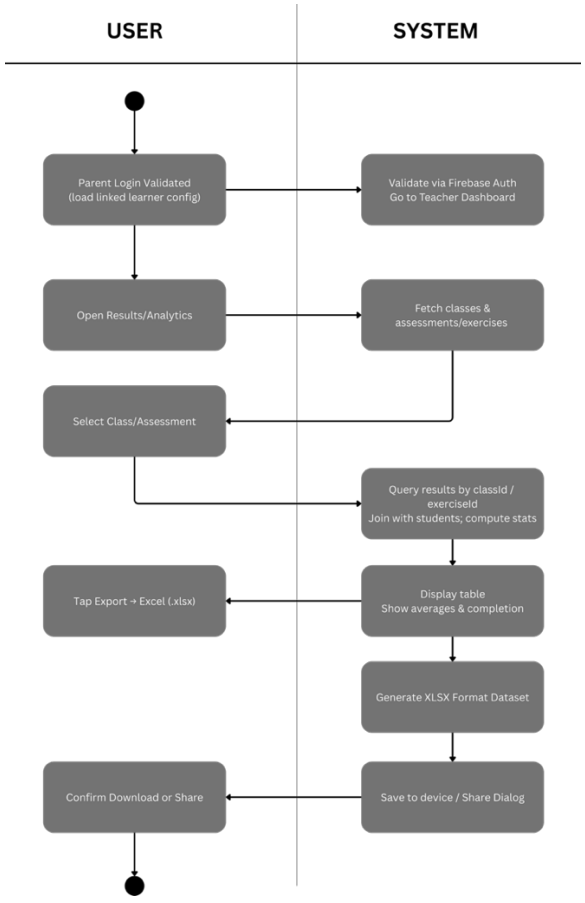
Figure 5. Learner Answers an Activity



This illustration explains how a student interacts with MATHtatag while answering a corresponding activity, starting with the parent entering using the Parent Key, followed by the system loading the student's account and corresponding activities, prompting the student to select an exercise assigned to him, acquire the associated information, which consist of questions, images, audio, Filipino Text-to-speech (TTS) directions, listen to the TTS, and answer each item using matching, reorder, or multiple choice. The system provides feedback, prompting whether the submitted answer is right or wrong. Incorrect answers can be attempted

again until the limit to maximum attempts has been reached, but when the student answers the questions or when limit to maximum attempts has been reached, the system would automatically compute the score, store the result in the database, and present the summary of the student's performance on the screen.

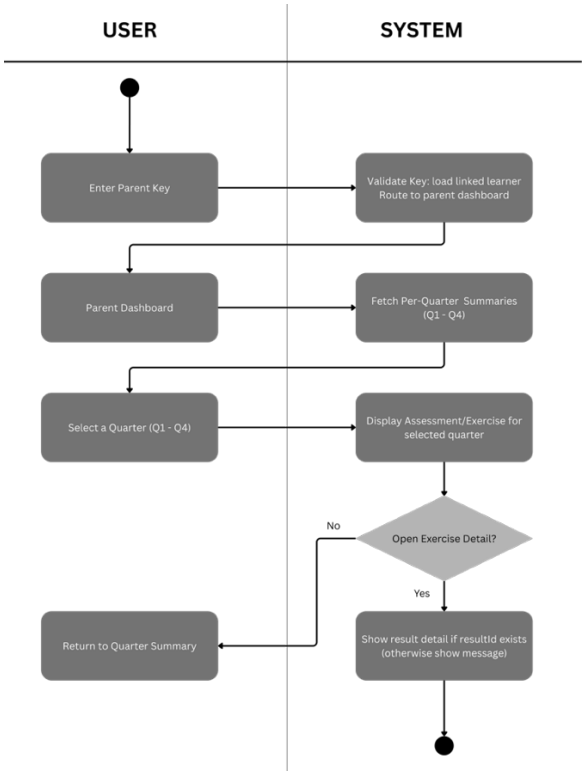
Figure 6. Teacher Views and Exports Results



The diagram depicts how teacher views and exports learner result in the system. The process starts when teacher accesses the system via Firebase Authentication and is taken to the dashboard where the classes and assessments they have access to are loaded. After displaying those, the teacher opens the results or analytics portion and selects a class or activity to view. The system queries the database, grabbing all the learner data for that class

and activity, joins with the actual class and exercise records, and calculates averages and completion rates. All of the results are displayed in table format so the teacher can see each learner's score and progress. When the teacher taps the "Export to Excel" option, the system generates an .xlsx file populated with the full dataset and asks the teacher to save or share. On success, the entire file has been saved or shared and a confirmation message can be seen.

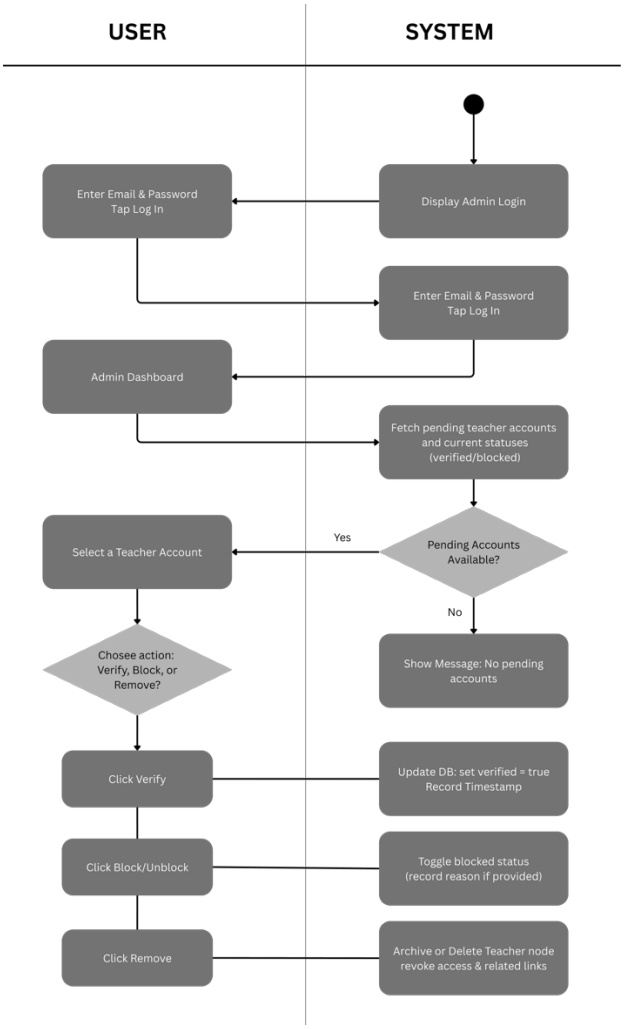
Figure 7. Parent Views Learner Progress



This diagram shows a parent checking on their child's progress using the app. It happens when the parent types in their Parent Key. The system verifies it is ours by loading the attached learner's account and sending us on our way to the parent dashboard. Once there, we can get a summary of our learner or by quarterly report (Q1–Q4). After picking select a quarter, the system collects and presents all associated activities or assessments for that

quarter. If the parent selects to open up an exercise, the system generates and presents the results, scores, and percent completed (if available). If no record exists, a message informs the parent. Our parent then returns to the quarterly summary to view other progress reports or to sign out.

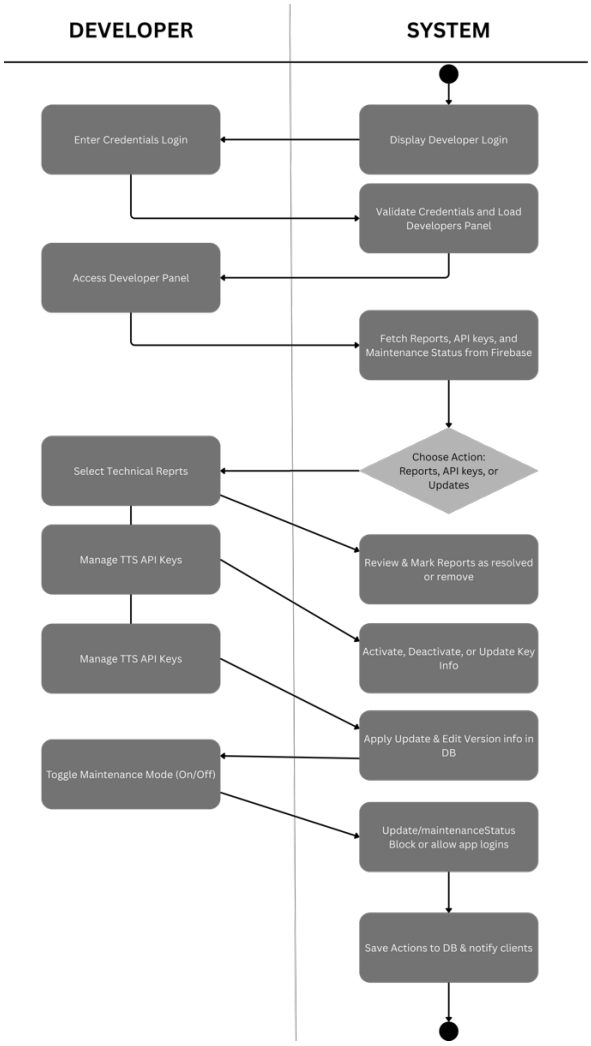
Figure 8. Admin Manages User Accounts



This diagram represents how an admin module can manage the teacher accounts. This process will start when the admin logs in through an email-and-password account. Afterward, the system will validate the submitted input and take them to their dashboard and retrieve all

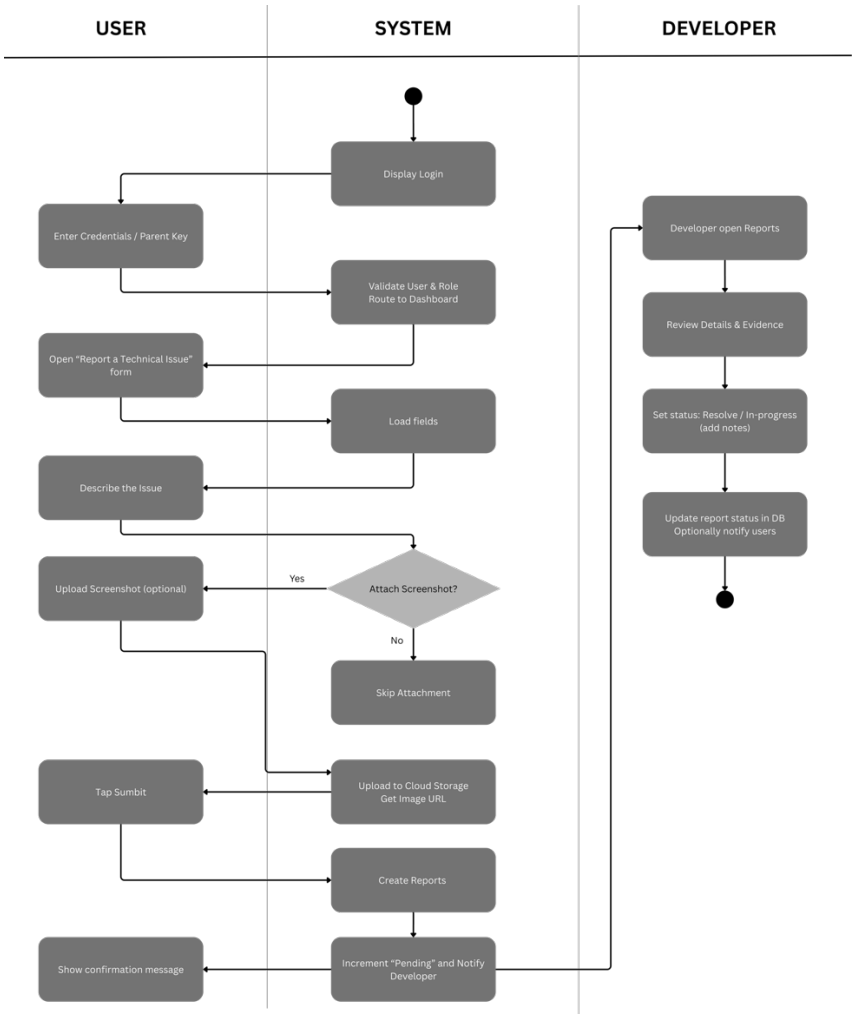
the pending teacher accounts and their verification/block status or not. If there aren't any pending accounts, a message is displayed. If there are, the admin will choose a teacher and choose to verify / block/unblock or delete the account. When verifying the account, the system updates the database setting it to "verified" and records this. If the admin blocks or unblock the teacher then the system will toggle the account's status and perhaps stores a reason. If the teacher is deleted, the system will delete or archive their account node and revoke permissions, and remove relevant links from the database.

Figure 9. System Developer Handles Maintenance and Updates



This diagram presents how a developer supports and manages updates for system. It all starts by logging in with their credentials. When the login is confirmed, the system will load the Developer Panel appropriately, then the Developer Panel will retrieve the most recent reports, TTS API keys, and maintenance settings from Firebase. After this, the developer will choose what to do, whether to look at reports, manage API keys, or proceed with updates to system. When developer views reports, the developer will either mark a report as resolved, or remove a completed report. When the developer manages the TTS keys, they will activate or deactivate, or update the TTS keys as seen fit, in order to maintain the Text-to-Speech function. When a system update uses the developer will make updates, change the version in database, and enable maintenance mode, if necessary. When maintenance is enabled email logins by users are temporarily disabled to ensure users do not log in and encounter interruption. Lastly, once the maintenance is complete and everyone else wraps up their work the system will update the database and notify the users when everything is back to normal, there will potentially be a refresh period time.

Figure 10. User Reports a Technical Issue

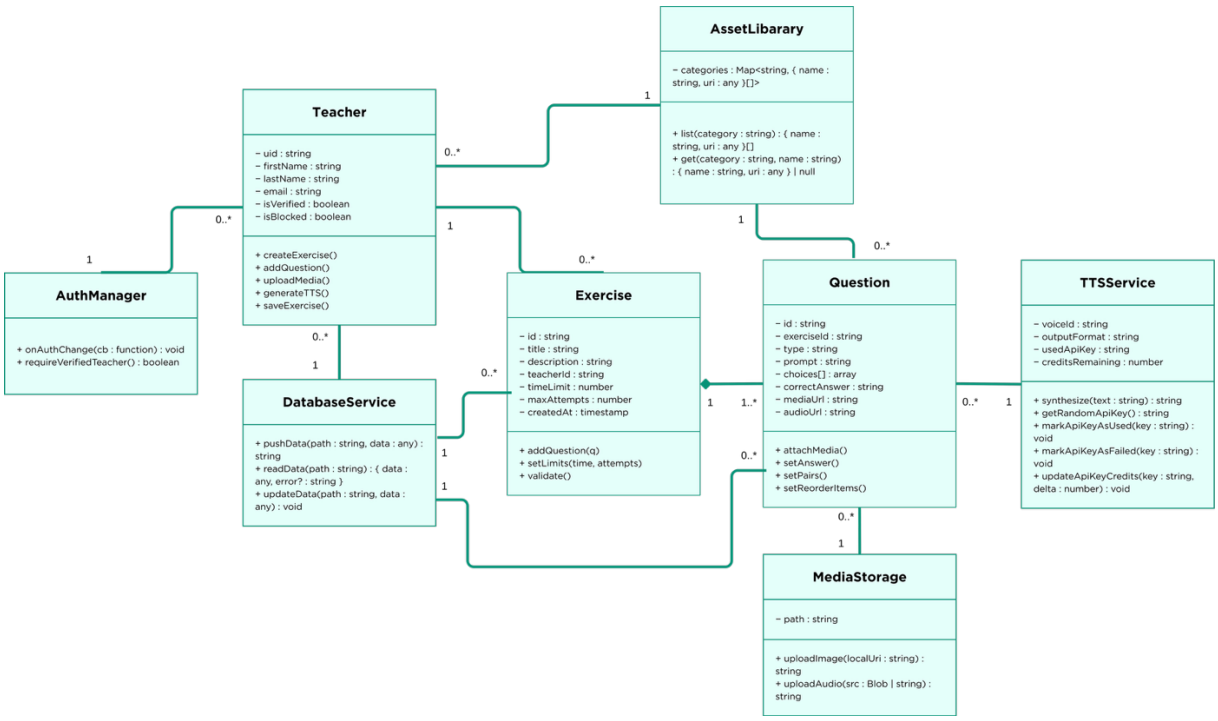


This diagram shows how Users report technical problems via the app. The process starts when user logs in using their credentials / Parent Key. Once logged in, the system checks the assigned role of the user and opens the appropriate dashboard. The user then chooses the form for “Report a Technical Issue” and completes the details of the problem and may attach a screenshot for clarity. If the user uploads an image, it’s stored in Cloud Storage and the system provides a link to it. With the form filled in, the user submits it, and the system saves the report to the database under a “pending” status and sends notification to a

developer. After receiving the report, the developer receives the report and checks on it by collecting the details and attachments, and then updating the status to “resolved” or “in-progress,” writing notes if necessary. Finally, the system updates the database and optionally sends the user a notification to let them know of the progress.

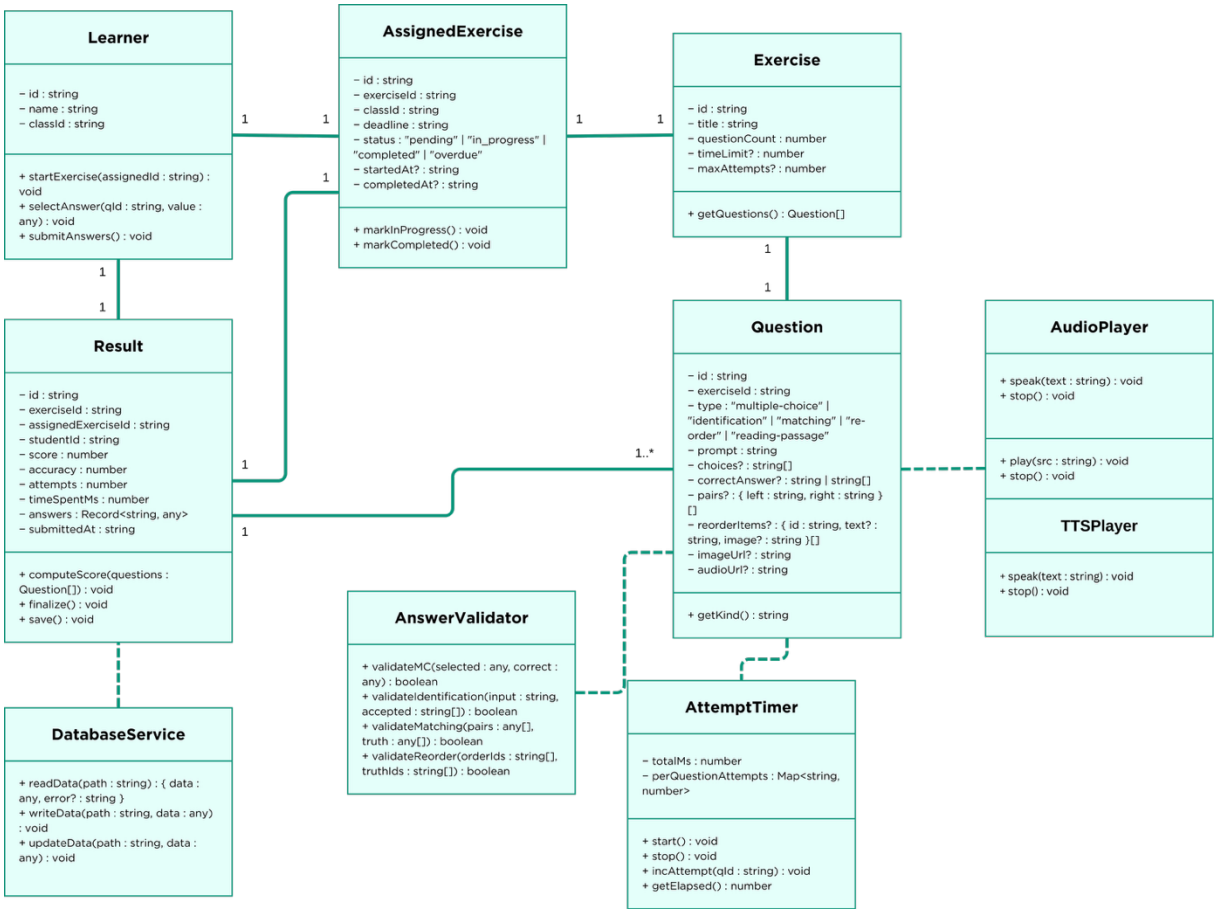
Class Diagram

Figure 11. UML Class Diagram of How Teacher Creates an Exercise



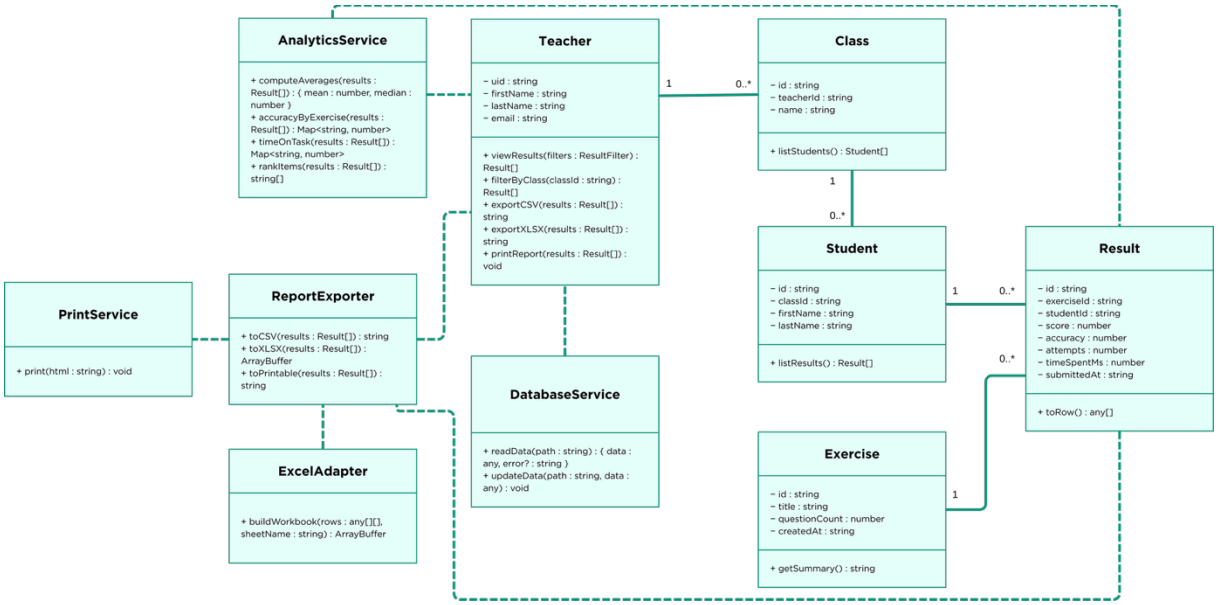
This diagram reflects the object relationship and how the classes interact when a verified teacher creates a new exercise. It indicates that the Teacher class composes an Exercise (which is composed of Questions), that the services for uploading media files and storing audio (using text-to-speech) are in the MediaStorage and TTSService class, and that the service for saving to firebase is in the DatabaseService class. The classes support authentication for selecting images include AuthManager and AssetLibrary.

Figure 12. UML Class Diagram of Learner Answers an Activity



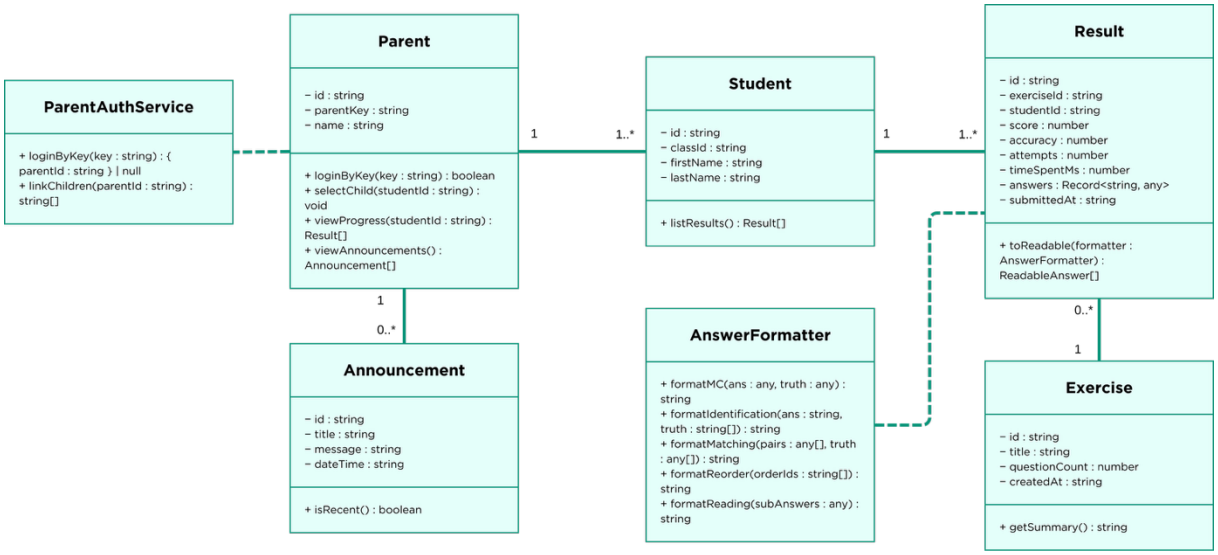
The structure shown in this diagram illustrates the relationships between classes when a learner answers an exercise. The Learner has an association with the AssignedExercise class which is associated with an Exercise class that has a multiple Question objects. The supporting classes, AnswerValidator, AudioPlayer, AttemptTimer, and the Result classes are the classes responsible for validating an answer, playing audio, keeping track of attempts, and determining the result, respectively. Finally, the DatabaseService class is the class that reads and writes learner information.

Figure 13. UML Class Diagram of Teacher Views and Exports Results



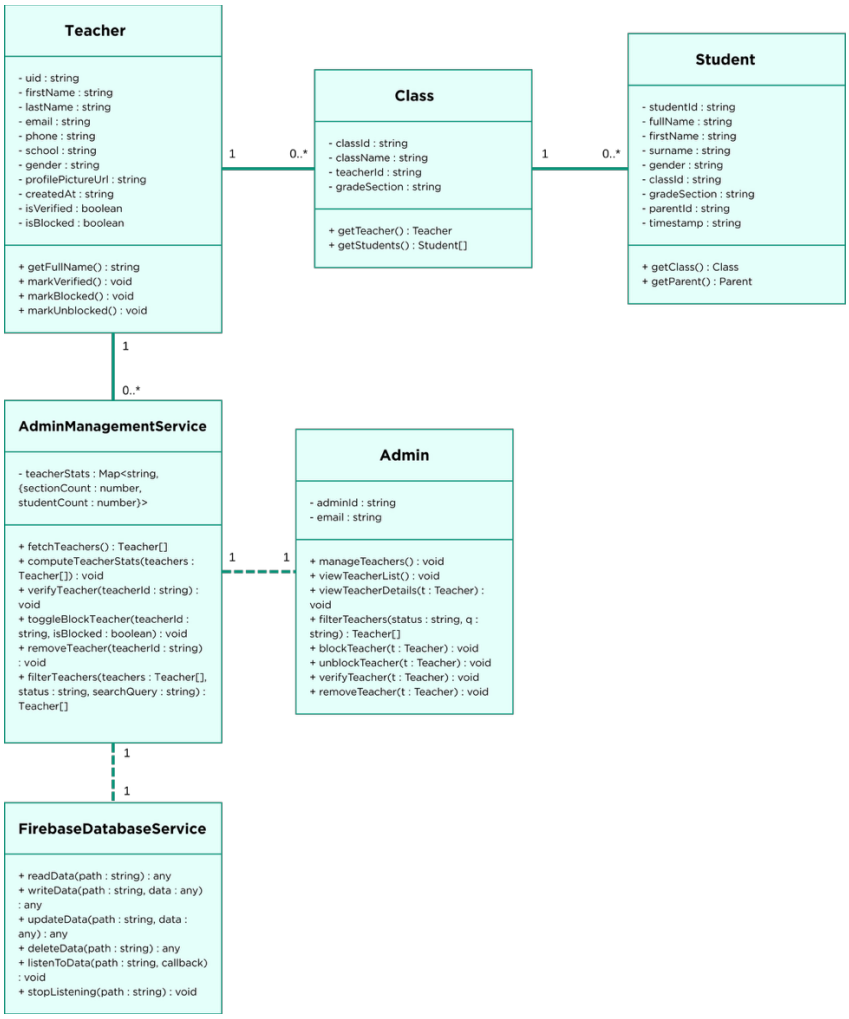
This diagram shows the process of a teacher retrieving, analyzing, and exporting the results of learners. The Teacher will access Class, Student, Exercise, and Result data via DatabaseService. AnalyticsService conducts the analysis, while ReportExporter, ExcelAdapter, and PrintService are responsible for producing CSV/XLSX and a printable report for classroom analytics.

Figure 14. UML Class Diagram of Parent Views Learner Progress



This diagram illustrates the classes and interactions that allow parents to track their child's learning progress together with announcements. The class Parent connects to one or multiple Student objects, each of which has a Result and Exercise data. The ParentAuthService verifies that the parent access is valid using a unique parent key, AnswerFormatter changes the stored answers into a readable version, and DatabaseService retrieves the results and announcements accomplished by the parent and student objects.

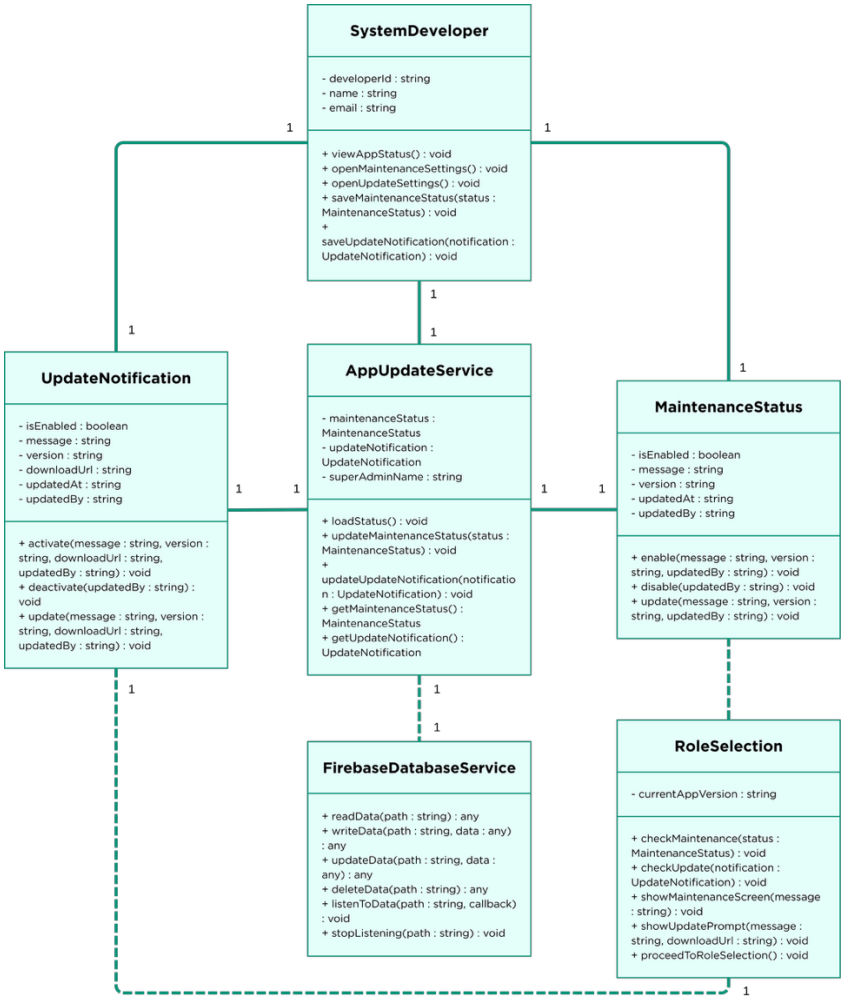
Figure 15. UML Class Diagram of Admin Manages User Accounts



This class diagram depicts how the Administrator manages accounts for teachers within this online school management system application. The Admin can view, verify, block, or remove and actions above accounts through this AdminManagementService which contains all logic regarding accounts. All Teachers will have their profile attributes and a status attribute, and live in a Class object that stores students; the Admin is able to judge based on the pupils enrolled what the teacher has been up to. This FirebaseDatabaseService helps with that, giving live access to stored data on the teacher. To round this all off, this class diagram

represents communication of one kind or another between the Admin, the account management functions and teacher, class and student data.

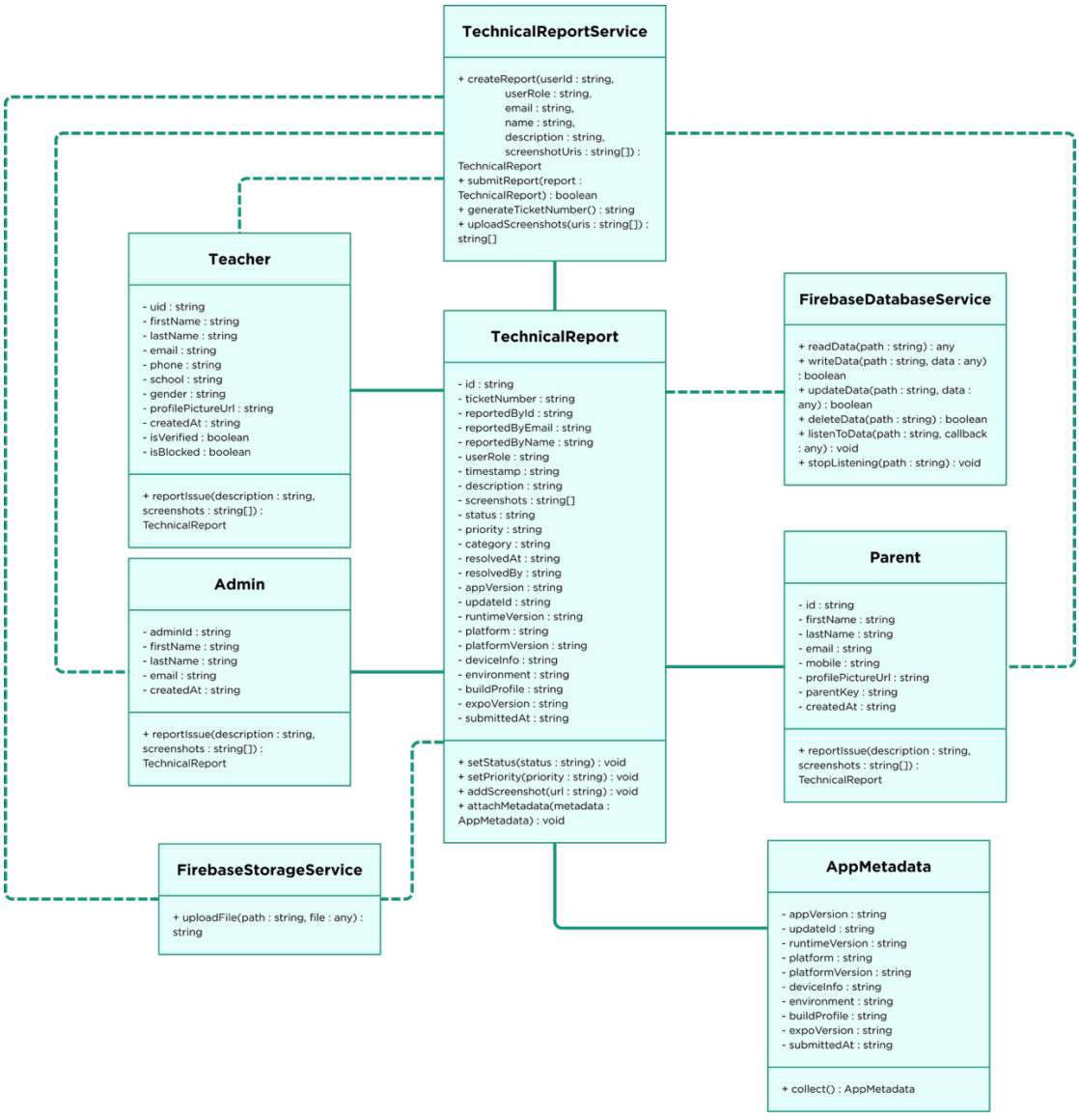
Figure 16. UML Class Diagram of System Developer Handles Maintenance and Updates



This class diagram illustrates the System Developer's roles in managing maintenance mode and update alerts in the app. The System Developer interacts with the AppUpdateService to load, update, and save maintenance and update alert configurations globally, saved as single instances of MaintenanceStatus and UpdateNotification. The states are written and read in real time via the FirebaseDatabaseService. The RoleSelection screen

queries these configurations to determine whether to let users into the system, or if they need to be prompted to update the app. Overall, the class diagram shows how the developer has the power to control application availability and update versions in a centralized service using Firebase.

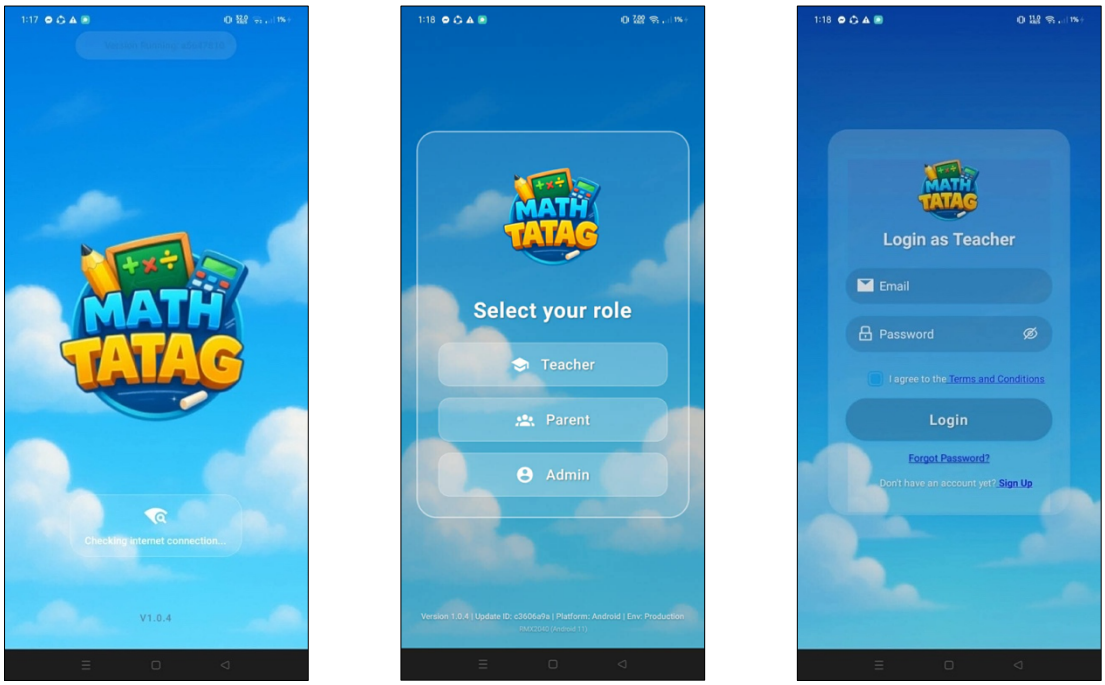
Figure 17. UML Class Diagram of User Reports a Technical Issue



This class diagram outlines the process of reporting technical issues by different users of the system; Teacher, Parent, and Admin. Each user is able to submit a report that gets inserted in the system as a TechnicalReport record that includes data on the technical issue that was encountered, as well as screenshots of the issue and AppMetadata that is automatically collected such as the app version and device used. The TechnicalReportService class takes care of submitting the report and creates the TechnicalReport record to upload report data and screenshots to storage through the FirebaseDatabaseService class and the FirebaseStorageService class. This allows the developer team to have all of the reported technical issues properly recorded, documented, stored, and available for review and troubleshooting.

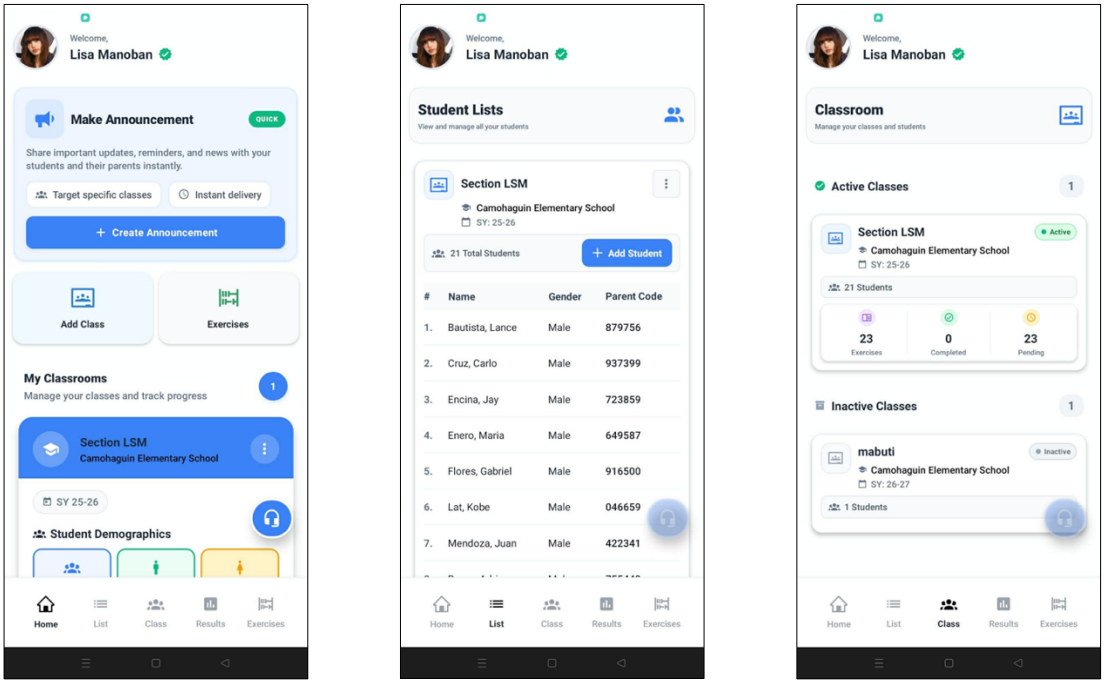
GUI Design

Figure 18. MATHtatag splash screen, role selection, and login UI



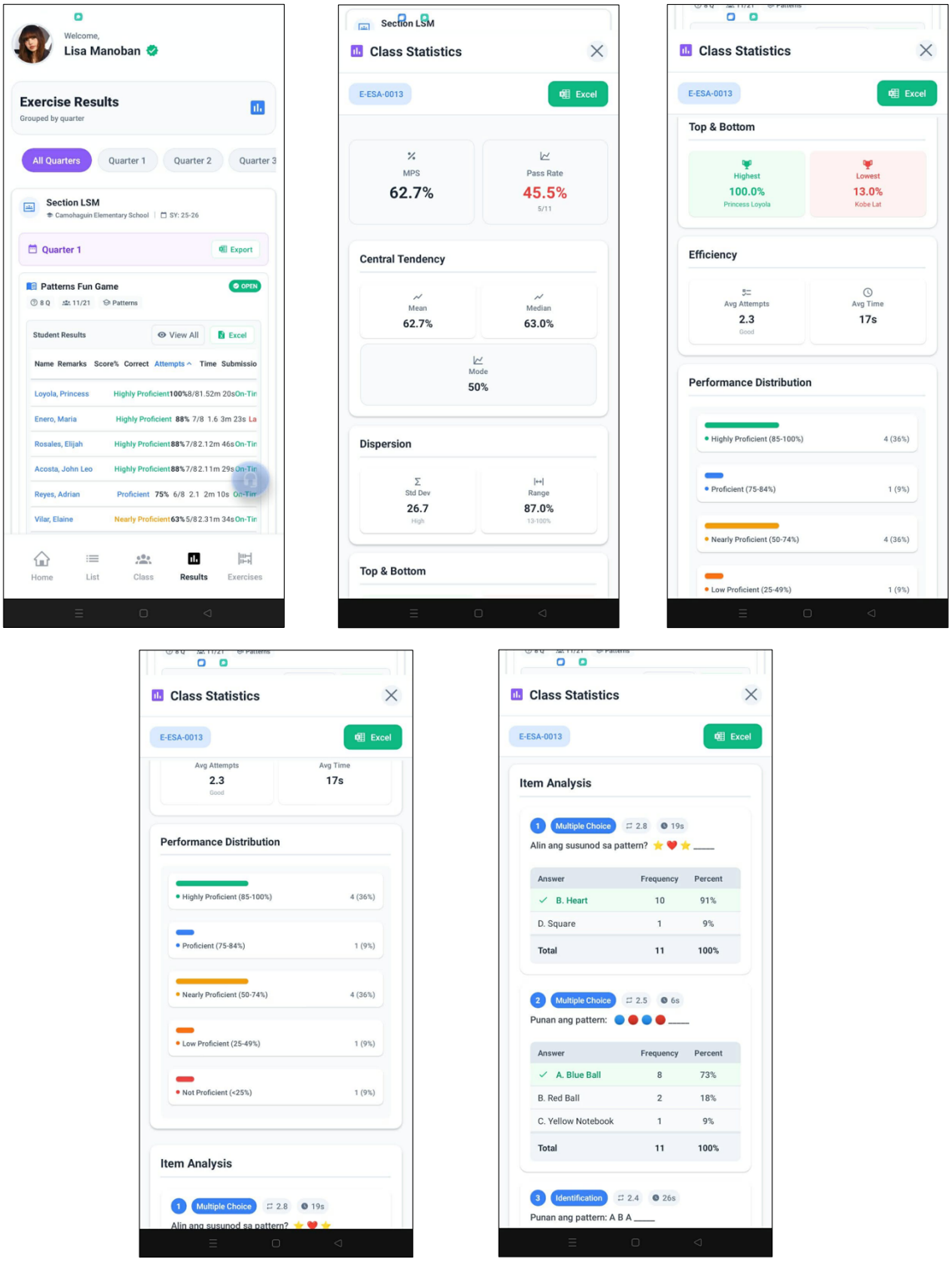
The GUI opens with a pristine splash screen and moves on to a simple screen of choice of roles, each leading off to similarly easy-to-walk-through login screens comprised of nothing more than simple input fields and the buttons to access them.

Figure 19. MATHtatag teacher interface showing announcements, student list, and classroom management screens.



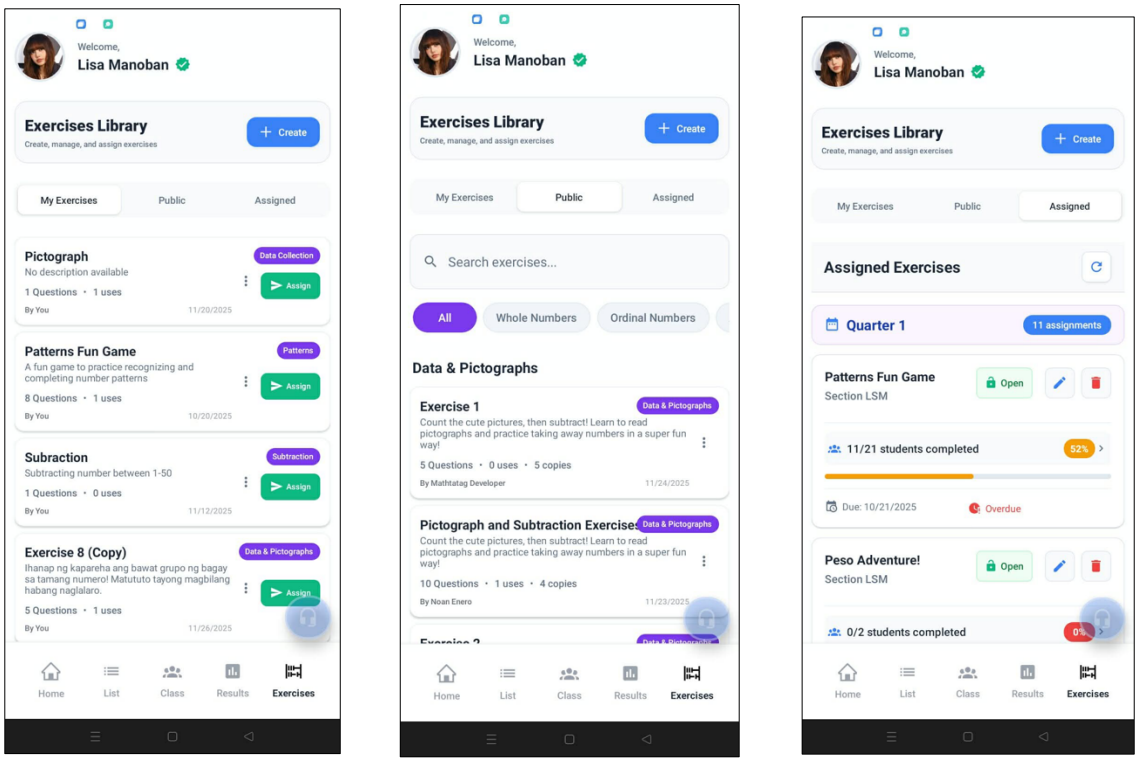
The dashboard for teachers has announcements, class lists and classroom status represented in clear blocks on a design with easy-to-understand card style components. Teachers are able to add students, manage their sections, navigate through the different tabs and have an efficient method of managing their classrooms with just one click per section.

Figure 20. MATHtatag teacher analytics interface showing exercise results, class statistics, and item analysis.



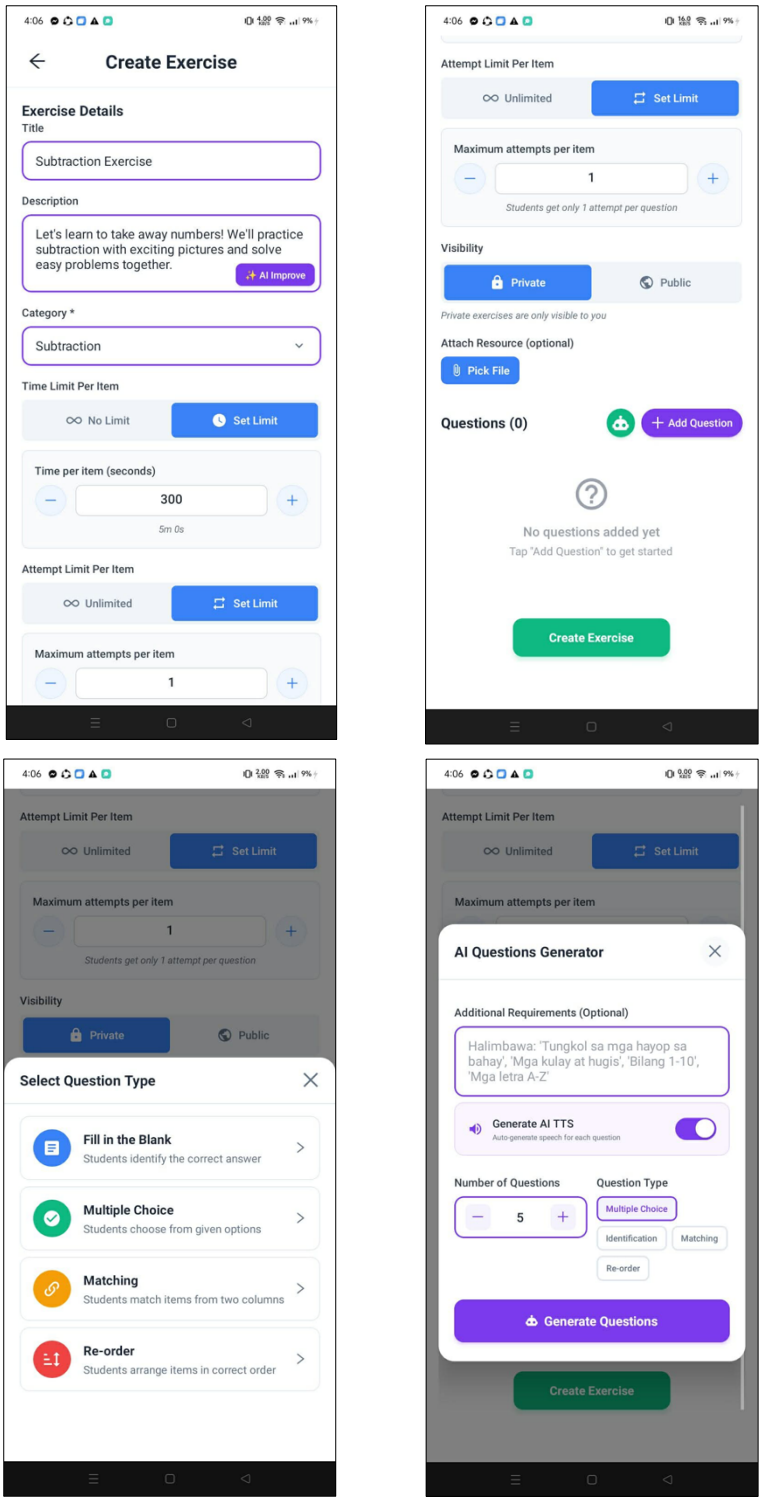
The analytics interface requires organizing exercise results, class stats and item analysis into panels that are frank and easy to interpret at a glance. Visual cues, charts, recap and summaries allow teachers to spot performance trends at a glance, and see learner strength and gaps. Interactive filters and exports support deeper review and data-driven decision making.

Figure 21. MATHtatag teacher interface for creating, viewing, and assigning exercises.



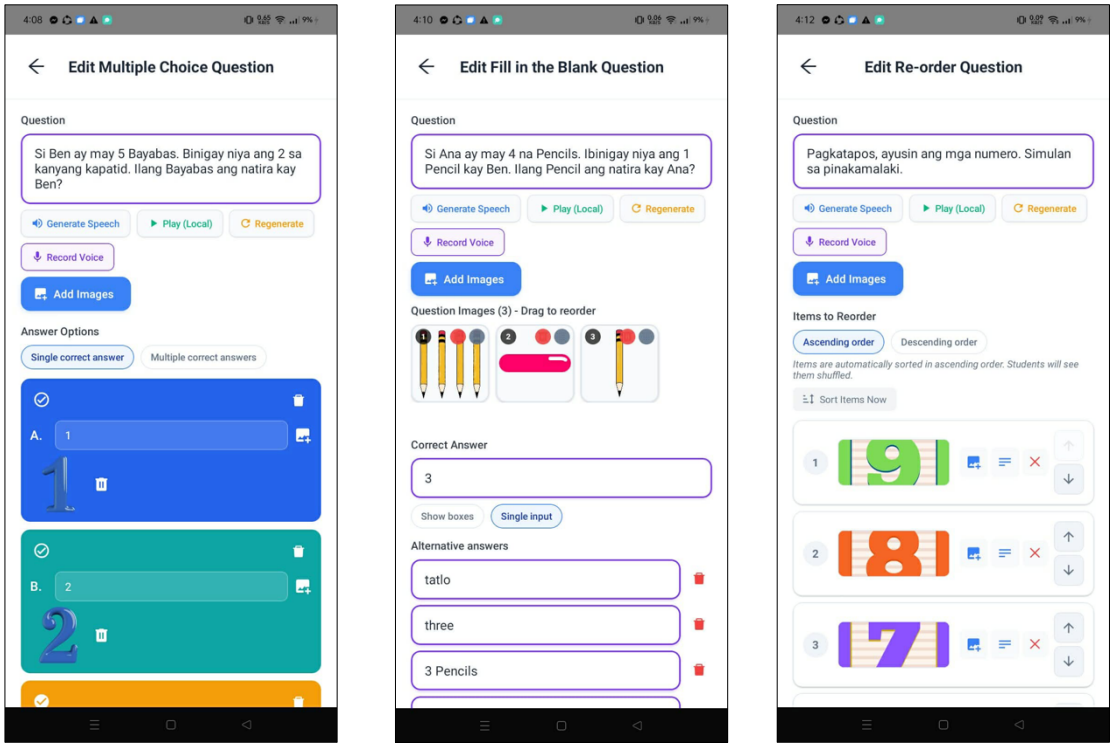
In the exercises library, teachers can find activities expertly organized within tabs for personal, public, and assigned content—always easy to browse or search. Each exercise card shows key information and quick-action buttons to open, assign, or edit it—and progress indicators and deadlines give teachers a glimpse of class completion status.

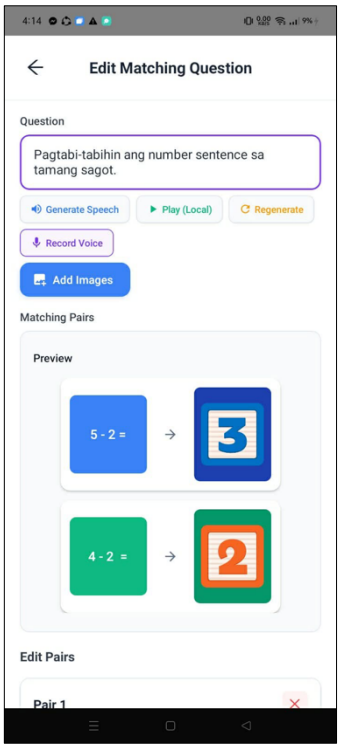
Figure 22. MATHtatag teacher interface for creating exercises and generating questions.

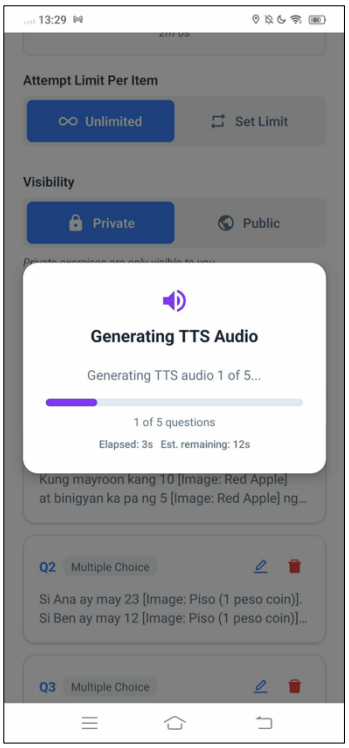


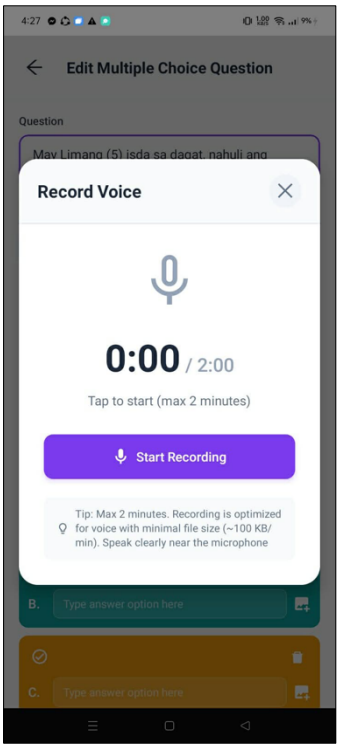
The exercise creator starts you out with fields for titles, descriptions, limits, and visibility, and then displays the questions themselves in a straightforward, form-like method. Teachers can go nice and old-school and type them all manually, or they can let the generator do it and get quick multiple-choice questions out of it. And once they do, they can get in and get out of all the other buttons and modals and build a complete activity in probably three clicks.

Figure 23. MATHtatag teacher tools for editing questions and generating TTS/audio.



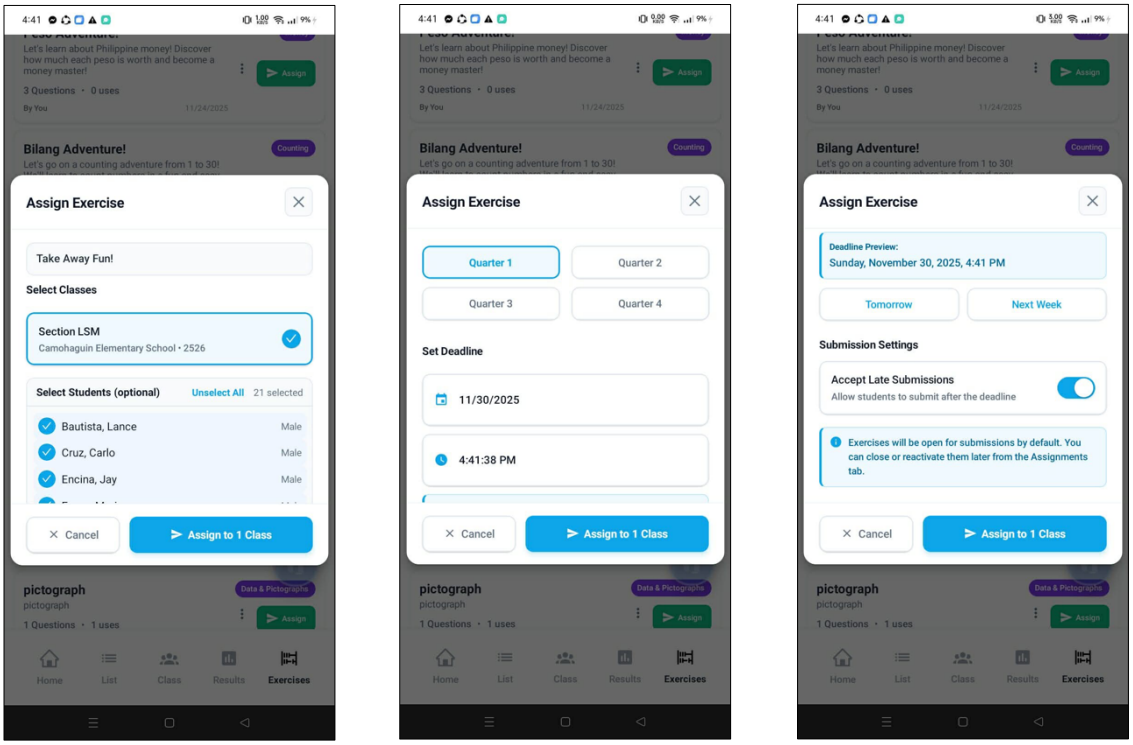






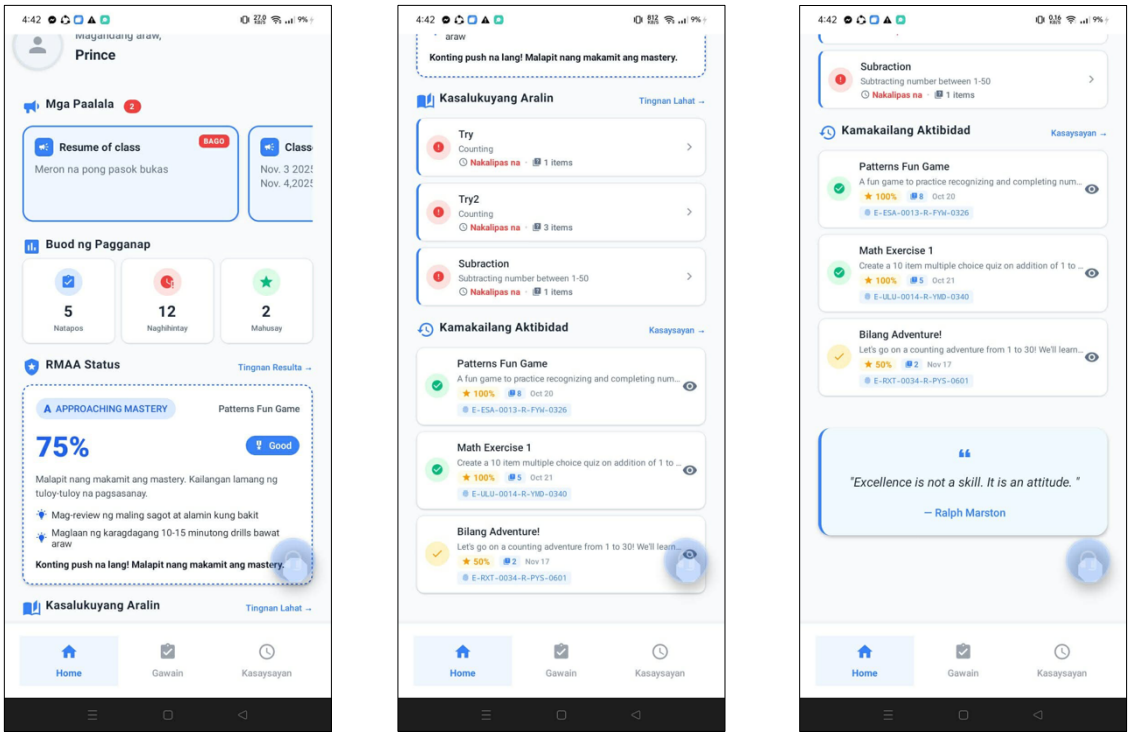
Each type of question is shown in a clear layout of fields that display text, images and answer inputs within the question editor, so teachers can easily create speech, play sound or record their own voice directly in the interface, and move, delete and adjust questions visually.

Figure 24. MATHtatag teacher view for exercise assignment and deadline settings.



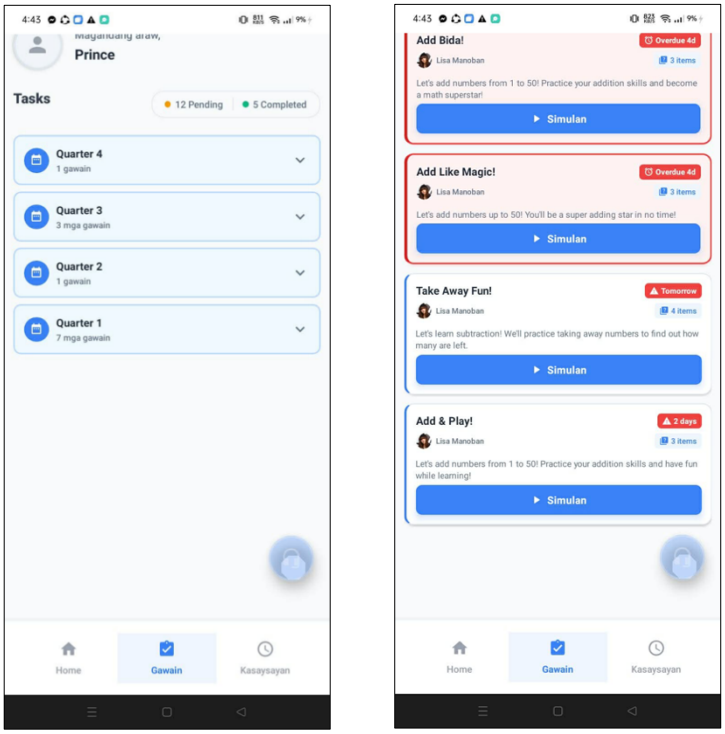
The assignment interface. Finally, when teachers want to assign students to exercises, they are presented with a clean modal in which they select classes/students, choose deadlines, and set rules. There's no confusing combination of switches here - date picking, quarter selection, and enabling late submissions are available via toggles and buttons. This section makes it easy for the teacher to quickly assign exercises without losing control over timing and requirements.

Figure 25. MATHtatag parent dashboard for monitoring student performance.



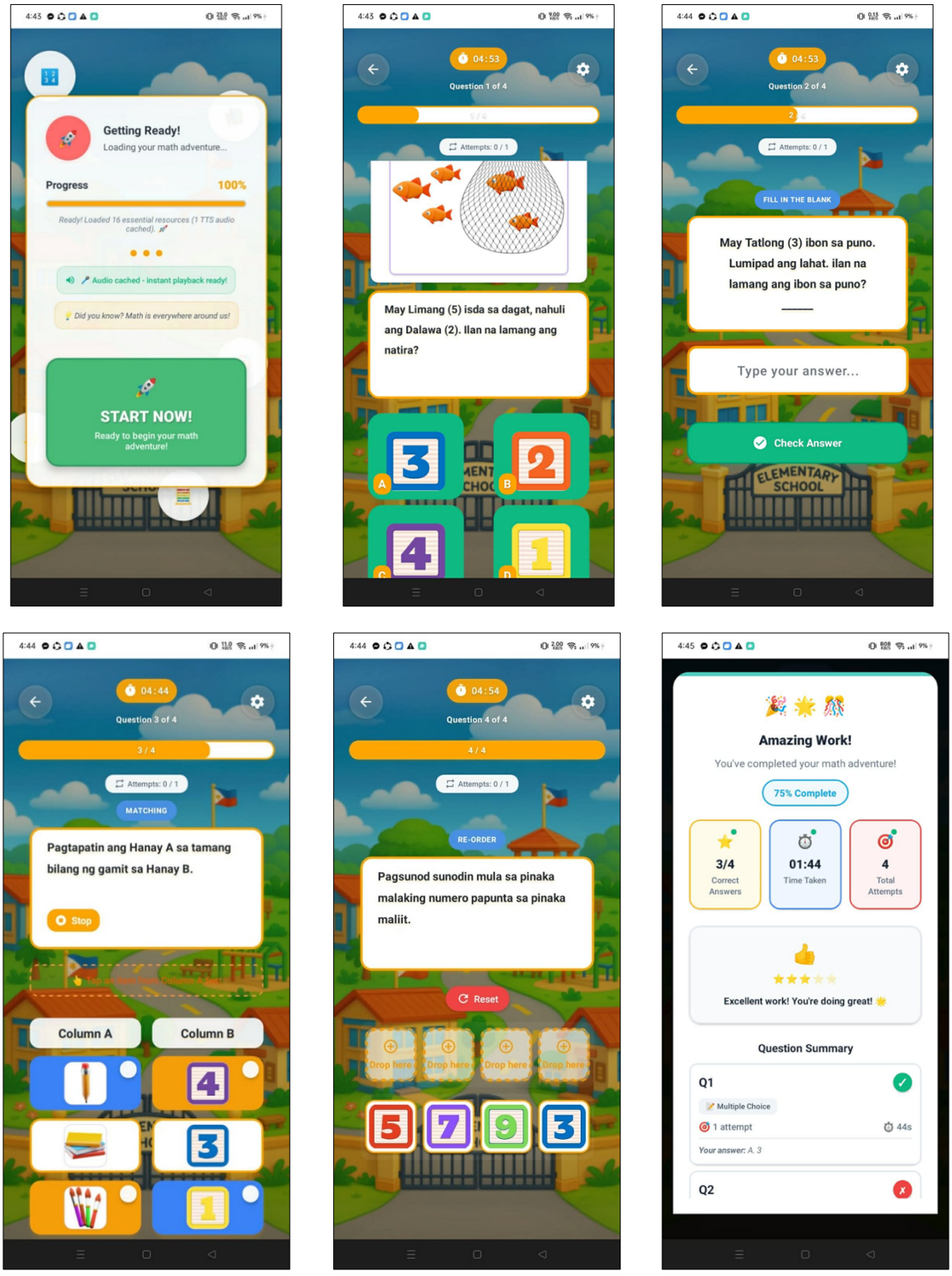
The parent dashboard features important things like reminders, summaries of progress, and RMA status in a simple card format layout. Parent can easily view ongoing lessons and lessons that their child has recently completed with indicators that display performance levels. The dashboard features easy-to-navigate tabs so that parents can eagerly view their child's learning with a few clicks.

Figure 26. MATHtatag parent view of learner tasks and activity deadlines.



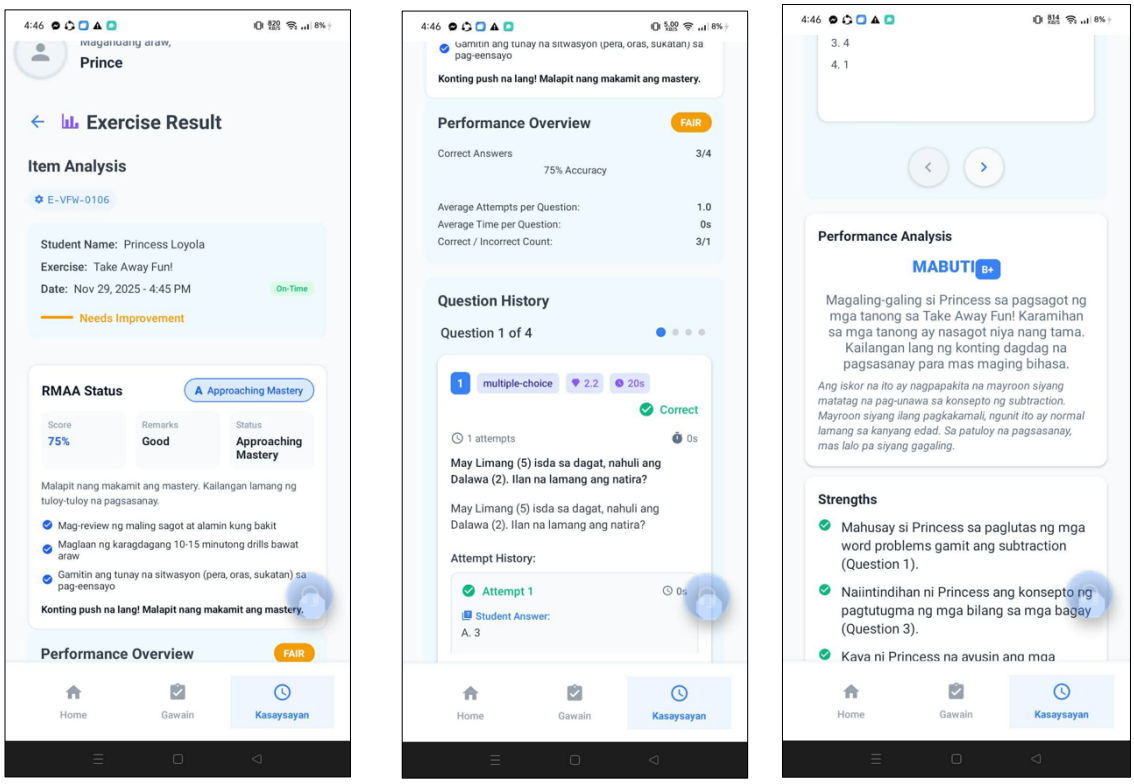
The parent task view displays each quarter in panels users can collapse to navigate quickly past assigned activities. Tasks are color-coded with indications of due dates, overdue status, and how many items there are. Parents can tap any activity card for details or to walk their child through it.

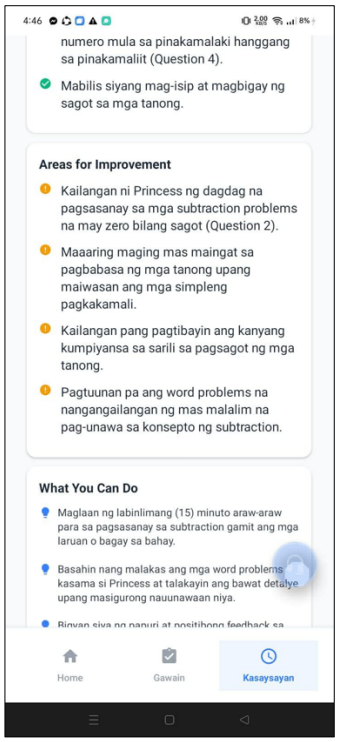
Figure 27. MATHtatag learner activity screens featuring interactive questions and performance results.

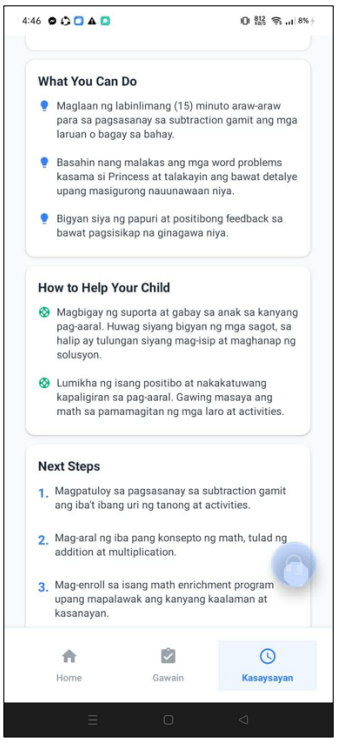


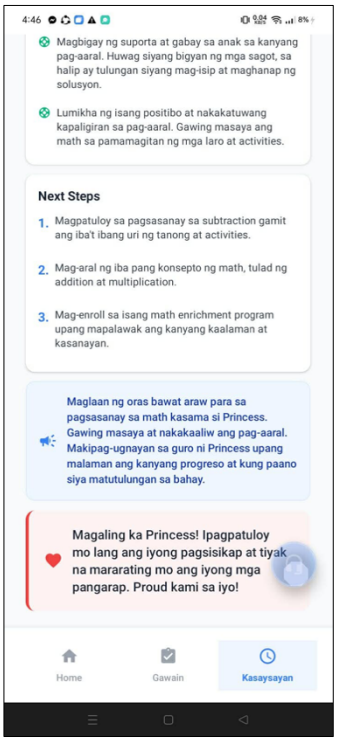
The learner activity screens make use of a bright, almost game-like look and feel where students answer interactive questions including multiple choice or fill-in-the-blank, matching, re-order. Timers and prompts set the pace, and a results screen gives a clear summary of the performance.

Figure 28. Parent interface showing detailed learner performance, item analysis, and guided recommendations.



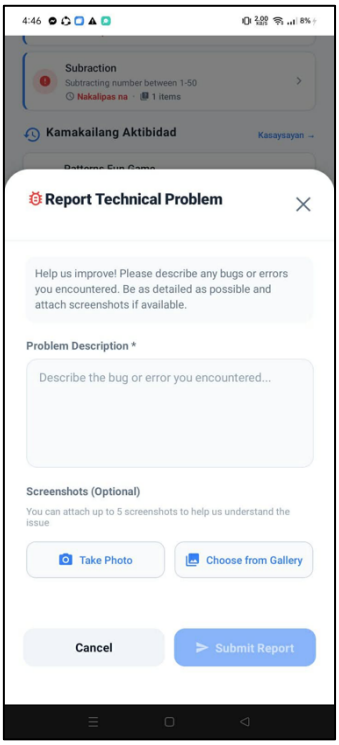






The parent interface provides an easy-to-read overview of student performance including item analysis, total score summaries and history of questions answered/attempted by students. The strengths, areas of weakness and recommendations for development will be organized into sections in a manner that will allow parents to easily support their child.

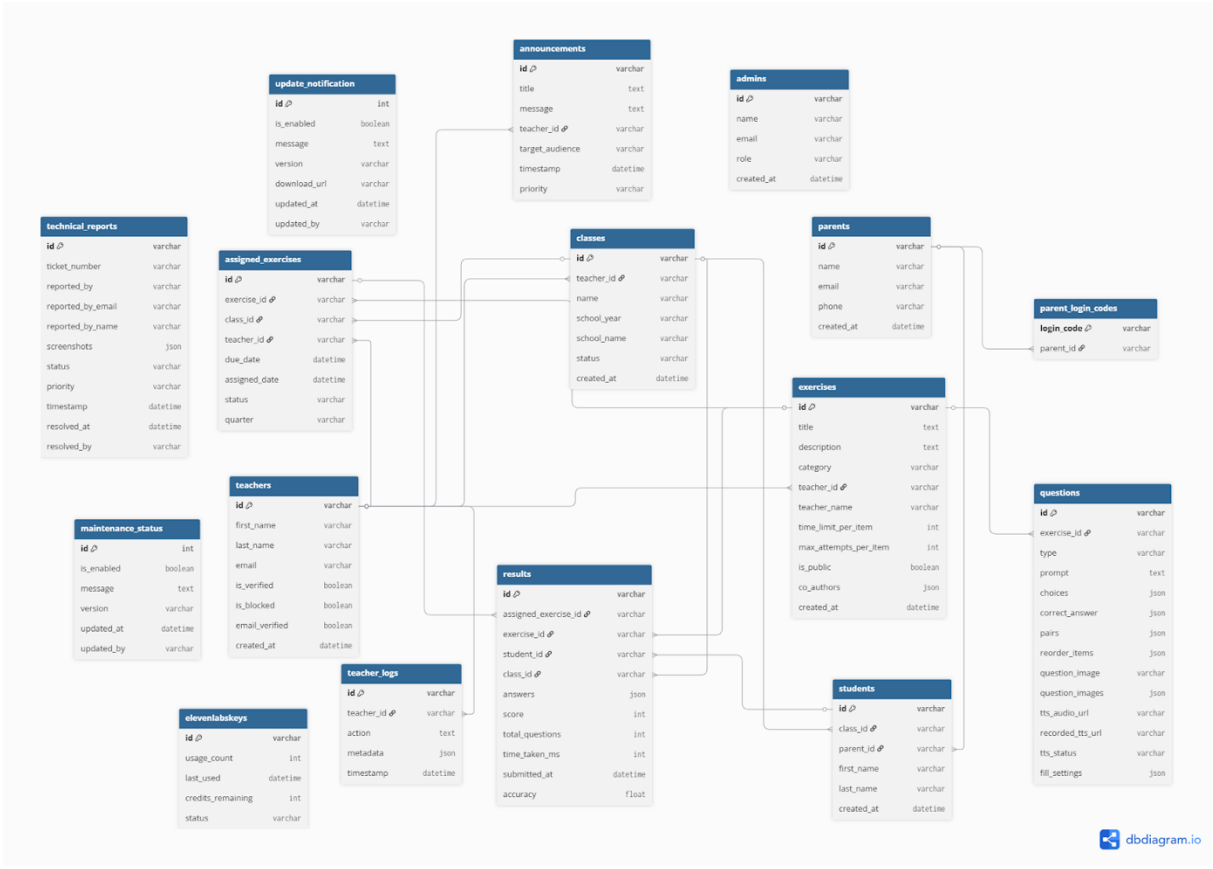
Figure 29. Technical problem reporting screen for parents and teachers in MATHtatag



The technical problem reporting screen contains a simple form allowing description of the issue and a photo upload if necessary. Convenient buttons allow taking a photo or uploading something from the gallery. Clear organization helps parents and teachers communicate technical issues.

Database Schema

Figure 30. Database Schema



Database schema representing all of the main data entities and relationships used within MATHtatag. Teachers directly manage classes, create exercises, assign exercises/assessments to learners, and see results collected in the results table. Students belong to a class and have parents associated, who can log in to the app with a unique parent code. Exercises consist of a number of questions. Announcements are made by teachers and either appear publicly or are targeted to their respective audients. In the case of an issue, the system also contains submitted technical reports that teachers use to resolve the issue. Tables for both maintenance history and update notification determine when the application is available and when users will receive prompts about version updates, respectively. Logs

indicating teacher activity are also stored in the database, and the app tracks usage of the ElevenLabs keys needed for text-to-speech. Relationships here is where real-time data exchanged occurs in the app, or to put it another way to provide real-time management of users, learning activities, and status of the platform itself.

JSON Tree

Figure 31. JSON Tree

```
/
├── system/
│   └── idCounters/        # Auto-incrementing ID counters
│       ├── PARENT/        # count: number, lastUpdated: timestamp
│       ├── TEACHER/
│       ├── STUDENT/
│       ├── EXERCISE/
│       └── ...
│
├── teachers/
│   └── {teacherId}/       # T-XXX-0001
│       ├── email
│       ├── firstName
│       ├── lastName
│       ├── emailVerified
│       ├── schoolCode
│       ├── classes: []
│       └── timestamp
│
├── students/
│   └── {studentId}/       # S-XXX-0001
│       ├── fullName
│       ├── firstName
│       ├── surname
│       ├── gender
│       ├── classId
│       ├── parentId
│       ├── gradeSection
│       └── timestamp
│
├── parents/
│   └── {parentId}/        # P-XXX-0001
│       ├── loginCode      # XXXX-XXXX format
│       ├── firstName
│       ├── lastName
│       ├── contactNumber
│       ├── address
│       ├── children: []   # Array of student IDs
│       └── infoStatus
│
├── classes/
│   └── {classId}/         # C-XXX-0001
│       ├── className
│       ├── section
│       ├── gradeLevel
│       ├── teacherId
│       ├── students: []
│       └── timestamp
│
├── exercises/
│   └── {exerciseId}/      # E-XXX-0001
│       ├── title
│       ├── description
│       ├── category
│       ├── teacherId
│       ├── teacherName
│       ├── timeLimitPerItem    # Time limit per question (seconds)
│       ├── maxAttemptsPerItem  # Max attempts per question
│       ├── questions: []
│       │   └── {
│       │       id: "E-XXX-0001-Q001",
│       │       type: "multiple-choice" | "identification" | "matching" | "re-order" |
│       │       question: "text",
│       │       questionImage: "url", # Single image
│       │       questionImages: [], # Multiple images (patterns)
│       │       answer: "..." | [], # String or array
│       │       options: [], # For multiple choice
│       │       optionImages: [], # Images for options
│       │       pairs: [], # For matching questions
│       │       reorderItems: [], # For re-order questions
│       │       ttsAudioUrl: "url", # AI-generated TTS
│       │       recordedTtsUrl: "url", # Teacher-recorded voice
│       │       ttsStatus: "idle" | "generating" | "ready" | "failed",
│       │       fillSettings: { # For identification questions
│       │       caseSensitive: boolean,
│       │       showBoxes: boolean,
│       │       altAnswers: [],
│       │       hint: "text"
│       │       }
│       │       }
│       ├── createdAt
│       ├── isPublic
│       └── coAuthors: []
│
├── assignedExercises/
│   └── {assignedId}/ # A-XXX-0001
│       ├── exerciseId
│       ├── classId
│       ├── teacherId
│       ├── dueDate
│       ├── assignedDate
│       └── status
│
├── results/
│   └── {resultId}/ # E-XXX-0001-R-ABC-0001
│       ├── studentId
│       ├── exerciseId
│       ├── classId
│       ├── answers: []
│       ├── score
│       ├── totalQuestions
│       ├── timeTaken
│       ├── submittedAt
│       └── accuracy
│
├── announcements/
│   └── {announcementId}/ # N-XXX-0001
│       ├── title
│       ├── message
│       ├── teacherId
│       ├── targetAudience # "all", "class", "student"
│       ├── timestamp
│       └── priority
│
└── parentLoginCodes/
    └── {loginCode}/ # Maps login codes to parent IDs
        └── parentId
```

The JSON tree illustrates the MATHtatag system hierarchical database structure utilizing the Firebase Realtime Database. The structures data for the system is organized into nodes for users, learning activities, and academic history. All teacher, student, and parent information is stored in separate user nodes that each carry their own role-based attributes, as well as associations to other entities that are important to that role such as class assignments, or the related family nodes that connect students to parents. Classes are the central link between teachers and learners. The exercises, assignedExercises, and results nodes maintain the materials for student learning and assessment. This allows teachers to assign the material for students to learn, and also maintains the assessment records, so the teacher can observe learners performance during the activities in real time. For effective communication with the school community, announcements are recorded into a node. The parentLoginCodes mapping into the nodes will provide a secured method for parents to identify themselves by unique codes into the MATHtatag app. Auto-incrementing IDs are managed by the system node to keep incremental and unique identification numbers throughout the coded structure This makes for good synchronization and usage of data across all roles of the mobile application.

Data Dictionary

System Table

Table 4.
Data Dictionary – System Table

Field Name	Data Type	Description	Format / Example
count	Integer	Latest incremental value used when generating new IDs.	5
lastUpdated	Timestamp	Last timestamp when counter was updated.	2025-02-18 14:21:05

Teachers Table

Table 5.
Data Dictionary – Teachers Table

Field Name	Data Type	Description	Format / Example
email	String	Teacher's registered email.	teacher@gmail.com
firstName	String	Teacher's first name.	–
lastName	String	Teacher's last name.	–
emailVerified	Boolean	Indicates if the teacher has verified their email.	true
schoolCode	String	Code representing the school.	PUP-LPZ
classes	Array (String)	IDs of classes assigned to this teacher.	["C-001-0001"]
timestamp	Timestamp	Date when teacher account was created.	2025-01-04 09:15:22

Students Table

Table 6.
Data Dictionary – Students Table

Field Name	Data Type	Description	Format / Example
fullName	String	Student's complete name.	–
firstName	String	First name of student.	–
surname	String	Last name of student.	–
gender	String	Student gender.	Male / Female
classId	String	Assigned class ID.	C-001-0001
parentId	String	Parent account linked to student.	P-001-0001
gradeSection	String	Grade and section.	Grade 1 - Malikhain
timestamp	Timestamp	Date added.	2025-02-01 10:22:55

Parents Table

Table 7.
Data Dictionary – Parents Table

Field Name	Data Type	Description	Format / Example
loginCode	String	Unique code for parent authentication.	A3F9-2KD1
firstName	String	Parent's first name.	–
lastName	String	Parent's surname.	–

contactNumber	String	Mobile number.	09123456789
address	String	Parent's address.	Brgy. Sta. Catalina, Lopez Quezon
children	Array (String)	List of linked student IDs.	["S-001-0001", "S-001-0002"]
infoStatus	String	Indicates if the profile is complete.	complete / incomplete

Classes Table

Table 8.
Data Dictionary – Classes Table

Field Name	Data Type	Description	Format / Example
className	String	Name of the class.	Grade 1 - Diamond
section	String	Section of class.	Diamond
gradeLevel	String	Level of the class.	Grade 1
teacherId	String	Assigned teacher.	T-001-0001
students	Array	Student IDs under this class.	["S-001-0001"]
timestamp	Timestamp	Date created.	2025-01-12 08:15:44

Exercises Table

Table 9.
Data Dictionary – Exercises Table

Field Name	Data Type	Description	Format / Example
title	String	Exercise title.	Subtracting Two-Digit Numbers
description	String	Brief description.	–
category	String	Tag for lesson classification.	Quarter 3 - Week 6
teacherId	String	ID of creator.	T-001-0004
teacherName	String	Creator's display name.	Ms. Dela Cruz
timeLimitPerItem	Integer	Time allowed per item.	30
maxAttemptsPerItem	Integer	Allowed attempts.	2
questions	Array of Objects	Contains question objects.	–
createdAt	Timestamp	Date created.	2025-03-12 14:40:10
isPublic	Boolean	Visible to other teachers.	false
coAuthors	Array	Teachers permitted to edit.	["T-001-0008"]

Exercise Questions Table

Table 10.
Data Dictionary – Exercise Questions Table

Field Name	Data Type	Description	Format / Example
id	String	Unique question ID.	E-001-0001-Q001
type	String	Type of question.	multiple-choice
question	String	Text of the question.	–
questionImage	String	URL of single image.	https://firebase…/image1.png
questionImages	Array	Multiple images.	["url1", "url2"]
answer	String/Array	Correct answer.	"12" or ["cat", "dog"]
options	Array	Choices for MCQ.	["12","14","15"]
optionImages	Array	Choices with images.	["url1","url2"]
pairs	Array	Matching pairs.	[{"A":"1","B":"One"}]
reorderItems	Array	Items to reorder.	["Step 1","Step 2"]
ttsAudioUrl	String	AI-generated voice recording.	URL
recordedTtsUrl	String	Teacher-recorded voice file.	URL
ttsStatus	String	TTS generation state.	"ready"
fillSettings	Object	Identification settings.	{caseSensitive: false}

Assigned Exercises Table

Table 11.
Data Dictionary – Assigned Exercises Table

Field Name	Data Type	Description	Format / Example
exerciseId	String	ID of exercise.	E-001-0001
classId	String	Class assigned.	C-001-0001
teacherId	String	Creator/assigner.	T-001-0008
dueDate	String	Deadline.	2025-03-20
assignedDate	Timestamp	Date assigned.	2025-03-15 09:22:10
status	String	Activity status.	active / closed

Results Table

Table 12.
Data Dictionary – Results Table

Field Name	Data Type	Description	Format / Example
studentId	String	Student who answered.	S-001-0003
exerciseId	String	Exercise answered.	E-001-0001
classId	String	Student class.	C-001-0001
answers	Array	Student answers.	[{"answer":"12","correct":true}]
score	Integer	Number correct.	8

totalQuestions	Integer	Total items.	10
timeTaken	Integer	Seconds spent.	150
submittedAt	Timestamp	Date submitted.	2025-03-15 14:01:44
accuracy	Float	Score percentage.	80.0

Announcements Table

Table 13.
Data Dictionary – Announcements Table

Field Name	Data Type	Description	Format / Example
title	String	Announcement title.	–
message	String	Content of message.	–
teacherId	String	Sender.	T-001-0001
targetAudience	String	Recipients.	all, class, student
timestamp	Timestamp	Date posted.	2025-03-10 08:30:10
priority	String	Priority level.	high / normal

Parent Login Codes Table

Table 14.
Data Dictionary – Parent Login's Table

Field Name	Data Type	Description	Format / Example
parentId	String	Maps login code to parent account.	P-001-0001

DEVELOPMENT METHODOLOGY

Development Methodology describes the processes, tools, and techniques that developer have applied in implementing MATHtatag mobile application. It describes how the system was planned, conceived, implemented, reviewed and validated for the duration of the capstone project. By choosing the right methodology for the project, the capstone group was able for successfully and responsively respond to requirements changes, user feedback and technical difficulties while keeping the application usable and working.

In Aside from system development, this section detailed how the application was implemented in the study. After stabilizing the deployment build, the MATHtatag app was given to the teacher and learners at Camohaguin Elementary School to test.

Since the study employed the pre-test and post-test design, the teacher first administered a pre-test, then give digital exercises through the MATHtatag application and let learners do them as regular assignment through their android devices. This was the intervention period where pupils practiced math with MATHtatag's apps interactive features and Filipino text-to-speech.

After MATHtatag was implemented, the teacher conducted a post-test to measure improvement. This pre-test - exercises - post-test sequence ensured that MATHtatag was embedded into the lesson, and allowed the researchers to see if it had had the desired effect in reinforcing basic numeracy skills.

Process Model

For the conception and development of MATHtatag, the researchers used the Agile Process Model. Agile was chosen as the most appropriate process model since it allows for iterative progress, rapid releases, and flexible design which suits the nature of a mobile educational system that has to respond quickly to suggestions, for example from teachers, students and parents.

Agile: the system was developed via short iterations that could be digested over a sprints. In each iteration there was planning, requirement refining, UI/UX design, coding along with testing all in the same cycle. Instead of building the entire system out in a single sweep, the capstone group built MATHtatag over time in working versions. At the end of each cycle, we delivered a functioning module of the app that might include: Teacher Login, Class Management, Exercise creation, Student Answering Interface, Parent Monitoring Dashboard, Voice and TTS Integration, Analytics. This allowed the capstone team to test features early to gather feedback immediately from partner teachers and expected end users.

The researchers followed the Agile model. This was the most convenient model to employ for the project because MATHtatag requires continuous iterations whenever we change the types of questions, improve the user interface considering the level of Grade 1 Learners, testing audio features and improving usability through observations done in the

classroom as the testing place of the system. Also, requirements can be updated by researchers whenever needed without interrupting the overall project development. Because of this, the capstone group has been able to work at a 'steady pace of progress', guarantee a high-quality improvement in each iteration, and produce a system that matches both user expectations and educational requirements under the MATATAG curriculum.

Development Tools

Table 15.
Development Tools

Category	Tool	Description
Programming Languages	TypeScript	Used as the main language for development. TypeScript provides type safety and improved code reliability when building complex mobile interfaces and real-time features.
	JavaScript (ES6)	Utilized alongside TypeScript for implementing app logic, UI components, and integration with third-party libraries.
Frameworks & Libraries	React Native	Core framework used to develop the mobile application for Android devices using a single codebase.
	Expo SDK (Expo 54)	Simplifies mobile development by providing built-in modules for camera, audio, file handling, navigation, and deployment. Enables real-time preview through Expo Go.
	Expo Router	A file-based navigation library used to manage screen transitions such as Login, Dashboards, and Exercise Answering.
	Reanimated & Gesture Handler	Enables smooth animations and touch interactions. Essential for interactive item-reordering and drag-and-drop question types.

Databases	Firebase Realtime Database	Provides real-time synchronization of class lists, exercises, student answers, analytics, and parent dashboards across devices.
Cloud Services & APIs	Firebase Authentication	Handles secure login and verification for teachers and admins. Supports email/password login and account validation.
	Firebase Storage	Stores exercise images, audio recordings, TTS files, and uploaded media used in questions.
	ElevenLabs Text-to-Speech API	Generates AI-powered voice instructions and narration for questions, improving accessibility for Grade 1 learners.
Integrated Development Environments (IDEs)	Visual Studio Code (VS Code)	The main IDE used for writing TypeScript code, debugging, and organizing project files. Provides extensions for formatting and linting.
Build and Testing Tools	Expo Go	Mobile app used for real-time testing of features during development without compiling APKs.
	Android Emulator	Software emulators used for testing UI responsiveness and device compatibility across platforms.
	EAS (Expo Application Services)	A cloud service from Expo that creates Android builds and streamlines the deployment process.
Version Control Tools	Git & GitHub	Used for version control, team collaboration, and tracking project changes. Ensures secure backup of the codebase.
Document & Asset Tools	Canva/Figma	Used to design UI mockups, icons, and other graphical assets used in the app.

TEST METHODOLOGY/PROCEDURES

Testing of MATHtatag took place throughout the entire development process so that all the different pieces of the system functioned properly as well as contributed to a pleasant learning experience for Grade 1 children and satisfied user needs. Testing activities occurred continuously through Agile sprints to catch issues early and iteratively tested to teacher, student and parent feedback.

As part of the study evaluation procedure, the deployment the system as an intervention was also rolled out using the pretest and posttest structure described above. The teacher administered a pretest, and then gave MATHtatag exercises for the learners to engage in as guided practice activity. These activities constitute the period of intervention after which the post-test is administered again, and result in the features of the app being embedded in an authentic learning activity.

To check the reliability and usability of functions of MATHtatag, as well as validating correctness, we conducted unit, integration, system, and user acceptance testing (UAT). These testing types are described below with the goals we were pursuing for each, aligned with the project's iterative development.

Unit Testing

Unit testing aimed primarily at checking individual functions, UI components and logic blocks in the application. These included validations of:

1. Input handling for the login and authentication

2. Modules for creating exercises (individual question types, options, and choice of reading of audio attachments)
3. Rending of questions in the student answering UI
4. Computation of scores and accuracy
5. Logic for rendering the parent's dashboard

Testing was done through console logs, component-level checks and live device testing using the Expo Go application.

Integration Testing

Integration testing phase assessed whether the various modules interacted with one another, typically those that made calls to Firebase real time or otherwise touched a module that needed multiple roles. These included tests on the proper integration of flows including:

1. Teacher - Class - Student assignment flow
2. Exercise creation - Storage upload - Student answering
3. Student submission - Result computation - Parent dashboard updates
4. Admin and Super Admin data viewing and management

This ensured that data moved smoothly across modules without errors.

System Testing

System tests were carried out to see if the complete MATHtatag app will worked properly as a complete mobile learning platform. The researchers tested it from beginning to end using actual Android operated Smartphones and emulators, ensuring that the main

features like Teacher Login, Parent Login via login code, Student Exercise Answering screens, exercise assignment retrieval, TTS playback and recorded audio output, class management, and real-time analytics worked. This phase will also checked the app's performance, including loading times, responsiveness of navigation, and stability during prolonged use, as well as its capacity to process images and audio files efficiently.

User Acceptance Testing (UAT)

The User Acceptance Testing phase was done among Grade 1 teachers together with parents of the partner schools to check MATHtatag for ease of use, clarity, and practical effectiveness in an actual educational setting. Teachers assessed the usability of creating exercises, assigning activities to the classrooms and watching over student progression, and navigating the self-portrait and teacher dashboard features. Parents tested the parent login, checked the accuracy of their child's scores together with attempts given in its display, and checked their general overview of performance in the dashboard. Findings from the User Acceptance Testing confirmed that MATHtatag system will fulfill its intended educational purpose, easy to pilot, and aligned with teacher and parent expectations of MATATAG Curriculum.

SYSTEM REQUIREMENTS

This part discussed the required hardware, software, and miscellaneous items necessary to build, test and run the MATHtatag mobile application. Since the system was built with React Native, Expo, and Firebase both the development machines and user devices had to meet some minimum performance requirements to ensure smooth operation and real-time processing.

Hardware Requirements

Table 16.
Hardware Requirements

Component	Minimum Requirement
Laptop/PC (For Developers)	• Minimum: Intel Core i5 / Ryzen 5, 8 GB RAM, 256 GB SSD Storage
	• Recommended: Intel Core i7 / Ryzen 7, 16 GB RAM, 512 GB or 1 TB SSD Storage
	• Stable internet connection for Firebase, API calls, and cloud syncing
Mobile Devices (For Testing and Use)	• Minimum: Android 8.0+, 2 GB RAM
	• Recommended: Android 11+, 4–6 GB RAM for smoother performance
	• Device with working microphone (for teacher audio recording and student playback)

Software Requirements

Table 17.
Software Requirements

Component	Minimum Requirement
Operating System	• Minimum: Windows 10 / macOS Catalina / Linux
	• Recommended: Windows 11 / macOS Ventura for better compatibility
Programming Languages	• TypeScript and JavaScript (ES6) as primary development languages
Frameworks & SDK	• React Native for cross-platform mobile development
	• Expo SDK 54 for building, running, and deploying the app

Backend Services	• Firebase Realtime Database for data storage
	• Firebase Authentication for secure login
	• Firebase Storage for media handling (images, audio, TTS files)
External APIs	• ElevenLabs Text-to-Speech API for generating voice instructions
IDE / Code Editor	• Visual Studio Code (VS Code) for writing and debugging code
Package Manager	• npm (Node Package Manager) for installing libraries and dependencies
Testing Tools	• Expo Go for real-time testing
	• Android Emulator for multi-device testing
Version Control	• Git and GitHub for code versioning and collaborative development

QUALITY PLAN

Quality Plan: The details the researcher described how they were assured that the MATHtatag mobile application would conform to the required standard of performance, usability, reliability and software quality in general. For this measure the researcher evaluated the application using the ISO 25010 Software Quality Model to provide systematic evaluation with respect to system quality across the eight major characteristics specified in the standard itself. The ISO standard was selected based on compatibility with implementation of educational technology standards, and for the systematic evaluation of mobile apps targeting teachers, students and parents.

These measures of quality were used throughout the entire development process but particularly in testing, system validation, and user acceptance testing. The impressions of

teachers and parents obtained as part of the system evaluation were also used to assess whether the quality goals for the system had actually been achieved.

Table 18.
ISO 25010 Quality Characteristics

ISO 25010 Quality Characteristic	Description in Relation to MATHtatag
1. Functional Suitability	Assessed if app performed requested functions correctly, and delivered curriculum aligned activities, helpful Filipino TTS. Also observed if results and performance summaries were correct and reliable.
2. Performance Efficiency	Assessed whether loading times, loading speed, nice and smooth performance even on average internet, little to no data usage. If score and progress reports instantly appeared.
3. Compatibility	Determined whether app installs with ease, works on a broad range of Android devices, is stable when used with other applications, and connects properly to onscreen apps such as TTS and online saving features.
4. Usability	Measuring how easy the app will be to navigate as a teacher / parent, or even by the Grade 1 learners themselves. Child-friendly design is checked, as well as whether the icons are clear, and whether the app seems to hold the potential for more engagement, confidence, home-school collaboration.
5. Reliability	Evaluated consistent system performance without crashing or losing data. Also checked stability during weak internet and the app's ability to recover after temporary disconnection.
6. Security	Assessed privacy protection, authorized access, and secure handling of learner information. Also checked whether the app collected only necessary school-related data.
7. Maintainability	Determined whether the app could be updated easily, issues could be reported conveniently, and whether the system remained stable after improvements or new updates.
8. Portability	Checking whether app installs easily on a range of devices (at least as it pertains to install for each reviewer), easily can be transferred to another phone, and whether it is stable for users with borderline technical skills.

EVALUATION PLAN

Evaluating the MATHtatag app will determine whether the project achieved its goals by gauging both quality of the system, and improvements of learners after using it. ISO 25010 standards were used to evaluate the system, where teachers together with parents rated areas such as usability, functionality, as well as reliability using the 5-point Likert scale. Their answers were computed using the **Weighted Mean (WM)** formula:

$$WM = \frac{(fst+fsl+fds+fsd+fna)}{N}$$

Where

fst = frequency of “Strongly Agree”

fsl = frequency of “Agree”

fds = frequency of “Neutral”

fsd = frequency of “Disagree”

fna = frequency of “Strongly Disagree”

N = the total number of respondents

Researchers used the 5-point Likert scale in assessing participants’ responses. This scale included these following codes and their corresponding meanings.

Table 19.
Likert Scale Interval and Description

Description	Scale	Interval
Strongly Agree	5	4.51 - 5.00
Agree	4	3.26 – 4.50
Neutral	3	2.51 – 3.25
Disagree	2	1.76 – 2.50
Strongly Disagree	1	1.00-1.75

To measure learning improvement, learners first took a Pretest to get their baseline score, used the MATHtatag learning activities, and then took the Posttest, which contained the same items. Their performance was computed using the **Percentage Score** formula:

$$\%\text{Score} = \frac{\text{Correct Answers}}{\text{Total Items}} \times 100$$

and their improvement was measured using the **Gain Score**:

$$\text{Gain} = \text{Posttest Score} - \text{Pretest Score}$$

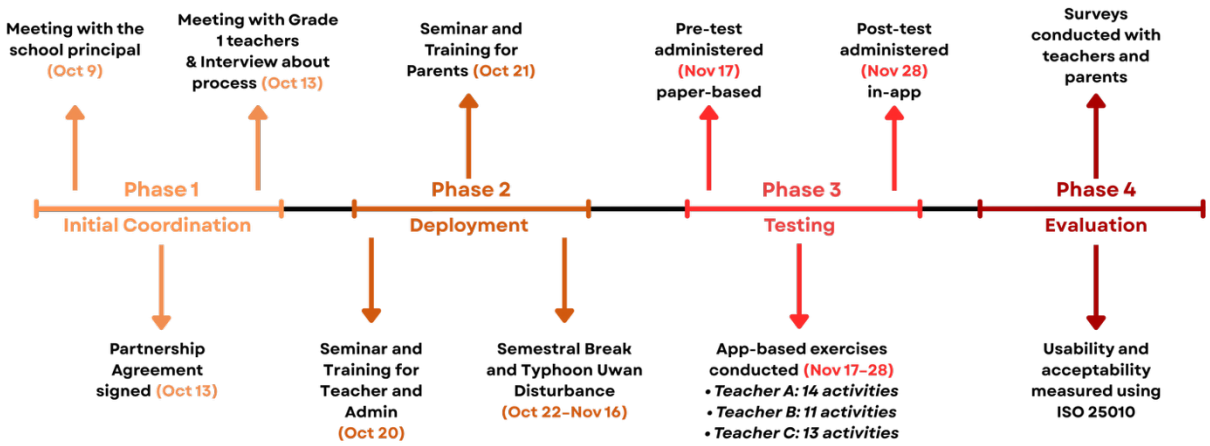
All results gathered from surveys, pretest, and posttest were examined to determine if MATHtatag system was effective, easy to use, and helpful in strengthening Grade 1 learners' mathematics skills.

Chapter 4
RESULTS AND DISCUSSION

The app MATHtatag was developed to help improve early math learning among young learners. It offers an easier and more engaging way to study through interactive activities with Filipino voice guidance. With the app, teachers may implement digital assessments with the ability for the parents of students to track progress through access to their student's digital assessment information and feedback, thus reducing delay due to the typical paper-and-pencil assessment process and facilitating better communication between home and school.

TIMELINE OF DEPLOYMENT

Figure 32. Timeline of MATHtatag Deployment and Data Collection

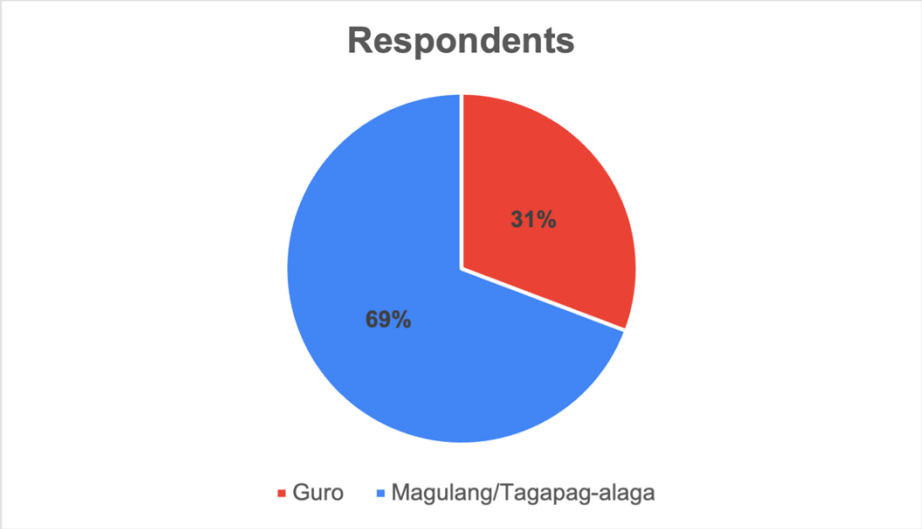


The timeline illustrates the sequence of events in deploying the MATHtatag application and collecting data for the study. It commences with this first contact with the school (October 9–13) where permissions, schedule, and preparations of delegated participants were completed. The deployment (October 21 through November 16) was succeeded by testing (November 17–28), which included running the paper-based pretest, the series of app-based exercises that the teacher had assigned to the class, and the posttest within the application.

The last phase involved just the day of evaluating all the data collected, with the pretest–posttest result comparisons, acceptability ratings of the system, and so on. The chronological action in the following timeline demonstrates a smooth and orderly process in addressing concerned preparations, keeping the app accessibility consistent throughout the course of the intervention, and gathering the results for evaluation.

THE DEMOGRAPHIC PROFILE OF THE RESPONDENTS

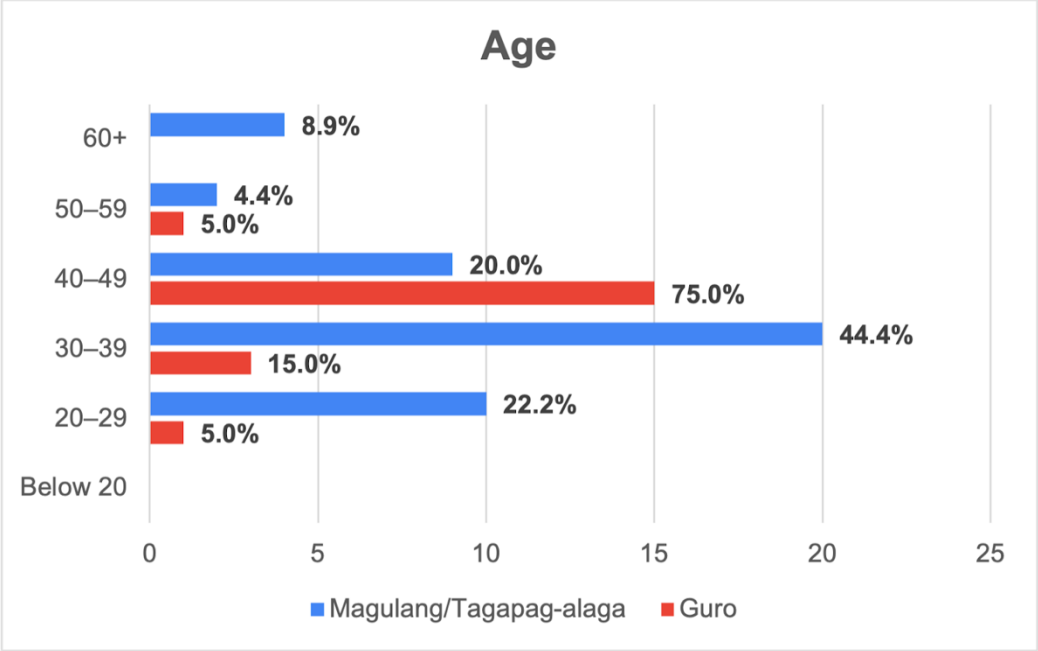
Figure 33. Demographic Profile of Respondent Role or User



In total, 65 people responded to this evaluation: 20 teachers (31%); 45 parents/guardians (69%). Teachers responded to how well MATHtatag could be used in the classroom, how well overall assessment was on task, and how well the curriculum fit. Parents thought about ease of app use at home and ways it supports learning for their child. "Including them both gives a more rounded picture of what the system offers as teachers and parents usually observe different facets of learner behavior and engagement". The Grade 1 learners were not considered respondents because they did not answer the ISO/IEC 25010 evaluation

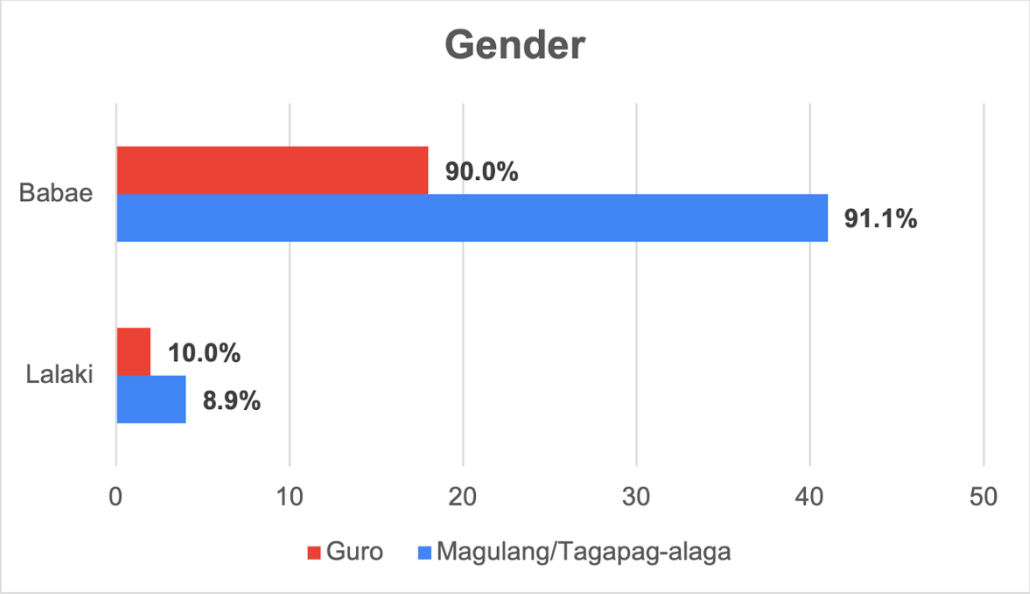
questionnaire. Instead, they were classified as participants, as they took part in the intervention through the pre-test, use of the MATHtatag application, learning activities, and post-test.

Figure 34. Demographic Profile of Respondent in terms of Age



Most parents/guardians were aged 30–39 (44.4%); the rest were aged 20–29 (22.2%) or 40–49 (20%), with a few older than 60 (8.9%). Teachers were more likely to be aged 40–49 (75%), but with others aged 30–39 (15%), 20–29 (5%), and 50–59 (5%). These are indicative of respondents having sufficient maturity and life experience to give realistically informed answers about a learning technology, here from a classroom perspective.

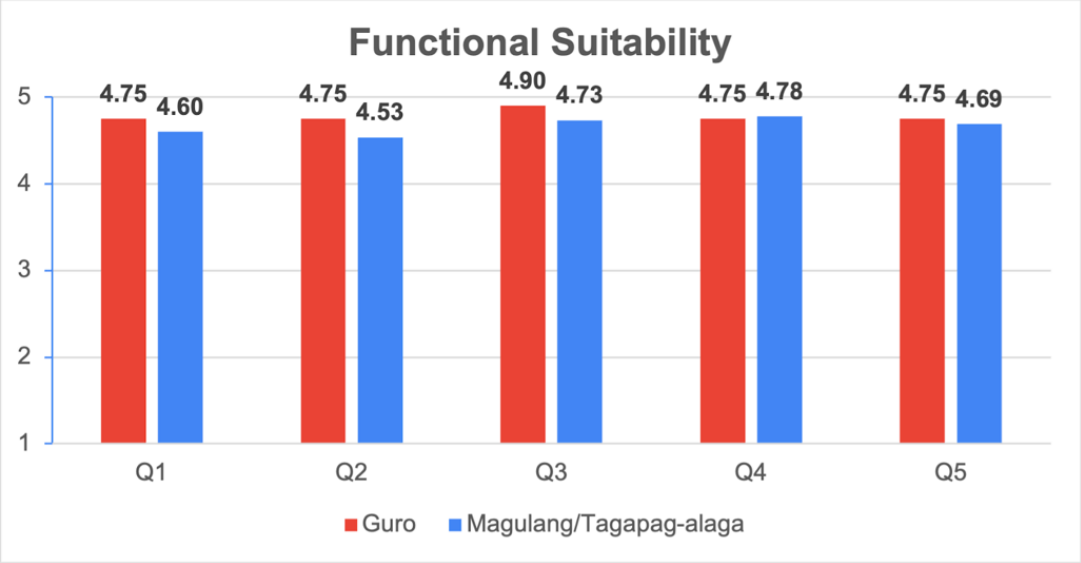
Figure 35. Demographic Profile of Respondent in terms of Gender



Both groups of respondents were overwhelmingly female: 91.1% of parents were female and 8.9% were male; likewise, 90% of teachers were female and 10% were male. This information reflects the predominance of women in elementary teaching and parenting roles, and is consistent with the prior finding that female guardians (usually mothers) parent or manage many young children's school-related activities and routine technology use, including the use of educational applications.

ISO 25010 SYSTEM EVALUATION

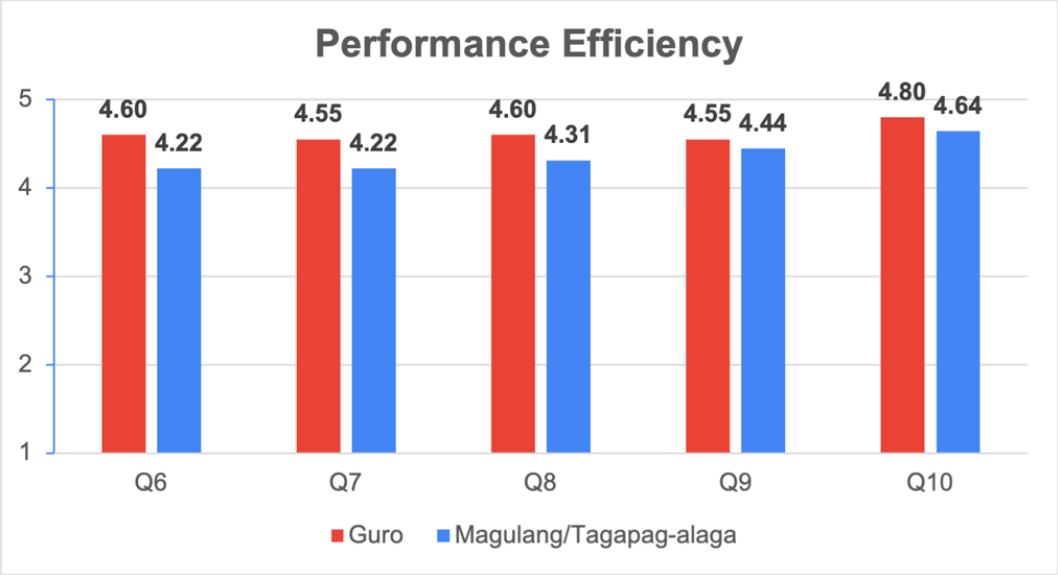
Figure 36. Evaluation Results Based on Functional Suitability



Looking at the Functional Suitability results, we have excellent results from teachers and parents, for a mean of 4.72, with teachers scoring 4.78 and parents 4.67, meaning the app does the functions it is supposed to do. Accordance with statements says users gave it strong scores around accuracy, appropriateness of features, curriculum alignment, the usefulness of text-to-speech, and reliability. Curriculum alignment and text-to-speech got above-average scores overall from users, which means that they found it a useful tool for teaching and learning.

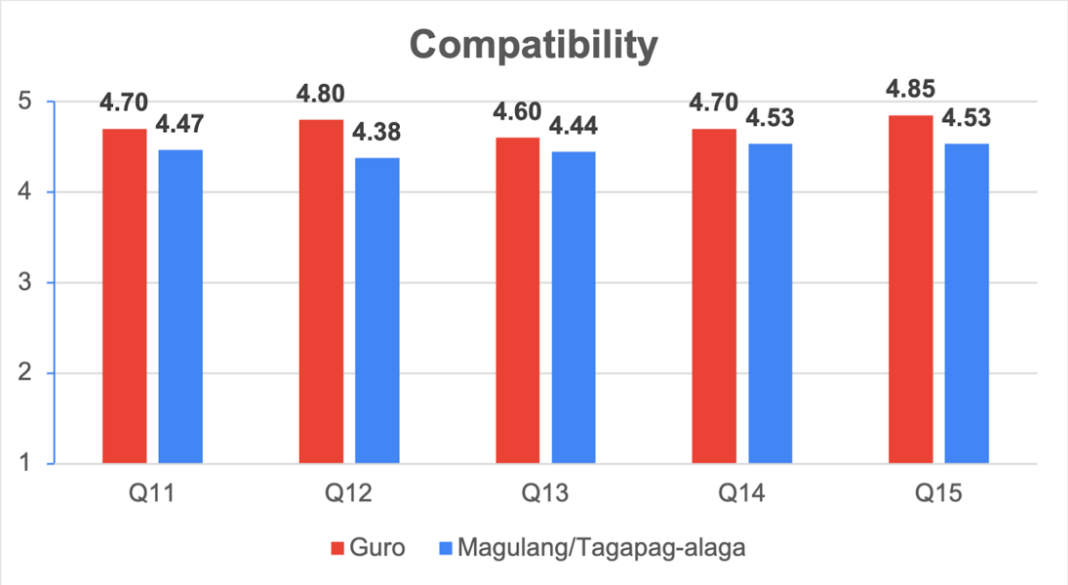
Overall our ratings here are solid, meaning the application meets functional requirements and users are supported in accomplishing their work successfully.

Figure 37. Evaluation Results Based on Performance Efficiency



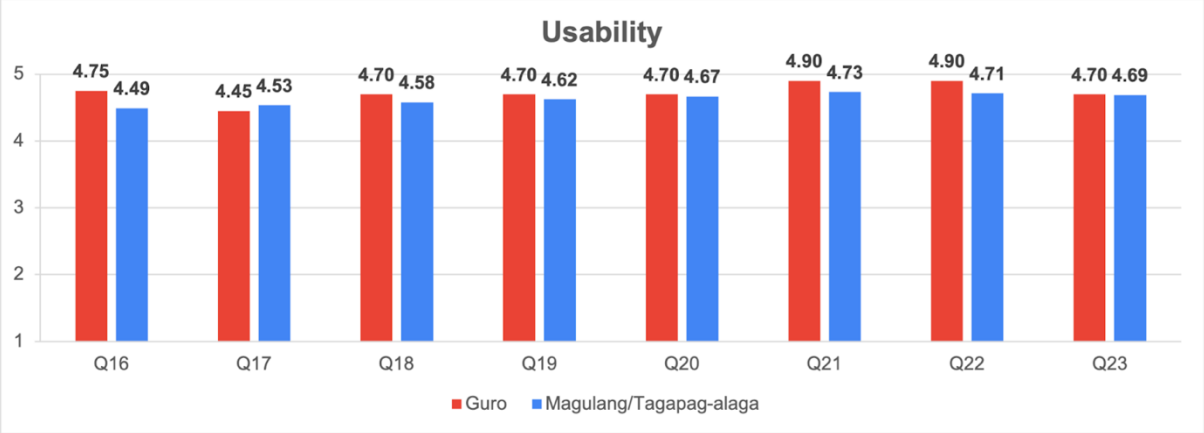
The results for Performance Efficiency were positive for both teachers and parents with a combined mean of 4.49 which indicates Agree interpretation. Teachers' mean score in this area was 4.62 (Strongly Agree) while parents' mean score was 4.37 (Agree). Our app is satisfactory in terms of speed, responsiveness and data usage. Both parents and teachers feel that the app 'loads quickly and stays responsive even on an average internet connection and runs smoothly when using several functions. They agree that the app does not slow down their device and uses very little data. The highest mean score in this category is for the fast display of progress report and scores which indicate that our users trust the feedback process and find it fast and reliable. Overall, we can conclude that our application provides the users with fast and responsive experience.

Figure 38. Evaluation Results Based on Compatibility



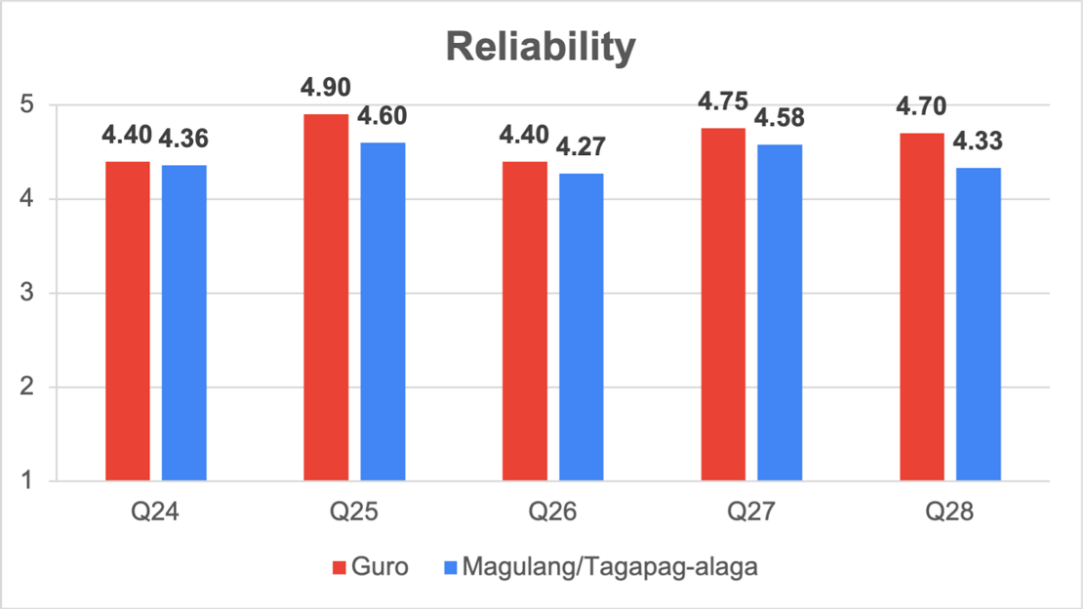
Compatibility results displayed a positive feedback from the teachers and the parents with an overall mean of 4.60, which is categorized under Strongly Agree agreed. The teachers rated this area 4.73 (Strongly Agree) while the parents rated it a 4.47 (Agree), which indicated that the app is compatible with varieties of Android devices. They agreed on the statement that using the app on their phones is easy, easy installation and easy to use, stable when other apps are opened on the phone at the same time, and connects properly to features like data saving and text-to-speech. The highest score is given to the ease of access and use item which showing that it is easy to open and use the app. The overall results shows that the application is compatible with a lot of Android devices and perform reliably.

Figure 39. Evaluation Results Based on Usability



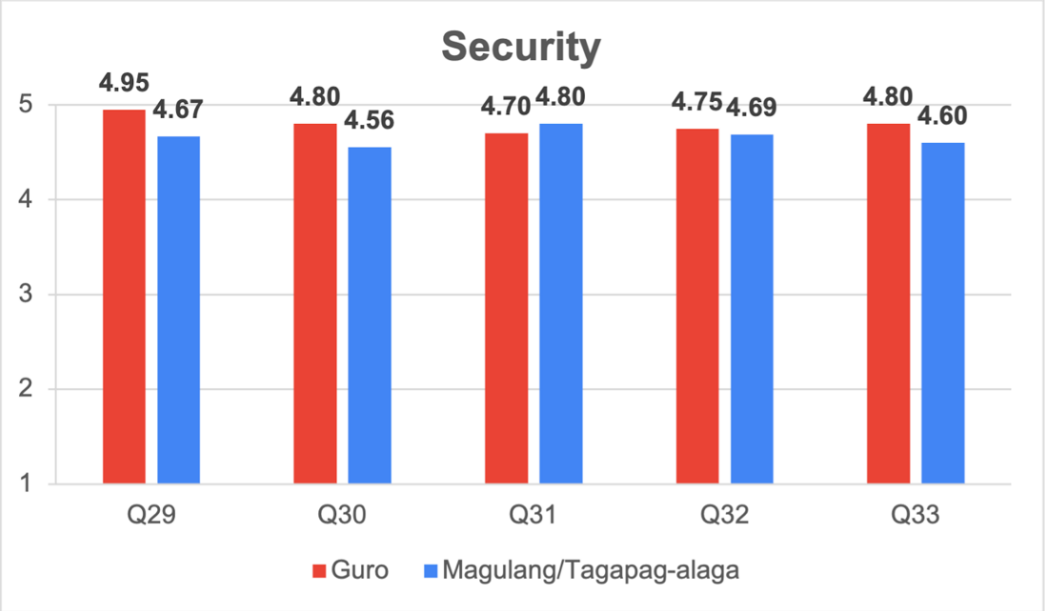
Findings also indicated that respondents were extremely positive about Usability, as reflected in the overall mean of 4.73 (Strongly Agree), where teachers giving a mean score of 4.78 (Strongly Agree), while parents gave 4.68 (Strongly Agree). This indicates that MATHtatag was simple to use, simple to move around through, as well befit for adults and children. They noted that the interface was clear, child centered, and manageable to use, so that learners can manage the app with very minimal help, that the design and layout help you to understand, and that the app keeps the learner's focus while they learn and also builds their confidence in mathematics. Both groups marked that the system enhances the cooperation of teachers, parents, and pupils together, together with increasing parents' involvement in learning. These results overall prove that this application is intuitive, fun, and very usable for all.

Figure 40. Evaluation Results Based on Reliability



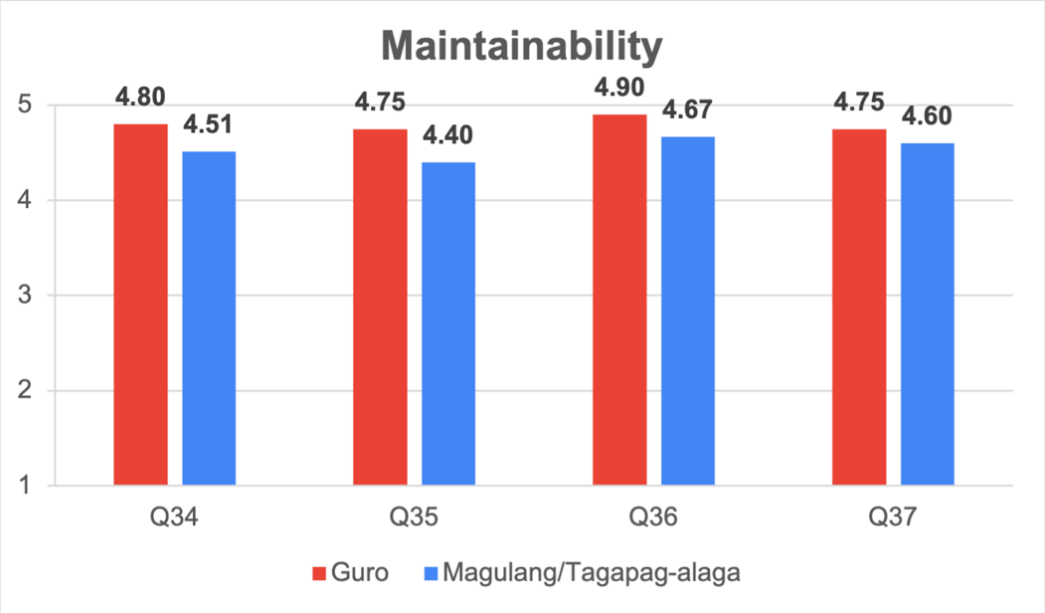
Findings for Reliability showed general consensus between the teachers and parents, with a combined mean of 4.53 interpreted as Strongly Agree. The teachers scored it at 4.63 (Strongly Agree) and the parents at 4.43 (Agree), showing that the application can be relied on in general use. They felt that the app did not crash, that it was able to correctly record learner's score and that the application was stable in length of use. Despite fluctuations in connection, both agreed that the application worked and could recover from outages. Overall, we can say that the application is reliable, stable and able to perform smoothly under different conditions.

Figure 41. Evaluation Results Based on Security



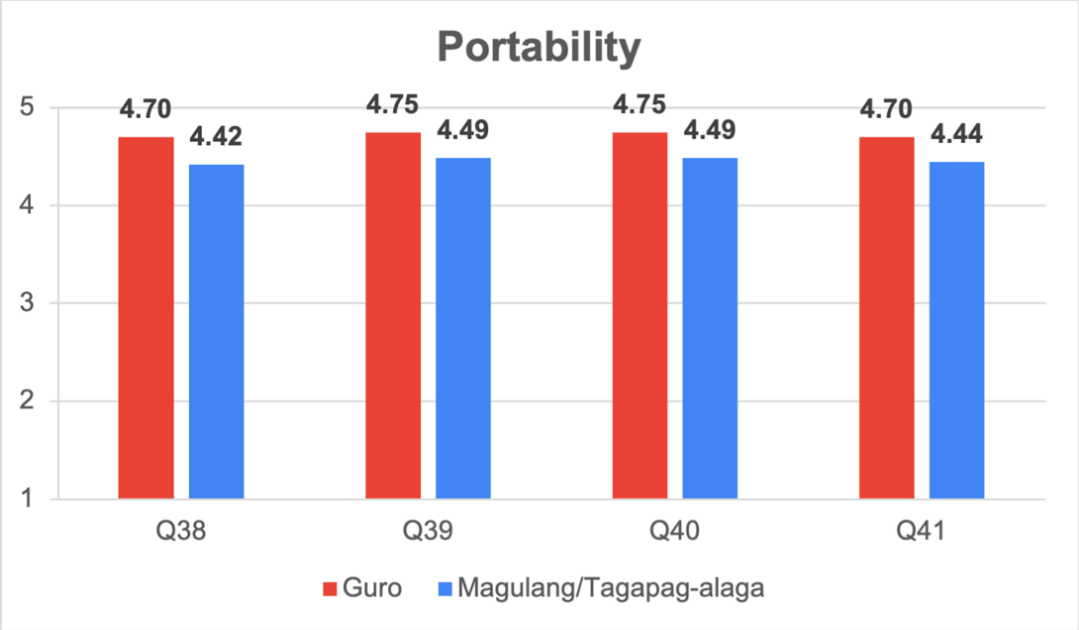
Findings for Security likewise showed very strong agreement from teachers and parents overall (combined mean 4.73, Strongly Agree), teachers scored 4.80 (Strongly Agree), and parents 4.66 (Strongly Agree). Parents and teachers feel confident in how well the app protects learner information. They agreed that personal data is safe, that only authorized users are able to access accounts, and that the system limits the collection of unnecessary information. They agreed that the app prevents misuse or unauthorized sharing of data and that use of system-generated IDs adds value. Results for this section confirmed that overall, the application is strong with respect to security features and handles learner information in an area that is safe.

Figure 42. Evaluation Results Based on Maintainability



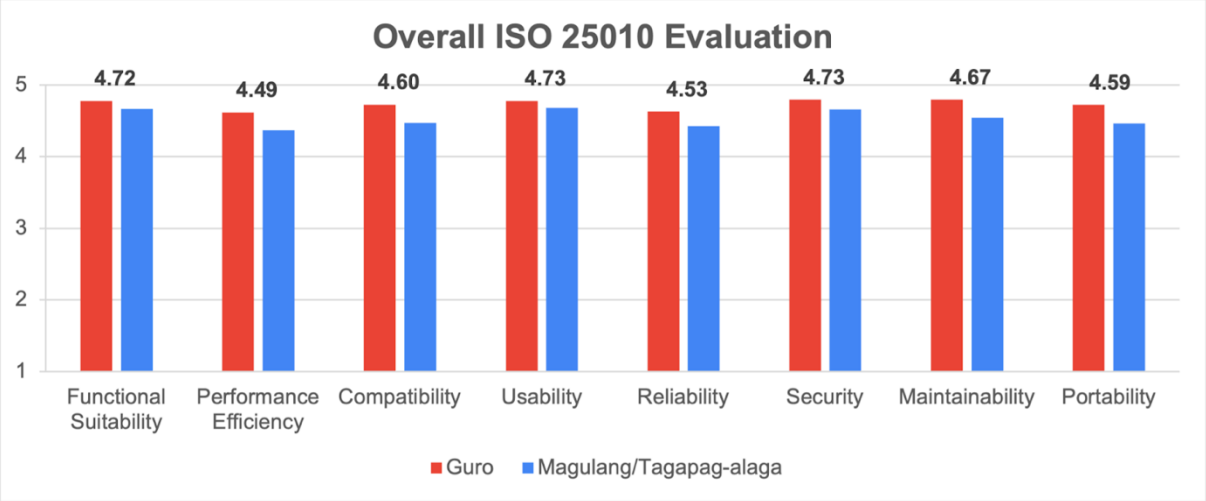
Maintainability has a positive result with the users confidence from both group, the teachers and parents with a mean of 4.67 which falls within the Strongly Agree. Teachers rated this 4.80 (Strongly Agree) and parents 4.55 (Strongly Agree). Users feel that the app can be updated and maintained as years go on. They agree that applying updates is easy to do. We can report any problems easily using the feedback feature. We feel that the system gets better over time with all the added features and upgrades. They agree that the app is stable and works well over updates. Altogether, these ratings indicates that the application is easy to maintain, allows flexibility for continued improvement and still works reliably for periods of time.

Figure 43. Evaluation Results Based on Portability



The portability results generally agreed by teachers and parents with the same rating of 4.59 (which is included in Strongly Agree). The teachers' scores reading 4.72 and can be described as Strongly Agree by the teachers; and the parents score reading 4.46, which we can described as Agree. It was indicated that the app works on many Android devices, and we could locate it in many places without any issues. We generally agreed that it was easy to install and use in at home and schools, that we could transfer it to another device as needed, and that it was simple enough for us even we had very little technical knowledge. This indicates that the application is portable and adaptable to various Android users.

Figure 44. Overall ISO 25010 Evaluation Results



Using the 5-point Likert scale, the teachers and the parents rated MATHtatag across the eight ISO 25010 software quality characteristics. The combined general mean of 4.63 (mean rating of teachers is 4.73 and mean rating of parents is 4.54) is interpreted as Strongly Agree. These ratings of the teachers and parents imply that users will perceive MATHtatag as a high-quality educational system. The very high ratings across the different categories showed that MATHtatag is functional, usable, reliable, secure, efficient, maintainable and portable. Overall, the ISO 25010 evaluation suggests that the application meets important software quality standards and effectively supports Grade 1 mathematics learning.

Overall, the ISO 25010 evaluation reveals that MATHtatag provides a reliable and user-friendly learning experience for Grade 1 students in learning mathematics. The high rating from teachers in addition to parents confirms that the system's design, features and performance align to the needs of real teachers and their students. These results support the general effectiveness of the application and indicate that MATHtatag is appropriate for further deployment and expansion to benefit more users.

PRETEST & POSTTEST EVALUATION RESULTS

To measure the extent to which MATHtatag had developed the mathematics performance of Grade 1 learners, a pretest and posttest, administered at Grade 1 level, were used in this section. The learners' prior knowledge was established through the pretest prior to using the app. For the posttest, teacher-led lesson activities and interactive activities that were assigned from MATHtatag and completed by them where used. The learners' scores before and after the intervention were compared in this section, while the degree of improvement based on descriptive statistics, gain scores and percentage increase are described.

Descriptive Statistics

Pretest Descriptive Statistics

Table 20.
Pretest Descriptive Statistics

Measure	Value
Lowest Score	**6**
Highest Score	**17**
Mean Score	**12.64444**
Mean Percentage Score (MPS)	**63.22222 %**

The result of the pretest indicates that the learners' initial performance varied widely before using the MATHtatag application. With scores from 6 to 17 out of 20, the learners have

competencies assessed that posed a problem for some but were easier for others even before the intervention. The computed mean score of 12.64, equivalent to an MPS of 63.22%, indicates a moderate level of mastery or a class that knew only in part the skills included in the test. In summary, the pre-test results reflect that learners were at a Moving Towards Mastery level with many opportunities to improve through teacher instruction and the interactive practice within MATHtatag.

Posttest Descriptive Statistics

Table 21.
Posttest Descriptive Statistics

Measure	Value
Lowest Score	**14**
Highest Score	**20**
Mean Score	**19.17778**
Mean Percentage Score (MPS)	**95.88889%**

The post-test yielded very good results. It shows that the learners' performance was substantially improved following the lessons and the use of the MATHtatag application. Scores went from a range of 14 to 20 (meaning that all learners scored high than the pretest, and none scored below the passing mean), Out of a scale of 20 the computed mean score of the learners is 19.18, equivalent to an MPS of 95.89. Based on the Department of Education's levels of mastery, an MPS rate of 86% - 95% lands under "Closely Approximating Mastery." 96% and above is categorized as "Mastered" (Department of Education, 2012). This means that the

learners exhibited very high levels of proficiency in the skills tested. It can be seen that the posttest results indicated that the combination of teacher-directed instruction supplemented with the follow-up activities using the MATHtatag application helped learners markedly improve their understanding of the Grade 1 Mathematics competencies.

Summary of Pretest and Posttest Score Distribution

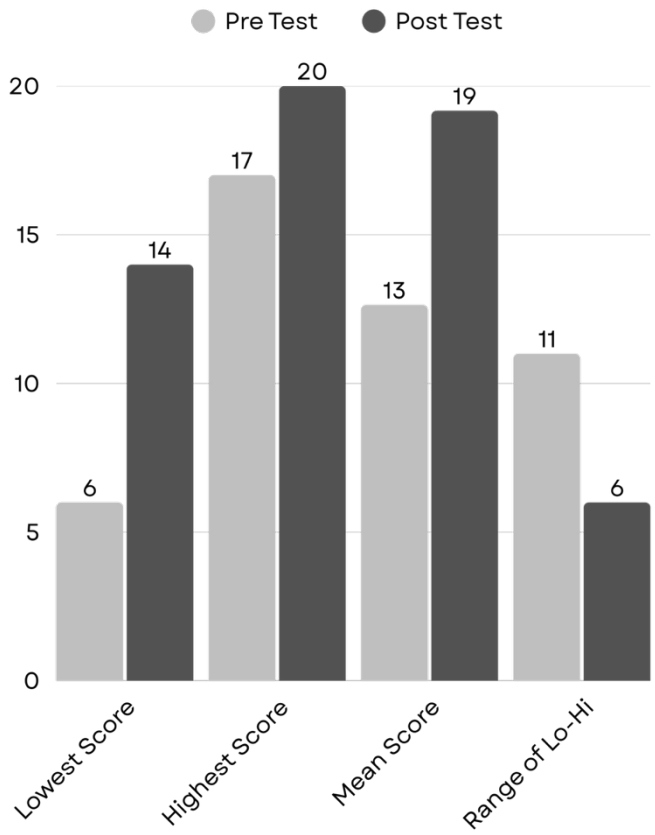
Table 22.
Pretest and Posttest Score Distribution

Score Range	Number of Learners (Pretest)	Number of Learners (Posttest)
0 – 5	0	0
6 – 10	13	0
11 – 15	19	3
16 – 20	13	42
TOTAL	**45**	**45**

The overall distribution of the scores presents itself as within the expected improvement from pretest to posttest. In the pretest, we see that 13 learners scored 6-10, 19 scored 11-15, and only 13 gained entry to score 16-20, showing that many learners are still developing their understanding of the relevant math language. In the posttest, no learner scored in the lower ranges, and only 3 are in the 11-15 range, while 42 learners score in the 16-20 range. This is a massive gain and suggests that essentially all of the learners operated at a high level after teacher-led lessons and MATHtatag activities. As a whole, this distribution suggests that this intervention effectively moved this class from developing to mastery level performance.

Range of Pretest and Posttest Scores

Figure 45. Range of Pretest and Posttest Scores



The range tells us how far apart the high and low scorers were in both tests. For the pretest, the high was 17 and the low 6. A range of 11 indicates the learners had different starting points. Then, after the teacher taught the learners and after they filled out their MATHtatag, they did the posttest. The high was 20 and the low was 14, meaning the range was smaller at 6. This tells us the learners' scores were closer together, and in this case that means that the students who struggled earlier improved. The change from 11–6 is that the intervention has narrowed the gap in learning and the kids who didn't do as well are more on par with their classmates now.

Section-Level Analysis

Section-Based Performance Comparison

Table 23.
Class A Results

CLASS A	Pre Test	Exercise														Post Test
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Mean Percentage Score	61.15%	93.2	88.9	93.3	69.1	75.6	77.5	94.3	82.9	56	84	100	76	76	96.7	96.20%
Passing Rate	38.46%	100	88.9	91.7	0	77.8	87.5	85.7	85.7	40	80	100	60	80	100	92.30%
Range of Highest-Lowest	9	2	2	3	1	1	1	2	2	3	2	0	3	1	1	6
Highest	16	15	5	5	7	4	4	5	5	4	5	5	5	4	5	20
Lowest	7	13	3	2	6	3	3	3	3	1	3	5	2	3	4	14
No. of Items	20	15	5	5	15	5	5	5	5	5	5	5	5	5	5	20

Table 24.
Class B Results

CLASS B	Pre Test	Exercise											Post Test
		1	2	3	4	5	6	7	8	9	10	11	
Mean Percentage Score	56.56%	66.3	57.1	73.8	83.1	68.3	61.7	70	75	80	86.7	86	96%
Passing Rate	12.50%	0	21.4	69.2	76.9	50	50	50	62.5	77.8	88.9	80	100%
Range of Highest-Lowest	11	2	4	4	4	4	4	3	3	2	2	3	2
Highest	17	7	5	5	5	5	5	5	5	5	5	5	20
Lowest	6	5	1	1	1	1	1	2	2	3	3	2	18
No. of Items	20	15	5	5	5	5	5	5	5	5	5	5	20

Table 25.
Class C Results

CLASS C	Pre Test	Exercise													Post Test
		1	2	3	4	5	6	7	8	9	10	11	12	13	
Mean Percentage Score	71.56%	82.6	92.4	69.3	70	82.9	88.6	83.1	67.3	80	92.3	93.8	92.3	98.3	97.20%
Passing Rate	69.23%	82.6	92.4	0	71.4	71.4	85.7	76.9	72.7	84.6	92.3	100	92.3	100	100%
Range of Highest-Lowest	10	3	3	1	5	4	3	4	4	2	2	1	2	1	2
Highest	17	3	3	7	5	5	5	5	5	5	5	5	5	5	20
Lowest	7	0	0	6	0	1	2	1	1	3	3	4	3	4	18
No. of Items	20	3	3	15	5	5	5	5	5	5	5	5	5	5	20

For all the Grade 1 classes there was improvement from pretest to posttest in light of the level of performance that the students started from, Class C the highest at (71.56% MPS), then Class A (61.15), Class B (56.56) for the apps based exercises there was marked improvement and despite the low actual starting point for Class B eventually kept pace with the others which meant that MATHtatag did indeed support learners when the ammo was less than ready to begin with. The learning gaps were closer together the more the intervention was undertaken. Ranges of 9–11-point scores on the pretest were compressed to be within the same range of 6 point scores all the way down on Class A, and only 2 points apart between Class B and C in posttest, meaning lower performing learners caught up with stronger peers. By the end of intervention all sections were at mastery performance level, Class A scoring 96.20% MPS (92.30% passing), with Classes B and C both scoring 96% and 97.20% MPSs with 100% passing rates. The consistent gains suggest continual use of MATHtatag helped learners extend the mastery, and produced balanced outcomes across being grades A to C,

strong support for an ISO 25010FunctionalSuitabilityhigh rating for the system, confirming MATHtatag does improve Grade 1 Mathematics learners.

Gain Score Analysis

Individual Gain Scores

Table 26.
Individual Gain Score

LEARNER NO.	PRETEST	POSTTEST	GAIN SCORE
L1	14	20	6
L2	10	20	10
L3	15	20	5
L4	9	20	11
L5	8	15	7
L6	13	20	7
L7	12	20	8
L8	6	18	12
L9	6	17	11
L10	10	20	10
L11	17	20	3
L12	11	20	9
L13	14	18	4
L14	14	20	6
L15	9	15	6
L16	8	20	12
L17	9	17	8
L18	15	20	5
L19	8	14	6
L20	11	20	9
L21	11	20	9
L22	7	20	13
L23	16	20	4
L24	16	20	4
L25	13	20	7

L26	16	20	4
L27	16	19	3
L28	13	20	7
L29	13	20	7
L30	16	20	4
L31	17	20	3
L32	17	20	3
L33	9	20	11
L34	16	20	4
L35	14	20	6
L36	15	20	5
L37	7	20	13
L38	11	17	6
L39	14	19	5
L40	17	18	1
L41	17	19	2
L42	16	18	2
L43	16	19	3
L44	14	20	6
L45	13	20	7

In the individual gain scores, we can see how much each learner improved from the pretest to the posttest. All those who participated - 45 learners - gained, meaning that each learner had an increase in their score after the teacher's lessons and after working on the practice activities of the MATHtatag application. We see that many learners also gained less than a total of 10, in the median to good range of 4 to 7 - this shows steady intervention throughout. Some learners had large gains as shown in the results of L2, L8, L9, L10, L22 and L37, of 10 points or higher - this indicated that the practice enabled them to internalize the competencies even more! On the other hand, there were learners who scored earlier high marks on pretest, so their gains were not as high; these learners were L11; L27; L31; L32; L40; L41; L42; L43 scores of just 1 -3 points gain but still gained - contributing to their strong

understanding. Overall, the gain score conforms to our expectation because the intervention was clearly a successful one on the whole - none of the students' scores decreased as evidenced by their pretests and posttests, and all of them showed gains in their mastery of the assessed skills.

Mean Gain Score

Table 27.
Mean Gain Score

Measure	Value
Mean Gain Score	**6.54**

The Mean Gain Score uses the pretest and posttest scores and the average of that difference among all learners is 6.54. This indicates that on average, the learner's scores increased by a little more than 6 points following the teachers' lessons and the interactive playtime inside the MATHtatag app. A 6.54-point gain in the 20-item test indicates that performance has improved, and that most learners were able to answer more items correctly in the posttest, and positively influenced their mastery of the Grade 1 mathematics included in the intervention. This steady improvement across almost all learners shows that classroom instruction and MATHtatag activities were effective in strengthening their learning in the lessons.

The increase in scores reflected in the mean gain scores is often reported as evidence of learning in educational literature, where gain scores are weighted against the total score possible to evaluate the magnitude of effect of an instructional intervention (Dimitrov & Rumrill, 2003).

Pretest–Posttest Improvement Percentage

Table 28.
Pretest and Posttest Improvement Percentage

Pretest Mean	Posttest Mean	Improvement %
12.64444	**19.17778**	**51.6696 %**

Pretest–Posttest Improvement Percentage shows an incredible gain by the learners. The group had a mean of 12.64 on the pretest and a mean of 19.18 on the posttest providing an impressive 51.67% gain. This indicates the big impact the Teacher led lessons and using MATHtatag application as a follow-up practice tool.

Improvement of more than half the original performance level indicates that not only did they reinforce existing knowledge, but that it was much higher than this after the treatment. In educational research, such a large gain is meaningful because it demonstrates that the learning strategy/tool has substantially boosted student learning that is clearly attributable to the effects of the treatment. Hattie (2009) asserts that noticeable increases in post-intervention performance are good indicators that the instructional approach produces learning gains for the learner.

In conclusion, the classroom instruction provided through in-class teaching and the use of MATHtatag's interactivity allows students to obtain the mathematical knowledge needed to be on grade level for their first grade year, resulting in improved test scores of over 51.67%.

SYSTEM / SOFTWARE TESTING RESULTS

System Testing was used to ensure that the MATHtatag mobile app performed as intended by way of its requirements. We verified the major features of the system: authentication, exercise management, learner assessment, performance monitoring and administrative controls. Each test was executed and assessed to ascertain that the system returned the successful results.

The test cases were developed for the purpose of determining if the actual system functioning is consistent with what was expected. A comparison was made between the expected and actual results and results were recorded as either a pass or a fail.

Authentication Module Test Cases

Table 29.
Authentication Module Test Cases

Test Case ID	Test Case Description	Expected Result	Actual Result	Test Result / Status
TC-AUTH-01	Login using valid teacher credentials	System allows access to Teacher Dashboard	Teacher Dashboard displayed successfully	Pass
TC-AUTH-02	Login using invalid teacher credentials	System displays an error message	Error message displayed correctly	Pass
TC-AUTH-03	Teacher login without email verification	System prevents login and shows verification notice	Login blocked with verification message	Pass
TC-AUTH-04	Parent login using valid Parent Key	System redirects to Parent Dashboard	Parent Dashboard displayed successfully	Pass

TC-AUTH-05	Parent login using invalid Parent Key	System displays invalid key message	Error message displayed correctly	Pass

Exercise Creation and Management Test Cases

Table 30.
Exercise Creation and Management Test Cases

Test Case ID	Test Case Description	Expected Result	Actual Result	Test Result / Status
TC-EX-01	Create a new math exercise	Exercise is saved successfully	Exercise saved and listed	Pass
TC-EX-02	Add Filipino text-to-speech to exercise	Audio is generated and attached	Audio generated and playable	Pass
TC-EX-03	Assign exercise to a class	Exercise is assigned to selected learners	Exercise assigned successfully	Pass
TC-EX-04	Edit an existing exercise	Changes are updated in the system	Exercise updated correctly	Pass

Student Exercise Answering Test Cases

Table 31.
Student Exercise Answering Test Cases

Test Case ID	Test Case Description	Expected Result	Actual Result	Test Result / Status
TC-STU-01	Start an assigned exercise	Exercise opens with instructions	Exercise opened successfully	Pass
TC-STU-02	Filipino TTS playback during exercise	Instructions are read aloud	Audio played correctly	Pass
TC-STU-03	Submit completed exercise	Answers are recorded and scored	Results saved successfully	Pass

TC-STU-04	Auto-submit when time limit ends	Exercise auto-submits	Auto-submit triggered correctly	Pass

Parent Dashboard Test Cases

Table 32.
Parent Dashboard Test Cases

Test Case ID	Test Case Description	Expected Result	Actual Result	Test Result / Status
TC-PAR-01	View learner performance summary	Performance data is displayed	Data displayed correctly	Pass
TC-PAR-02	View exercise results	Scores and completion status shown	Results shown accurately	Pass
TC-PAR-03	Update parent profile information	Profile updates saved	Information updated successfully	Pass

Admin and System Management Test Cases

Table 33.
Admin and System Management Test Cases

Test Case ID	Test Case Description	Expected Result	Actual Result	Test Result / Status
TC-ADM-01	Verify a teacher account	Teacher status updated to verified	Teacher verified successfully	Pass
TC-ADM-02	Block a teacher account	Teacher access is restricted	Access blocked correctly	Pass
TC-ADM-03	Enable maintenance mode	System displays maintenance notice	Maintenance notice displayed	Pass
TC-ADM-04	Display update notification	Update alert shown during startup	Update alert displayed correctly	Pass

Chapter 5
SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

This chapter states the overall conclusion of the study, including the summary of major findings of the evaluation of the system used, pretest-posttest results of the grades 1 learners, stated conclusions according to the specific objectives and recommendations for improvement of the system, implementation and future researches.

Summary of the Findings

The study focused on developing and assessing the acceptance of MATHtatag, a mobile application designed to help Grade 1 mathematics learning through teacher assistance in assessments, Filipino text-to-speech, and parent monitoring features. For this research, the researchers applied a one-group pretest–posttest approach and involved 45 Grade 1 learners from three sections, as well as their teachers and parents acting as system evaluators. Evaluation data were gathered through pretest and posttest score improvements and an ISO 25010-based acceptability survey that was administered to teachers and parents as it measured learners' performance improvement together with the general software quality of the developed app.

A overall mean rating of 4.63, interpreted as Strongly Agree, shows acceptability based on the ISO 25010 results. Functional Suitability obtained a mean of 4.72 meaning lessons, activities, and Filipino text-to-speech worked correctly and supported the MATATAG Curriculum. The Performance Efficiency mean was 4.49, indicating consistent loading of lessons and responsiveness, although parents stated that it was slower on older devices. The mean for Compatibility was 4.60 meaning that the MATHtatag app was usable on most of the

respondents' Android phones. An Usability mean of 4.73 indicates that the interface was simple, child-friendly and stimulating. The Reliability mean was 4.53 meaning that the app runs consistently and helps learners save progress. Security received 4.73, indicating that the respondents had strong hopes that the system guaranteed privacy and protection. Maintainability was 4.67 meaning the app can be updated and improved. Portability had a mean of 4.59 indicating that the app could easily be installed, opened, and transferred to other Android-supported devices.

As for the learning outcomes, the pretest–posttest results indicate that there is a large improvement in the performance of the learner with the increased percentage mean scores from 63.22% (pretest) to 95.89% (posttest). A certainty in improvement is established. The Mean Gain Score of 6.54 and the improvement rate, 51.67% indicate an increase in mastery after the teacher-led lessons and mastery through effortful and repeated practices. During the pretest majority of the learners got a score of 11–15, and on the other hand, after the intervention, majority 42 out of the 45 learners reached the 16–20 mastery, therefore enough proof that MATHtatag is a tool in learning Grade 1 Mathematics Competencies.

Overall, MATHtatag is an effective and high quality educational tool that met software design standards, promoted meaningful learning gains, and allowed learners to achieve mastery in mathematics, with the help of teacher or other instructor instruction and app-based activities.

Conclusion

Considering the findings results from this study, these conclusions that are directed by the study objectives are recommended. A mobile application was developed, where teachers could create and manage mathematics assessments aligned to MATATAG Curriculum for Grade 1 learners. The Filipino text-to-speech feature can assist the learners in understanding and answering questions, and following instructions, as evidenced by the functional suitability and usability scoring high. The accurate real-time analytics and progress dashboards make it convenient for teachers and parents to check on learner performance. The app strengthens the collaboration among the teachers, parents, and learners, as indicated in the positive usability responses, which has brought the conveniences to the parents in spending at home with their children. The system is acceptable based on ISO 25010 measures, as shown by the overall mean of 4.63 indicates strong agreement with all quality categories. The app is effective for performance improvement of the learners in mathematics, as shown with how much the posttest results improved from the pretest scores.

In addition, things like support for teachers and parents were also present in the system. Features like the Gemini AI feature, which helped teachers come up with assessment questions more quickly could help make the process of lesson preparation on curriculum simpler as well, and assessment questions can become more streamlined and easier for the teacher. Features such as excel export features, that let the teacher save and archive list of learners and their records also make it more useful to go back and see how learners have progressed, and to keep documents back at the school. Supporting the parents, simple recommendations would be provided to the parents that made it possible for them to understand what their child was learning, and help provide additional support at home.

Overall, the results confirm that the system increased instructional accessibility, assessment processes, and student performance, demonstrating that this is a feasible solution for early-grade mathematics instruction.

Recommendations

Considering the findings and conclusions of this study, these recommendations are presented to guide the further improvement of MATHtatag system, the application of findings in schools, provide implications for future research and provide recommendation to practices: These recommendations are aimed at improving performance, enhancing accessibility, enrich user experience and broaden the educational impact of the application.

1. For better accuracy and broader implications of the findings in any future work, future analyses should use a larger sample population and more than one site. The future work will also assist researchers in evaluating MATHtatag's long-term results on students.

2. Offline support and optimization for older, low-end Android devices to let more learners access the app and gain from it seamlessly.

3. iOS and web versions plus stronger data backup/syncing so we can update the app securely and get more reliable analytics.

4. Schools should adopt MATHtatag as a complement to their teaching and run sessions for teachers and parents to help them understand how to make best use of its dashboards, analytics, and Filipino text-to-speech features.

5. Educational leaders may support the integration of digital assessment tools like MATHtatag to strengthen data-driven instruction and enrich early mathematics learning in primary education.

6. Policies may be developed to encourage the adoption of accessible, localized learning technologies that support learners with different needs, especially in early-grade numeracy.

With the implementation of these recommendations, MATHtatag is on track to achieve an enhanced level of functionality, efficiency, and scalability, thereby increasing its utility as both a resource for teaching mathematics in early grades, and also aiding the creation of collaborative networks for teachers, parents and students.

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APPENDICES

APPENDIX 1

APPROVAL SHEET

This capstone project **MATHtatag: A Mobile Application for Online Assessment and Performance Monitoring with Filipino Text-to-Speech for Grade 1 in Mathematics** prepared and submitted by Cedick L. Artiaga, Engelbert R. Morales, Ken P. Cas, and Lloyd Jernell C. Loteriña in partial fulfilment of the requirements for the degree, Bachelor of Science in Information Technology has been:

Recommended for Approval and Acceptance:

Date:

ICON C. OBMERGA _____
Faculty-in-Charge

Approved by the Capstone Project Defense Panel:

MARIE ANDREA E. ZURBANO, MIT _____

ROSARIO D. ANULAO, MIT _____

MARK VENCE V. DUNGCA _____

Accepted for the Department of Information Technology:

_____ _____
ICON C. OBMERGA
Chairperson

Accepted for the College of Computer and Information Sciences:

_____ _____
DR. LOURDES B. AVILA
Academic Head

APPENDIX 2

DATA GATHERING INSTRUMENT

RESEARCH QUESTIONNAIRE

Capstone Title: MATHtatag: A Mobile Application for Online Assessment and Performance Monitoring with Filipino Text-to-Speech for Grade 1 in Mathematics

Proponent: Ken P. Cas, Lloyd Jernell C. Loteriña
Cedick L. Artiaga, Engelbert R. Morales

Degree Program: Bachelor of Science in Information Technology

---

Dear Respondent,

We, the researchers taking BSIT and currently conducting our research study entitled: "MATHtatag: A Mobile Application for Online Assessment and Performance Monitoring with Filipino Text-to-Speech for Grade 1 in Mathematics". In line with this may we request a little time to test and evaluate the proposed system to conclude its acceptability in attaining the objectives of this study.

In compliance with the Data Privacy Act of 2012 (Republic Act No. 10173), we are committed to protecting your personal information. The data collected through this survey will be used solely for academic and research purposes related to the said study.

Kindly rate the following characteristics of the applications by placing a check (✓) mark on the space provided.

Five Likert Scale:

1 – Strongly Disagree
2 – Disagree
3 – Neutral
4 – Agree
5 – Strongly Agree

---

1. Respondents Profile:

Respondents': ☐ Teacher ☐ Parent/Guardian
Name (Optional): _____ Age: _____
Sex: ☐ Male ☐ Female ☐ Other

2. Software Acceptability and Features

ISO 25010 Category	Survey Questions	1	2	3	4	5
Functional Suitability	1. The app performs its intended functions accurately.					

	2. The features of the app are appropriate for my role and needs.					
	3. The activities and lessons are aligned with the Grade 1 MATATAG Curriculum.					
	4. The Filipino text-to-speech feature helps me and learners understand math questions and instructions more clearly.					
	5. The app provides correct and reliable results about learner performance.					
Performance Efficiency	6. The app loads quickly and responds without noticeable delay.					
	7. The app performs well on devices with average internet connection.					
	8. The app runs smoothly even when several functions are used at once.					
	9. The app uses only a small amount of internet data and does not slow down our device.					
	10. Progress reports and scores appear immediately after each activity.					
Compatibility	11. The app works well on our Android mobile phone.					
	12. The app is easy to install and simple to set up.					
	13. The app remains stable when used together with other applications.					
	14. The app connects properly to the internet features, such as saving data and using text-to-speech.					
	15. The app is easy to access and use on my device.					
Usability	16. The app is easy for me to use and navigate.					
	17. My learner can use the app easily with minimal assistance.					
	18. The design and layout are simple and suitable for both adults and young learners.					
	19. The interface, buttons, and icons are simple and child-friendly.					
	20. The app makes learning mathematics more effective and engaging for the learner					
	21. The app encourages cooperation/collaboration among teachers, parents, and learners.					
	22. The app improves the learner's confidence and interest in mathematics.					
	23. The app increases my involvement in my learner's learning activities.					

Reliability	24. The app works consistently without crashing, freezing, and losing data during use.					
	25. The learner's scores and progress are always recorded and displayed/retrieved accurately.					
	26. The app functions properly even when the internet connection is weak.					
	27. The app remains stable even after long periods of use.					
	28. The app can recover after a temporary loss of internet connection.					
Security	29. The learner's personal information is protected by the system.					
	30. Only authorized users can access their accounts and data.					
	31. The app only records learners' names for school use and does not collect other personal data such as addresses or photos.					
	32. The system protects information from misuse or unauthorized sharing.					
	33. The app uses system-generated IDs to protect learners' privacy, while names are recorded only for school monitoring purposes.					
Maintainability	34. The app can be updated easily to improve its features.					
	35. I can report errors or issues conveniently using the feedback or report feature.					
	36. The app has continuous improvements to provide better learning experiences.					
	37. The app keeps working well whenever it is updated					
Portability	38. The app is easy to install on different Android devices.					
	39. The app can easily be adapted for use of other schools or home setting.					
	40. The app can be transferred to another device without difficulty.					
	41. The app can easily be used by users even with limited technical skills.					

Additional Comments and Suggestions:

_____

_____

CERTIFICATION

This is to certify that the questionnaire prepared by the researchers, namely: **KEN P. CAS, LLOYD JERNELL C. LOTERIÑA, CEDICK L. ARTIAGA, ENGELBERT R. MORALES** in relation to the research entitled, **"MATHTATAG: A MOBILE APPLICATION FOR ONLINE ASSESSMENT AND PERFORMANCE MONITORING WITH FILIPINO TEXT-TO-SPEECH FOR GRADE 1 IN MATHEMATICS"**, has been checked and validated.

Signed this day 13th of October, 2025 at Polytechnic University of the Philippines – Lopez, Quezon Campus.

Certification Issued By:

Rosario D. Anulao, MIT

IT Instructor

CERTIFICATION

This is to certify that the questionnaire prepared by the researchers, namely: **KEN P. CAS, LLOYD JERNELL C. LOTERIÑA, CEDICK L. ARTIAGA, ENGELBERT R. MORALES** in relation to the research entitled, **"MATHTATAG: A MOBILE APPLICATION FOR ONLINE ASSESSMENT AND PERFORMANCE MONITORING WITH FILIPINO TEXT-TO-SPEECH FOR GRADE 1 IN MATHEMATICS"**, has been checked and validated.

Signed this day 10th of October, 2025 at Polytechnic University of the Philippines – Lopez, Quezon Campus.

Certification Issued By:

MARK VENCE V. DUNGCA
IT Instructor

CERTIFICATION

This is to certify that the questionnaire prepared by the researchers, namely: **KEN P. CAS, LLOYD JERNELL C. LOTERIÑA, CEDICK L. ARTIAGA, ENGELBERT R. MORALES** in relation to the research entitled, **"MATHtatag: A Mobile Application for Online Assessment and Performance Monitoring with Filipino Text-to-Speech for Grade 1 in Mathematics"**, has been checked and validated.

Signed this day 10th of October, 2025 at Polytechnic University of the Philippines – Lopez, Quezon Campus.

Certification Issued By:

Assic. Prof. Marie Andrea E. Zurbano

Laboratory Head

APPENDIX 3

DECLARATION ON THE USE OF ARTIFICIAL INTELLIGENCE

We, the researchers of the capstone project entitled “MATHtatag: A Mobile Application for Online Assessment and Performance Monitoring with Filipino Text-to-Speech for Grade 1 in Mathematics,” declare that Artificial Intelligence (AI) tools were used in limited and specific parts of this project. Within the MATHtatag application, the Gemini API was integrated to generate text for parent-facing summaries and feedback, while the ElevenLabs API was used to provide Filipino text-to-speech for questions, instructions, and learning support. These features were developed and integrated by the researchers to support Grade 1 learners, teachers, and parents.

During the preparation of this capstone project, ChatGPT and Grammarly were used only as supplementary tools for selected writing and language tasks. ChatGPT was used to help improve the clarity, wording, and organization of selected sentences, while Grammarly was used to check grammar, spelling, punctuation, and sentence clarity. Perplexity was used as a supplementary tool for locating possible sources and references. These tools were not fully relied upon to write the capstone project, conduct the research, analyze the data, or make conclusions. The main content of the manuscript was developed, reviewed, and finalized by the researchers based on their research work, gathered data, verified sources, and approved methodology.

All AI-assisted outputs were treated only as suggestions and were reviewed, edited, and verified by the researchers before use. The researchers acknowledge that AI tools may provide inaccurate or incomplete information and therefore take full responsibility for the

accuracy, integrity, originality, analysis, interpretation, conclusions, and recommendations presented in this capstone project. We affirm that the use of AI was limited to the purposes stated above and that the final capstone project represents the collective work, judgment, and responsibility of the researchers.

KEN P. CAS
Researcher

LLOYD JERNELL C. LOTERIÑA
Researcher

CEDICK L. ARTIAGA
Researcher

ENGELBERT R. MORALES
Researcher

APPENDIX 4
CLIENT FORMS AND REPORTS

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OFFICE OF THE VICE PRESIDENT FOR CAMPUSES
LOPEZ QUEZON CAMPUS

BAGONG PILIPINAS

CERTIFICATE OF DEPLOYMENT

This is to certify that the capstone project entitled **"MATHtatag: A Mobile Application for Online Assessment and Performance Monitoring with Filipino Text-to-Speech for Grade 1 in Mathematics"** has been successfully deployed on October 20, 2025, and is currently in use by all participating stakeholders, namely the teacher, parents, and school administrator of the partner institution.

This deployment marks the completion of the system's implementation phase and confirms that the application is operational, functional, and ready for evaluation and feedback by its intended users.

SIGNED BY:

KEN P. CAS
Lead Researcher

LLOYD JERNELL C. LOTERIÑA
Researcher

CEDICK L. ARTIAGA
Researcher

ENGELBERT R. MORALES
Researcher

Noted by:

ICON C. OBMERGA
Capstone Adviser
PUP- Lopez, Quezon

Approved by:

ANTONIO J. MEDRANO
Head Teacher I
Camohaguin Elementary School

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POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY
LOPEZ CAMPUS

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October 09, 2025

Dear Sir Medrano,

We, the undersigned Bachelor of Science in Information Technology students from the Polytechnic University of the Philippines – Lopez Campus, would like to express our intent to introduce and deploy our capstone project entitled "MATHtatag: A Mobile Application for Online Assessment and Performance Monitoring with Filipino Text-to-Speech for Grade 1 in Mathematics."

The purpose of this project is to assist teachers in conducting online assessments and tracking learners' performance efficiently, while also providing accessible learning support through Filipino Text-to-Speech technology for young learners.

As part of our development and evaluation process, we would like to request permission to visit Camohaguin Elementary School to conduct a short presentation and consultation with your Grade 1 teachers. During this visit, we will demonstrate the current version of our mobile application and gather insights or feature requests that may further enhance its functionality and suitability for early childhood education.

We believe that your valuable feedback and collaboration will greatly contribute to improving the system's usability and alignment with the actual needs of both teachers and learners.

We hope for your kind consideration and approval to coordinate with your teaching staff at a date and time most convenient for you. Should you have any questions or require additional information, please feel free to reach us at ken.panlilio.cas@gmail.com.

Thank you very much for your time and support. We look forward to the opportunity to collaborate with your esteemed institution.

Sincerely yours,

KEN P. CAS
Lead Researcher

LLOYD JERNELL C. LOTERIÑA
Researcher

CEDICK L. ARTIAGA
Researcher

Engelbert R. Morales
ENGELBERT R. MORALES
Researcher

Approve by:

ICON C. OBMERGA
Faculty Adviser

MARIEL M. TAÑOLA 10-18-2025

MAY M. ERIA 10-13-2025

NOAH C. ENERO 10-15-2025

I. MEDRANO 10-09-25

PUP A. Mabini Campus, Anonas Street, Sta. Mesa, Manila 1016
Trunk Line: 335-1787 or 335-1777
Website: www.pup.edu.ph | Inquiries: https://bit.ly/PUPSINTA

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PARTNERSHIP AGREEMENT

This Partnership Agreement (the "**Agreement**") is made on the **13th day of October, 2025** and entered into by and between the

POLYTECHNIC UNIVERSITY OF THE PHILIPPINES – LOPEZ, QUEZON CAMPUS *represented by the research team composed of* ***KEN P. CAS, LLOYD JERNELL C. LOTERIÑA, CEDICK L. ARTIAGA,*** *and* ***ENGELBERT R. MORALES****, hereinafter referred to as the "****Researchers****".*

and

CAMOHAGUIN ELEMENTARY SCHOOL*, represented by* ***Mr. ANTONIO I. MEDRANO, Head Teacher I and Officer-In-Charge****, hereinafter referred to as the "****Partner School****".*

The Partnership

This agreement establishes a formal partnership between the Researchers and the Partner School for the purpose of deploying and evaluating the Capstone Project entitled **"MATHtatag: A Mobile Application for Online Assessment and Performance Monitoring with Filipino Text-to-Speech for Grade 1 in Mathematics."**

The project aims to enhance teaching and learning experiences by enabling teachers and pupils to conduct online assessments, monitor learner performance, and improve comprehension through Filipino text-to-speech (TTS) technology. Both parties recognize the shared goal of advancing digital education and fostering technological innovation in the classroom.

The Researchers

The Researchers agree to undertake the following responsibilities during the implementation and testing of the MATHtatag system:

- Deploy, maintain, and support the application within the school premises during the testing and evaluation period.
- Shoulder all operational expenses related to the project, including but not limited to deployment, maintenance, database hosting, text-to-speech services, and Gemini API usage.
- Provide technical support, troubleshooting, and system updates to ensure smooth operation throughout the duration of the project.
- Safeguard all data collected and ensure full compliance with the app's Privacy Policy and Terms and Conditions.

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- Use any data collected solely for educational and research purposes, with no intent of commercial use.

The Researchers shall retain intellectual property rights over the system while granting the Partner School the right to use it for educational and research purposes during the partnership.

The Partner School

The Partner School, represented by its administration and faculty, agrees to:

- Allow the deployment and evaluation of the MATHtatag application in its Grade 1 Mathematics classes.
- Facilitate communication and coordination between the Researchers and the teachers involved in the study.
- Ensure that the data gathered from the application is used strictly for academic and research purposes.
- Provide feedback and recommendations to help the Researchers improve the application's performance and usability.

The Partner School shall not be required to shoulder any cost during the initial deployment and evaluation period. Should the school wish to continue using the system beyond the deployment phase, responsibility for operational costs (including database hosting, text-to-speech, and Gemini API service fees) shall transfer to the school.

Despite this, the Researchers commit to provide free patches, updates, and technical assistance throughout **School Year 2025–2026** to support the system's reliability and educational value.

Intended Users

The MATHtatag application is intended for:

- **Primary Users:** Grade 1 pupils and their respective Mathematics teachers at Camohaguin Elementary School.
- **Secondary Users:** School administrators and the Researchers, for monitoring, evaluation, and data analysis during the study.

All users shall be provided access limited to the educational purpose and duration of this partnership.

Privacy Policy

The Researchers affirm that the MATHtatag system complies with the **Data Privacy Act of 2012 (Republic Act No. 10173)**. All data gathered will be handled with strict confidentiality and integrity. Specifically:

- Only non-sensitive academic data such as assessment results, completion times, and performance records will be collected.
- No personal identifiers such as photos, addresses, or full names of pupils will be made public or shared

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externally.

- All collected information will be encrypted, stored securely, and accessible only to authorized personnel.
- Data will be used exclusively for research and academic purposes, and will be deleted upon project completion or at the request of the Partner School.

Terms and Conditions

By participating in this partnership and using the MATHtatag system, both the Researchers and the Partner School agree to the following terms:

- The application shall be used solely for educational and research purposes and may not be sold, distributed, or modified for commercial intent.
- Teachers and students are expected to use the application responsibly and in accordance with the school's ICT policies.
- The Researchers maintain ownership of the system's intellectual property but grant the school a free educational license during the partnership.
- Any misuse, tampering, or unauthorized duplication of the software will result in termination of the agreement.
- The Researchers reserve the right to withdraw system access should any breach of the agreement occur.

Duration and Validity

This Partnership Agreement takes effect upon signing and remains valid for the duration of the deployment and evaluation of the MATHtatag system. Either party may terminate the agreement by providing written notice at least fifteen (15) days prior to the intended termination date, ensuring all data and ongoing activities are properly concluded.

Legal Binding

This Agreement serves as a legally binding document between the Researchers and the Partner School and shall be governed by the laws of the Republic of the Philippines. Both parties acknowledge that they have read, understood, and voluntarily agreed to the terms contained herein.

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BAGONG PILIPINAS

IN WITNESS WHEREOF, the parties hereunto affix their signatures this **13th day of October 2025**, at **Camohaguin Elementary School, Gumaca, Quezon**.

SIGNED BY:

For the Researchers:

KEN P. CAS
Lead Researcher

LLOYD JERNELL C. LOTERIÑA
Researcher

CEDRICK L. ARTIAGA
Researcher

Engelbert R. Morales
ENGELBERT R. MORALES
Researcher

For the Partner School:

ANTONIO I. MEDRANO
Head Teacher I
Officer-In-Charge
Camohaguin Elementary School

Noted by:

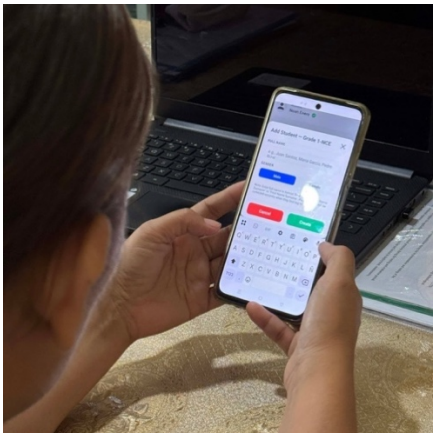
ICON C. OBMERGA
Capstone Adviser
PUP–Lopez, Quezon

Documentation

Initial Coordination

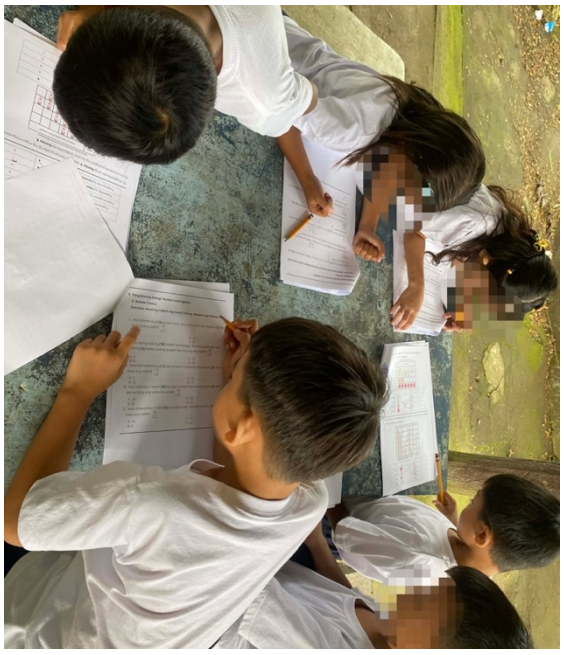
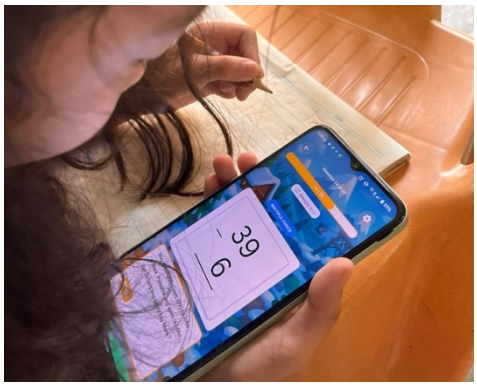
Phase 1 established communication with the partner school through meetings with the principal and Grade 1 teachers, concluding with the signing of the partnership agreement.

Deployment



Phase 2 provided training and seminars for teachers, administrators, and parents to prepare them for system use.

Testing and Evaluation

Phase 3 and 4 implemented the full classroom testing of MATHtatag—from pre-test, app-based exercises, and post-test administration to collecting ISO 25010-based feedback on usability and system effectiveness.

APPENDIX 5
EVALUATION TOOL, TEST DOCUMENTS AND TEST RESULTS

PRE TEST

Name: _____

Section: _____

1. **Unang Bahagi: Data and Probability**

   A. Multiple Choice.

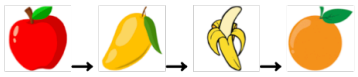
   Directions: Basahing mabuti ang bawat tanong. Bilugan ang tamang sagot.

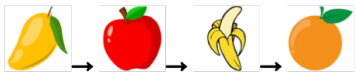
Mga Paboritong Prutas ng mga Grade 1 – Rose

PRUTAS	BILANG NG MGA PRUTAS					
SAGING						
MANGGA						
DALANDAN						
MANSANAS						

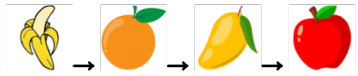
1. Ayon sa pictograph, apat **(4)** na bata ang may paborito ng Mansanas , anim **(6)** ay Mangga , dalawang **(2)** ay Dalandan , at tatlong **(3)** ay Saging .

   Alin ang tamang pagkakasunod-sunod ng bilang ng paboritong prutas ng mga bata mula sa pinakamarami hanggang sa pinakakaunti?

A. 

C. 

B.

D. 

2. Anim **(6)** na bata ang may paborito ay Mangga

at tatlo **(3)** ay Saging

. Kung paghahambingin ilang bata ang lamang ng pagkagusto sa Mangga kaysa sa Saging? **(Gamitin ang pagbawas sa pagsagot.)**

A. 1
C. 3
B. 2
D. 4

Mga Paboritong Gamit sa Paaralan ng mga Grade 1 – Rose

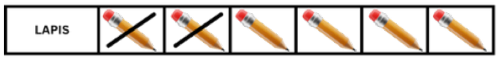
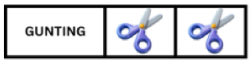
GAMIT	BILANG NG MGA GAMIT
LAPIS	
RULER	
GUNTING	
LIBRO	

3. Tingnan ang pictograph. Aling gamit sa paaralan ang pinakapaborito ng mga mag-aaral sa Grade 1 – Rose?

   A. Gunting
   C. Libro
   B. Lapis
   D. Ruler

4. Alin naman ang mga gamit ang may magkasingdaming bilang ang pabirito ng mga mag-aaral?

   A. Libro at Ruler
   C. Lapis at Ruler
   B. Gunting at Libro
   D. Gunting at Ruler

5. May anim **(6)** na pumili ng Lapis  at dalawa **(2)** ay Gunting  . Kung paghahambingin ilan ang higit na dami ng Lapis kaysa sa Gunting? **(Gamitin ang pagbawas sa pagsagot.)**

   A. 5
   C. 3
   B. 4
   D. 7

1

B. Matching Type.

Direction: Basahin at unawain ang bawat tanong. Tingnan mabuti ang mga pictograph. Itugma o iguhit ang linya mula sa bawat item papunta sa tamang bilang ayon sa ipinapakitang datos.

6. Bilangin ang hayop na nasa pictograph. Paghambingin ang **Hanay A** ayon sa bilang nito sa **Hanay B.**

A		B
Ibon	• •	𝍸
Pusa	• •	𝍸 | | | | |
Unggoy	• •	| | | |
Palaka	• •	𝍸 | |

7. Pagtambalin ang **Hanay A** ayon sa tinutukoy na bilang ng prutas papunta sa tamang bilang sa **Hanay B.**

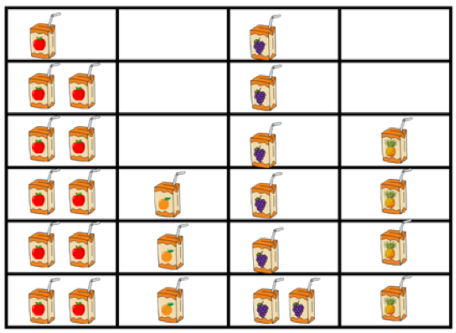
Mga Paboritong Prutas ng mga Grade 1 – Rose

PRUTAS	BILANG NG PRUTAS					
MANGGA						
SAGING						
DALANDAN						
MANSANAS						

A		B
Mangga	• •	2
Saging	• •	6
Dalandan	• •	4
Mansanas	• •	1

8. Pagtambalin ang **Hanay A** ayon sa tinutukoy na "flavor" ng juice papunta sa tamang bilang sa **Hanay B.**

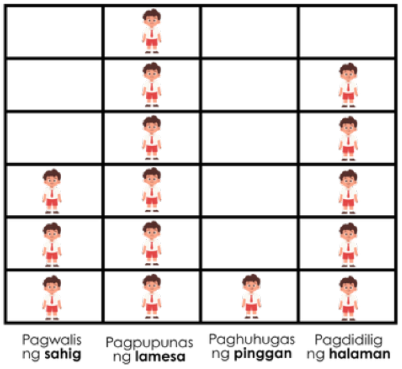
Mga Juice Boxes na Nabenta sa Isang Linggo sa Tindahan



A		B
Mansanas	• •	3
Dalandan	• •	4
Ubas	• •	11
Pinya	• •	7

9. Pagtambalin ang **Hanay A** ayon sa tamang bilang ng mga mag-aaral na gumagawa sa bawat gawaing bahay papunta sa **Hanay B.**

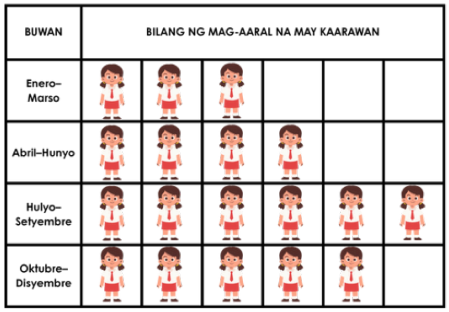
Mga Gawaing-Bahay na Ginagawa ng mga Grade 1 – Rose



A		B
Pagwawalis ng **sahig**	• •	5
Paghuhugasan ng **pinggan**	• •	6
Pagdidilig ng **halaman**	• •	1
Pagpupunas ng **lamesa**	• •	3

10. Pagtambalin ang **Hanay A** ayon sa tamang bilang ng mga mag-aaral mula sa bawat pangkat ng buwan papunta sa **Hanay B.**

Bilang ng mga Mag-aaral Ayon sa Buwan ng Kaarawan



A		B
Enero–Marso	• •	6
Abril–Hunyo	• •	4
Hulyo–Setyembre	• •	3
Oktubre–Disyembre	• •	5

2. Pangalawang Bahagi: Numbers and Algebra

A. Multiple Choice.

Directions: Basahing mabuti ang bawat tanong. Bilugan ang tamang sagot.

1. May labinlimang **(15)** laruan si Ana. Kinuha ni Maria ang dalawa **(2)**. Ilan na lang ang natira? $\begin{array}{r} 15 \\ -\ 2 \\ \hline \end{array}$

   A. 13 C. 15
   B. 5 D. 20

2. May labing-walong **(18)** basket ng mangga. Dinala sa palengke ang tatlong **(3)** basket upang ipagbili. Ilan na lang ang natira? $\begin{array}{r} 18 \\ -\ 3 \\ \hline \end{array}$

   A. 6 C. 15
   B. 8 D. 10

3. May labindalawang **(12)** ibon sa puno. Lumipad ang dalawa **(2)**. Ilan na lang ang natira? $\begin{array}{r} 12 \\ -\ 2 \\ \hline \end{array}$

   A. 9 C. 12
   B. 10 D. 11

4. May tatlumpu't siyam **(39)** na lobo si Mark. Pumutok ang anim **(6)** dito. Ilan na lang ang natira kay Mark? $\begin{array}{r} 39 \\ -\ 6 \\ \hline \end{array}$

   A. 33 C. 14
   B. 20 D. 15

5. May dalawampu't anim **(26)** na lapis si Lara. Nawala ang lima **(5)**. Ilan na lang ang natira? $\begin{array}{r} 26 \\ -\ 5 \\ \hline \end{array}$

   A. 10 C. 21
   B. 4 D. 14

B. Fill in the blank.

Direction: Basahin ang bawat sitwasyon. Pagkatapos, isulat ang tamang sagot sa patlang upang ipakita kung ilan ang natira ayon sa kuwento.

6. May **7** na mangga sa mesa. Kumuha si Nanay ng **2** upang kainin. Ilang mangga ang natira sa mesa? _____
7. May **9** na lapis sa kahon. Kumuha si Tina ng **1** na gagamitin sa pagsusulat. Ilang lapis ang naiwan sa kahon? _____
8. May **4** na ibon sa sanga ng puno. Lumipad ang **lahat** ng ibon papunta sa kabilang puno. Ilang ibon ang natira sa sanga? _____

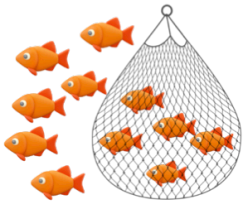
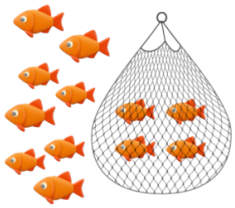
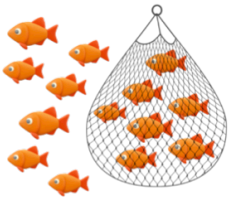
C. Matching Type.

Direction: Basahin ang bawat kwento o larawan. Itugma o iguhit ang linya mula sa bawat sitwasyon papunta sa tamang pangungusap ng pagbabawas.

9. Itugma ang kwento sa **Hanay A** tamang subtraction sentence sa **Hanay B**.

A		**B**
Kwento		**Number Sentence**
May 5 na bunga sa puno; nahulog ang 2.	• •	7 – 4 = 3
May 7 na bola; tinanggal ang 4.	• •	9 – 5 = 4
May 9 na isda; 5 ang umalis.	• •	6 – 3 = 3
May 6 na lapis; nawala ang 3.	• •	5 – 2 = 3

10. Pagtambalin ang **Hanay A** at **Hanay B** mula sa bawat larawan ng mga isda papunta sa tamang pangungusap ng pagbabawas.

A		B
	•	14 - 7 = 7
	•	11 - 5 = 6
	•	12 - 4 = 8

CERTIFICATION

This is to certify that the questionnaire prepared by the researchers, namely: **KEN P. CAS, LLOYD JERNELL C. LOTERIÑA, CEDICK L. ARTIAGA, ENGELBERT R. MORALES** in relation to the research entitled, **"MATHTATAG: A MOBILE APPLICATION FOR ONLINE ASSESSMENT AND PERFORMANCE MONITORING WITH FILIPINO TEXT-TO-SPEECH FOR GRADE 1 IN MATHEMATICS"**, has been checked and validated.

Signed this 19th day of November, 2025 at Aliji Elementary School, Tagkawayan I District Tagkawayan, Quezon.

Certification Issued By:

CARLITO C. ALOC
Master Teacher II

CERTIFICATION

This is to certify that the questionnaire prepared by the researchers, namely: **KEN P. CAS, LLOYD JERNELL C. LOTERIÑA, CEDICK L. ARTIAGA, ENGELBERT R. MORALES** in relation to the research entitled, **"MATHTATAG: A MOBILE APPLICATION FOR ONLINE ASSESSMENT AND PERFORMANCE MONITORING WITH FILIPINO TEXT-TO-SPEECH FOR GRADE 1 IN MATHEMATICS"**, has been checked and validated.

Signed this day 17th of November, 2025 at Polytechnic University of the Philippines – Lopez, Quezon Campus.

Certification Issued By:

Citadel E. Cas
Master Teacher I

CERTIFICATION

This is to certify that the questionnaire prepared by the researchers, namely: **KEN P. CAS, LLOYD JERNELL C. LOTERIÑA, CEDICK L. ARTIAGA, ENGELBERT R. MORALES** in relation to the research entitled, **"MATHTATAG: A MOBILE APPLICATION FOR ONLINE ASSESSMENT AND PERFORMANCE MONITORING WITH FILIPINO TEXT-TO-SPEECH FOR GRADE 1 IN MATHEMATICS"**, has been checked and validated.

Signed this day 13th of November, 2025 at Polytechnic University of the Philippines – Lopez, Quezon Campus.

Certification Issued By:

Noan C. Enero
Grade 1 – Teacher II

CERTIFICATION

This is to certify that the questionnaire prepared by the researchers, namely: **KEN P. CAS, LLOYD JERNELL C. LOTERIÑA, CEDICK L. ARTIAGA, ENGELBERT R. MORALES** in relation to the research entitled, **"MATHTATAG: A MOBILE APPLICATION FOR ONLINE ASSESSMENT AND PERFORMANCE MONITORING WITH FILIPINO TEXT-TO-SPEECH FOR GRADE 1 IN MATHEMATICS"**, has been checked and validated.

Signed this day 13th of November, 2025 at Polytechnic University of the Philippines – Lopez, Quezon Campus.

Certification Issued By:

May M. Eria

Grade 1 – Teacher I

CERTIFICATION

This is to certify that the questionnaire prepared by the researchers, namely: **KEN P. CAS, LLOYD JERNELL C. LOTERIÑA, CEDICK L. ARTIAGA, ENGELBERT R. MORALES** in relation to the research entitled, **"MATHTATAG: A MOBILE APPLICATION FOR ONLINE ASSESSMENT AND PERFORMANCE MONITORING WITH FILIPINO TEXT-TO-SPEECH FOR GRADE 1 IN MATHEMATICS"**, has been checked and validated.

Signed this day 13th of November, 2025 at Camohaguin Elementary School – Gumaca, Quezon.

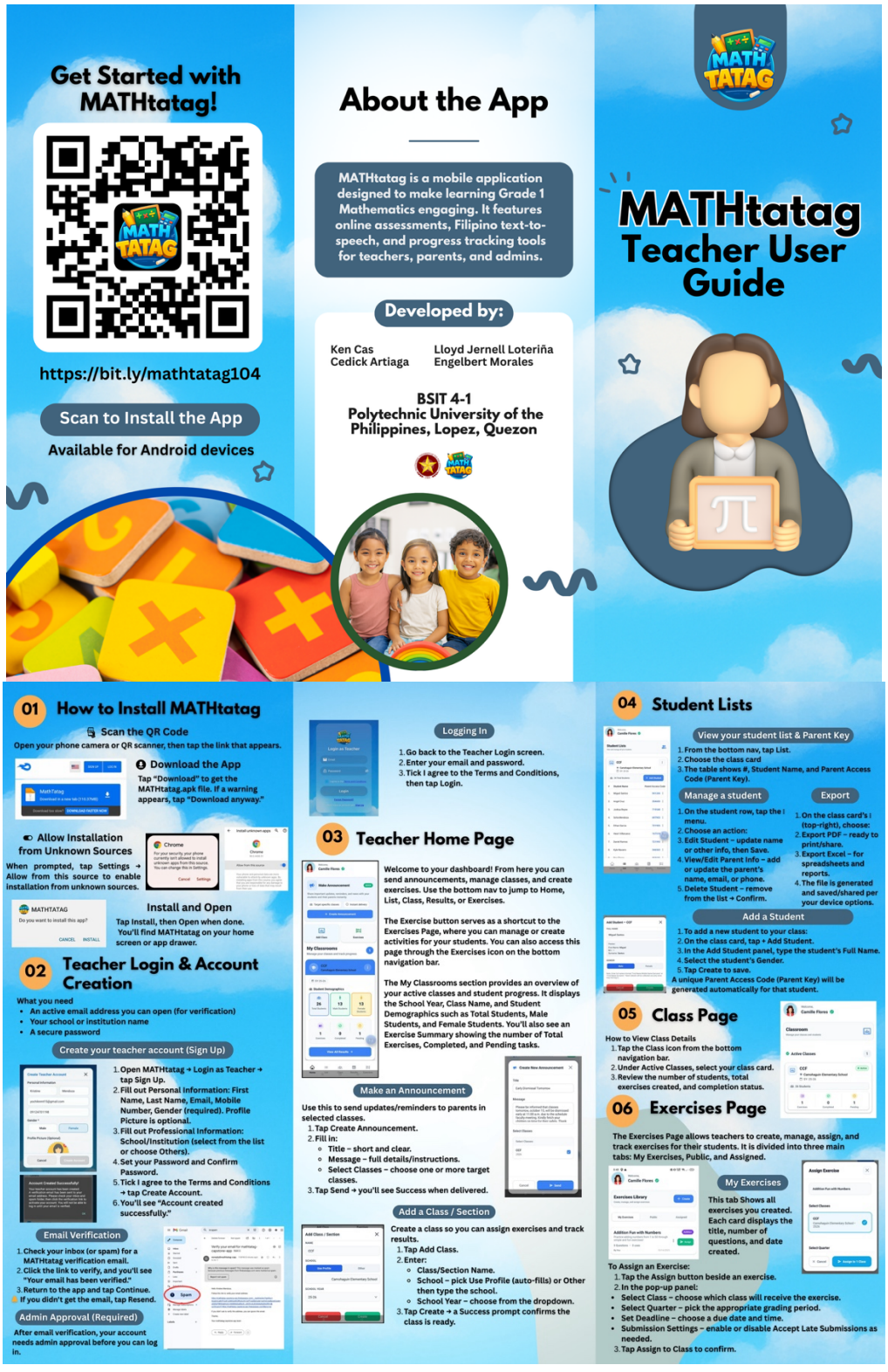
Certification Issued By:

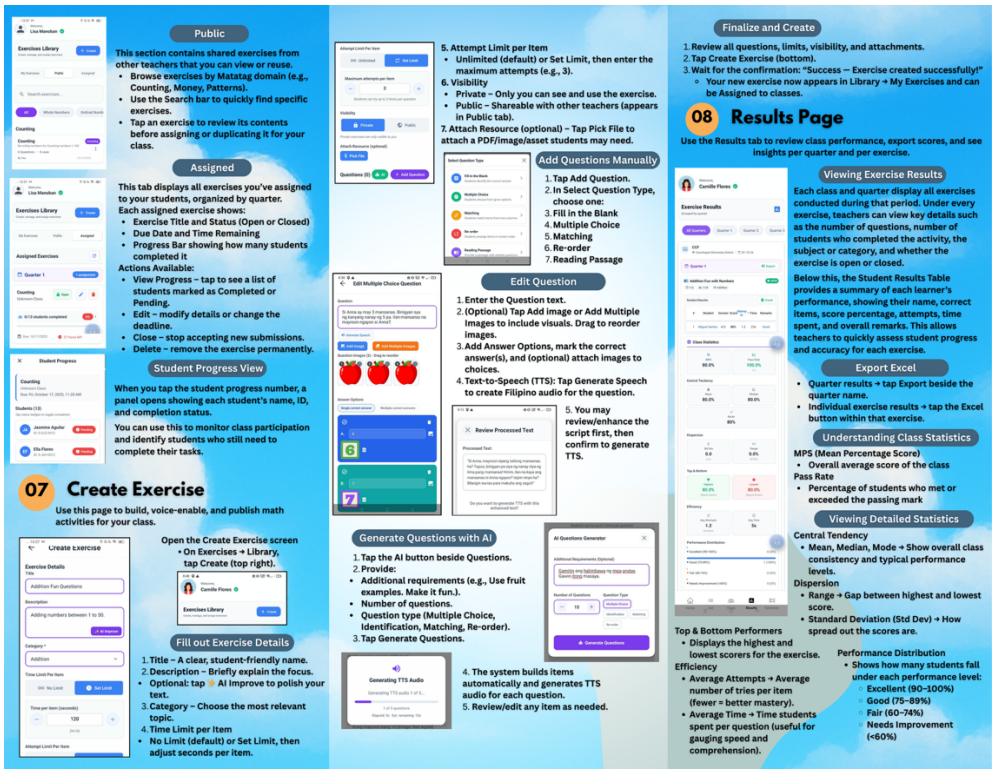
Madel M. Tañola
Grade 1 – Teacher II

APPENDIX 6

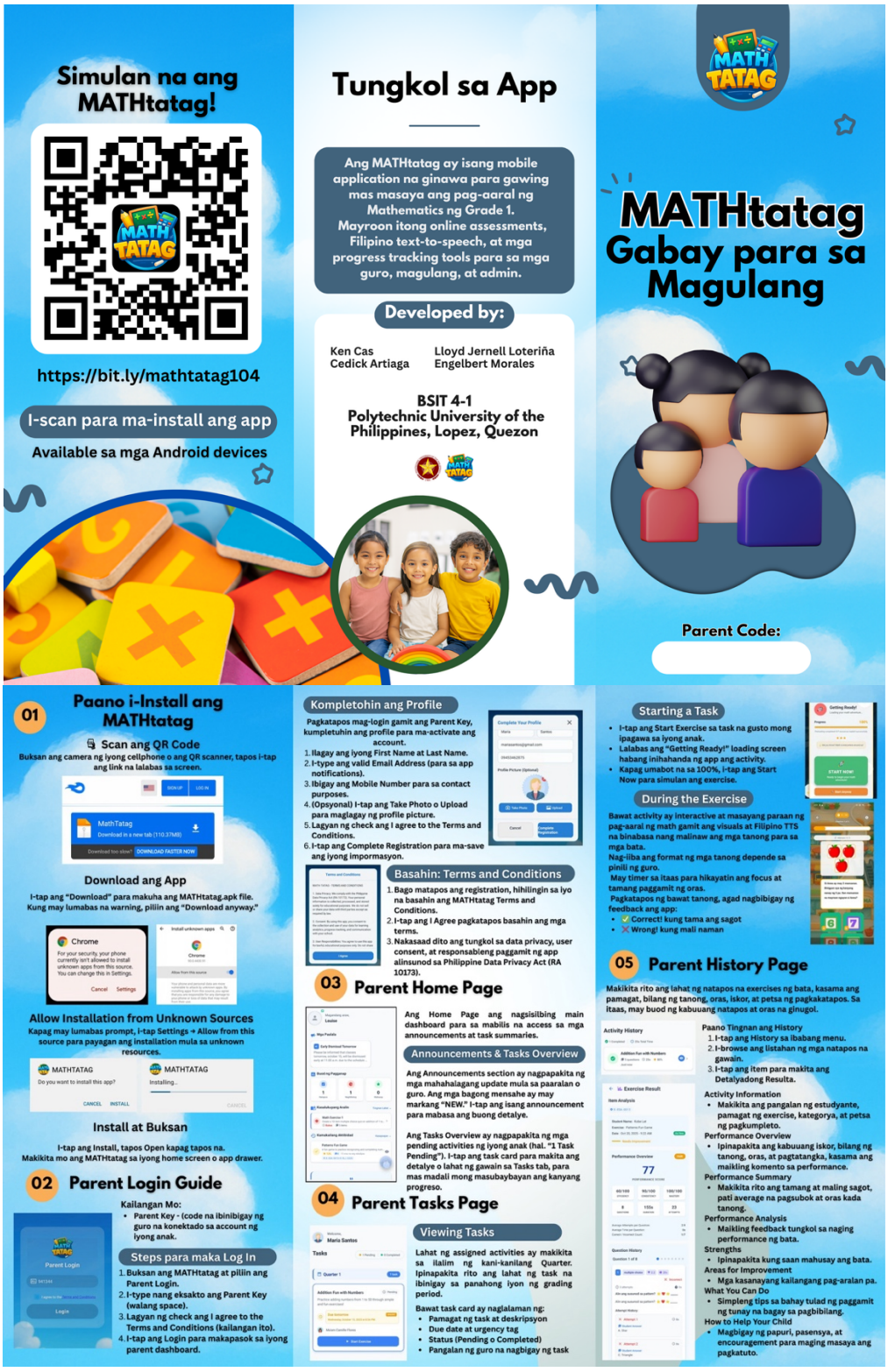
USER'S MANUAL

ADMIN User Guide


Get Started with MATHtatag!
https://bit.ly/mathtatag104
Scan to Install the App
Available for Android devices
About the App
MATHtatag is a mobile application designed to make learning Grade 1 Mathematics engaging. It features online assessments, Filipino text-to-speech, and progress tracking tools for teachers, parents, and admins.
Developed by:
Ken Cas
Cedick Artiaga
Lloyd Jernell Loteriña
Engelbert Morales
BSIT 4-1
Polytechnic University of the Philippines, Lopez, Quezon
MATHtatag Teacher User Guide
01 How to Install MATHtatag
Scan the QR Code
Open your phone camera or QR scanner, then tap the link that appears.
Download the App
Tap "Download" to get the MATHtatag.apk file. If a warning appears, tap "Download anyway."
Allow Installation from Unknown Sources
When prompted, tap Settings → Allow from this source to enable installation from unknown sources.
Install and Open
Tap Install, then Open when done. You'll find MATHtatag on your home screen or app drawer.
02 Teacher Login & Account Creation
What you need
An active email address you can open (for verification)
Your school or institution name
A secure password
Create your teacher account (Sign Up)
1. Open MATHtatag → Login as Teacher → tap Sign Up.
2. Fill out Personal Information: First Name, Last Name, Email, Mobile Number, Gender (required). Profile Picture is optional.
3. Fill out Professional Information: School/Institution (select from the list or choose Others).
4. Set your Password and Confirm Password.
5. Tick I agree to the Terms and Conditions → tap Create Account.
6. You'll see "Account created successfully."
Email Verification
1. Check your inbox (or spam) for a MATHtatag verification email.
2. Click the link to verify, and you'll see "Your email has been verified."
3. Return to the app and tap Continue.
If you didn't get the email, tap Resend.
Admin Approval (Required)
After email verification, your account needs admin approval before you can log in.
Logging In
1. Go back to the Teacher Login screen.
2. Enter your email and password.
3. Tick I agree to the Terms and Conditions, then tap Login.
03 Teacher Home Page
Welcome to your dashboard! From here you can send announcements, manage classes, and create exercises. Use the bottom nav to jump to Home, List, Class, Results, or Exercises.
The Exercise button serves as a shortcut to the Exercises Page, where you can manage or create activities for your students. You can also access this page through the Exercises icon on the bottom navigation bar.
The My Classrooms section provides an overview of your active classes and student progress. It displays the School Year, Class Name, and Student Demographics such as Total Students, Male Students, and Female Students. You'll also see an Exercise Summary showing the number of Total Exercises, Completed, and Pending tasks.
Make an Announcement
Use this to send updates/reminders to parents in selected classes.
1. Tap Create Announcement.
2. Fill in:
Title – short and clear.
Message – full details/instructions.
Select Classes – choose one or more target classes.
3. Tap Send → you'll see Success when delivered.
Add a Class / Section
Create a class so you can assign exercises and track results.
1. Tap Add Class.
2. Enter:
Class/Section Name.
School – pick Use Profile (auto-fills) or Other then type the school.
School Year – choose from the dropdown.
3. Tap Create → a Success prompt confirms the class is ready.
04 Student Lists
View your student list & Parent Key
1. From the bottom nav, tap List.
2. Choose the class card
3. The table shows #, Student Name, and Parent Access Code (Parent Key).
Manage a student
1. On the student row, tap the ⋮ menu.
2. Choose an action:
3. Edit Student – update name or other info, then Save.
4. View/Edit Parent Info – add or update the parent's name, email, or phone.
5. Delete Student – remove from the list + Confirm.
Export
1. On the class card's ⋮ (top-right), choose:
2. Export PDF – ready to print/share.
3. Export Excel – for spreadsheets and reports.
4. The file is generated and saved/shared per your device options.
Add a Student
1. To add a new student to your class:
2. On the class card, tap + Add Student.
3. In the Add Student panel, type the student's Full Name.
4. Select the student's Gender.
5. Tap Create to save.
A unique Parent Access Code (Parent Key) will be generated automatically for that student.
05 Class Page
How to View Class Details
1. Tap the Class icon from the bottom navigation bar.
2. Under Active Classes, select your class card.
3. Review the number of students, total exercises created, and completion status.
06 Exercises Page
The Exercises Page allows teachers to create, manage, assign, and track exercises for their students. It is divided into three main tabs: My Exercises, Public, and Assigned.
My Exercises
This tab Shows all exercises you created. Each card displays the title, number of questions, and date created.
To Assign an Exercise:
1. Tap the Assign button beside an exercise.
2. In the pop-up panel:
Select Class – choose which class will receive the exercise.
Select Quarter – pick the appropriate grading period.
Set Deadline – choose a due date and time.
Submission Settings – enable or disable Accept Late Submissions as needed.
3. Tap Assign to Class to confirm.

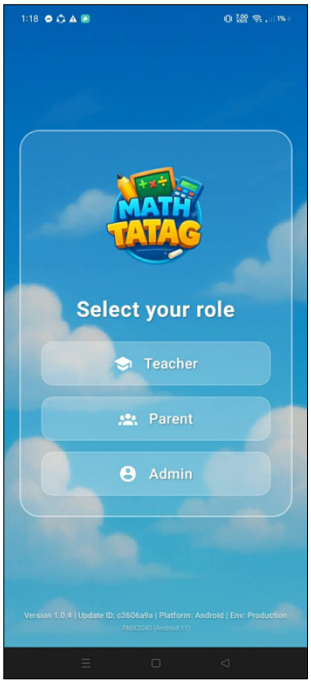


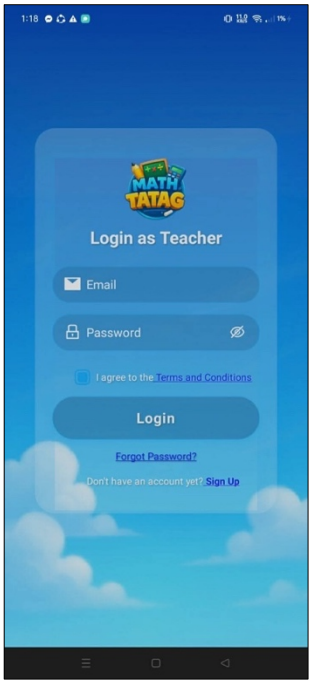
TEACHER User Guide

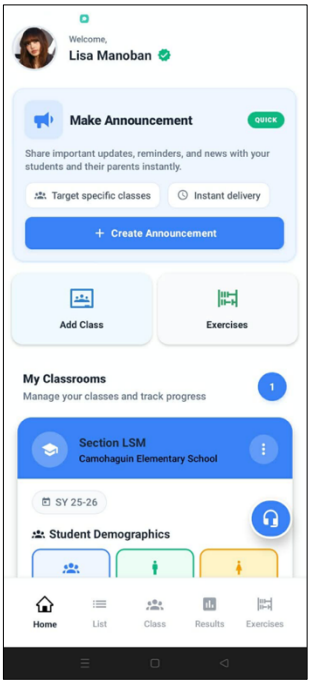


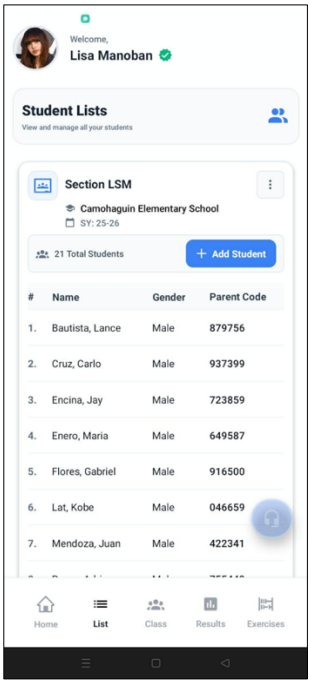
PARENT User Guide

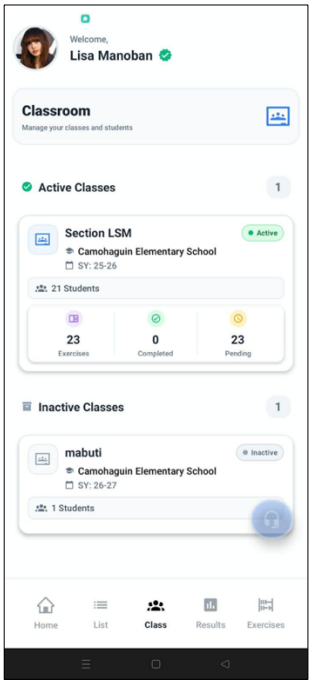
APPENDIX 7
SAMPLE GENERATED OUTPUTS

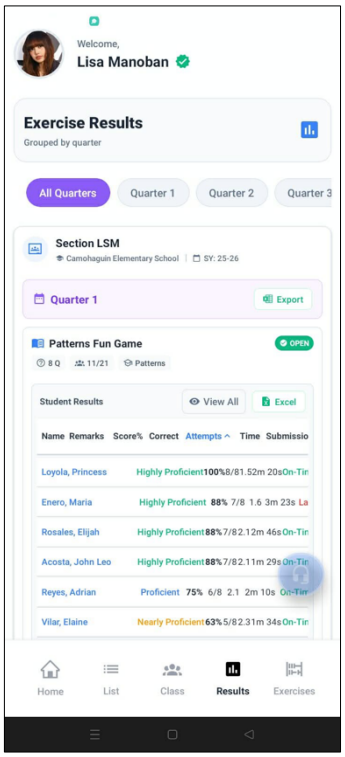


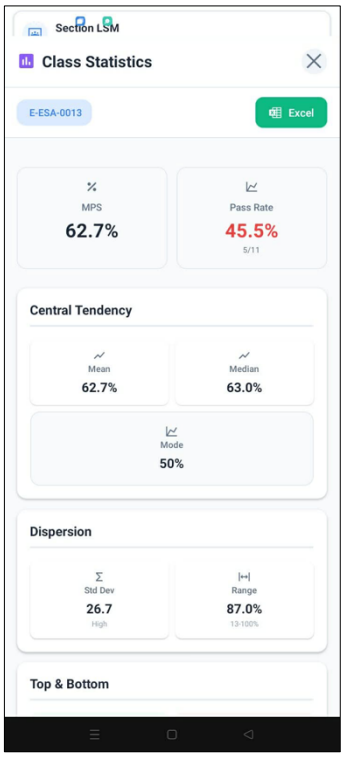


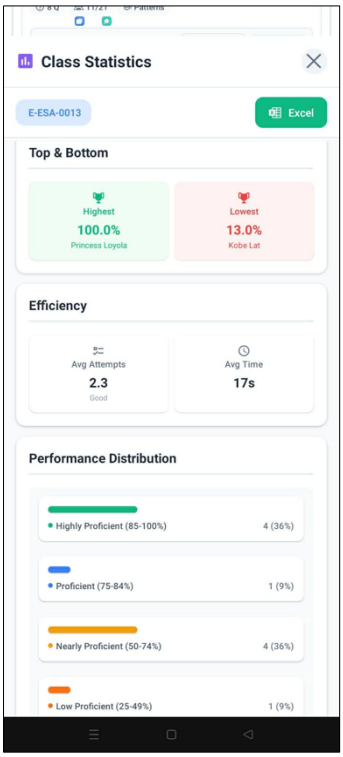


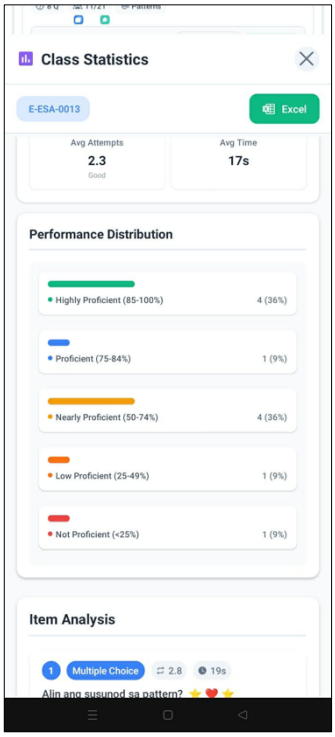


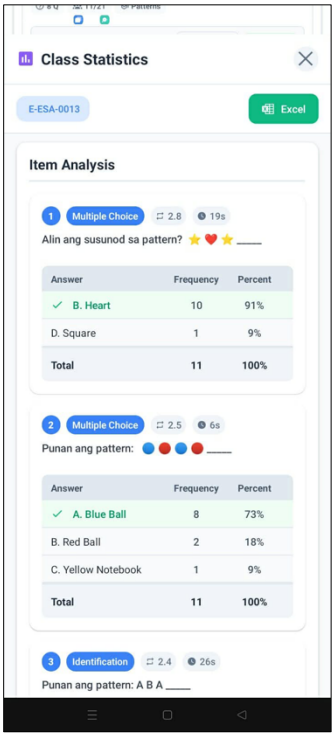


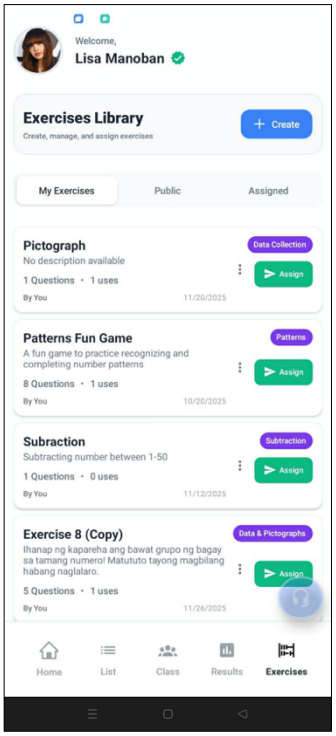

Welcome,
Lisa Manoban
Exercise Results
Grouped by quarter
All Quarters
Quarter 1
Quarter 2
Quarter 3
Section LSM
Camohaguin Elementary School
SY: 25-26
Quarter 1
Export
Patterns Fun Game
OPEN
8 Q
11/21
Patterns
Student Results
View All
Excel
Name Remarks Score% Correct Attempts Time Submissio
Loyola, Princess Highly Proficient 100% 8/8 1.52m 20s On-Tir
Enero, Maria Highly Proficient 88% 7/8 1.6 3m 23s La
Rosales, Elijah Highly Proficient 88% 7/8 2.12m 46s On-Tir
Acosta, John Leo Highly Proficient 88% 7/8 2.11m 29s On-Tir
Reyes, Adrian Proficient 75% 6/8 2.1 2m 10s On-Tir
Vilar, Elaine Nearly Proficient 63% 5/8 2.31m 34s On-Tir
Home
List
Class
Results
Exercises

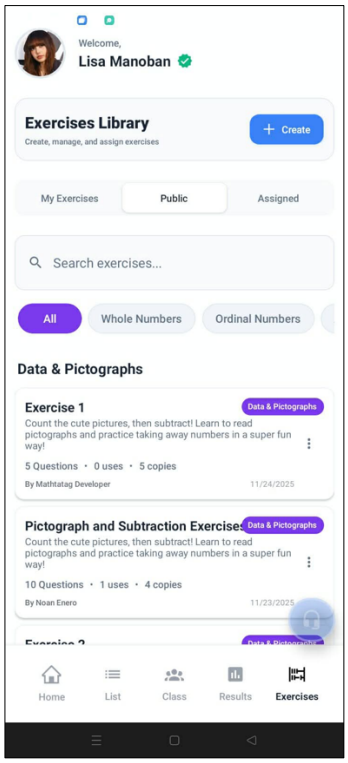

Section LSM
Class Statistics
E-ESA-0013
Excel
MPS
62.7%
Pass Rate
45.5%
5/11
Central Tendency
Mean
62.7%
Median
63.0%
Mode
50%
Dispersion
Std Dev
26.7
High
Range
87.0%
13-100%
Top & Bottom

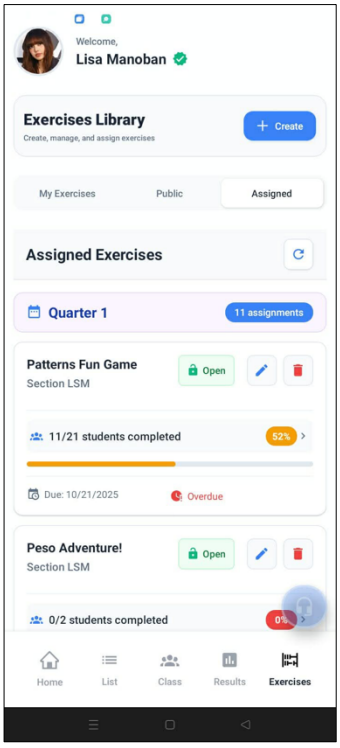

Class Statistics
E-ESA-0013
Excel
Top & Bottom
Highest
100.0%
Princess Loyola
Lowest
13.0%
Kobe Lat
Efficiency
Avg Attempts
2.3
Good
Avg Time
17s
Performance Distribution
Highly Proficient (85-100%) 4 (36%)
Proficient (75-84%) 1 (9%)
Nearly Proficient (50-74%) 4 (36%)
Low Proficient (25-49%) 1 (9%)

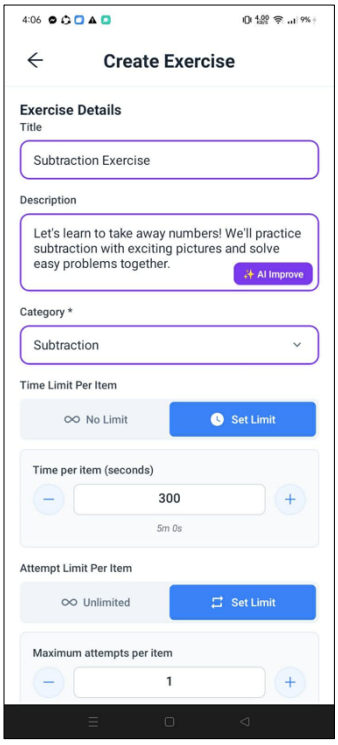

Class Statistics
E-ESA-0013
Excel
Avg Attempts
2.3
Good
Avg Time
17s
Performance Distribution
Highly Proficient (85-100%) 4 (36%)
Proficient (75-84%) 1 (9%)
Nearly Proficient (50-74%) 4 (36%)
Low Proficient (25-49%) 1 (9%)
Not Proficient (<25%) 1 (9%)
Item Analysis
1 Multiple Choice 2.8 19s
Alin ang susunod sa pattern?

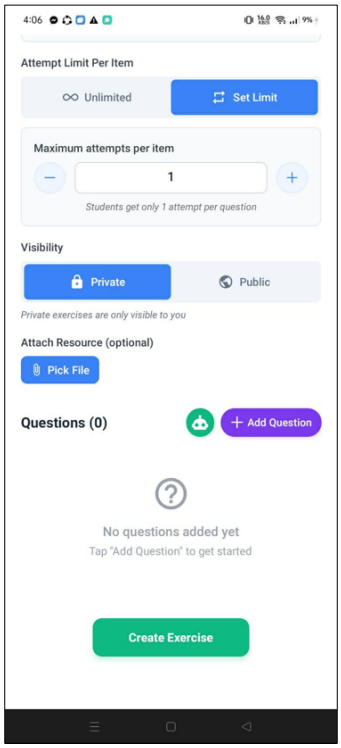

Class Statistics
E-ESA-0013
Excel
Item Analysis
1 Multiple Choice 2.8 19s
Alin ang susunod sa pattern?
Answer Frequency Percent
B. Heart 10 91%
D. Square 1 9%
Total 11 100%
2 Multiple Choice 2.5 6s
Punan ang pattern:
Answer Frequency Percent
A. Blue Ball 8 73%
B. Red Ball 2 18%
C. Yellow Notebook 1 9%
Total 11 100%
3 Identification 2.4 26s
Punan ang pattern: A B A ____

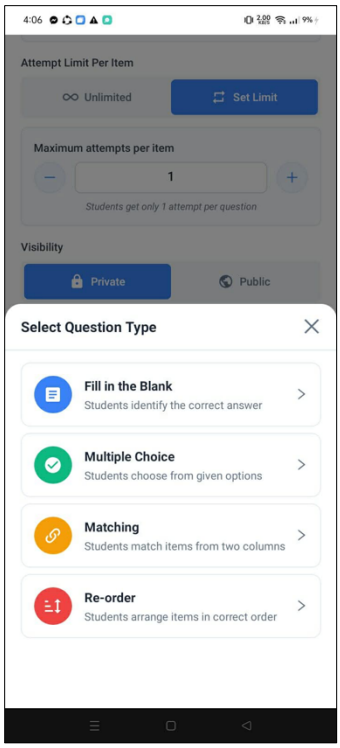

Welcome,
Lisa Manoban
Exercises Library
Create, manage, and assign exercises
Create
My Exercises
Public
Assigned
Pictograph
Data Collection
No description available
1 Questions • 1 uses
Assign
By You
11/20/2025
Patterns Fun Game
Patterns
A fun game to practice recognizing and completing number patterns
8 Questions • 1 uses
Assign
By You
10/20/2025
Subraction
Subtraction
Subtracting number between 1-50
1 Questions • 0 uses
Assign
By You
11/12/2025
Exercise 8 (Copy)
Data & Pictographs
Ihanap ng kapareha ang bawat grupo ng bagay sa tamang numero! Matututo tayong magbilang habang naglalaro.
5 Questions • 1 uses
Assign
By You
11/26/2025
Home
List
Class
Results
Exercises

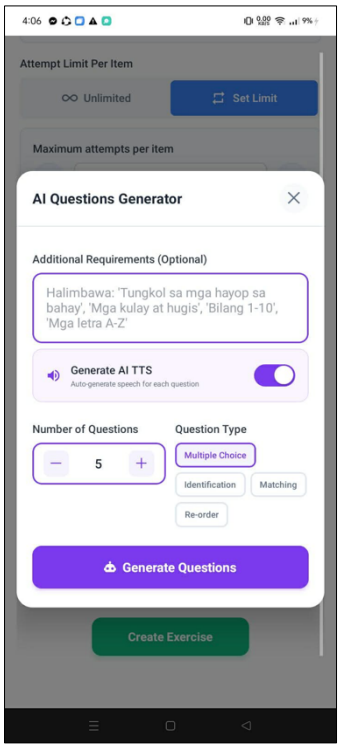

Welcome,
Lisa Manoban
Exercises Library
Create, manage, and assign exercises
Create
My Exercises
Public
Assigned
Search exercises...
All
Whole Numbers
Ordinal Numbers
Data & Pictographs
Exercise 1
Data & Pictographs
Count the cute pictures, then subtract! Learn to read pictographs and practice taking away numbers in a super fun way!
5 Questions • 0 uses • 5 copies
By Mathtatag Developer
11/24/2025
Pictograph and Subtraction Exercises
Data & Pictographs
Count the cute pictures, then subtract! Learn to read pictographs and practice taking away numbers in a super fun way!
10 Questions • 1 uses • 4 copies
By Noan Enero
11/23/2025
Home
List
Class
Results
Exercises

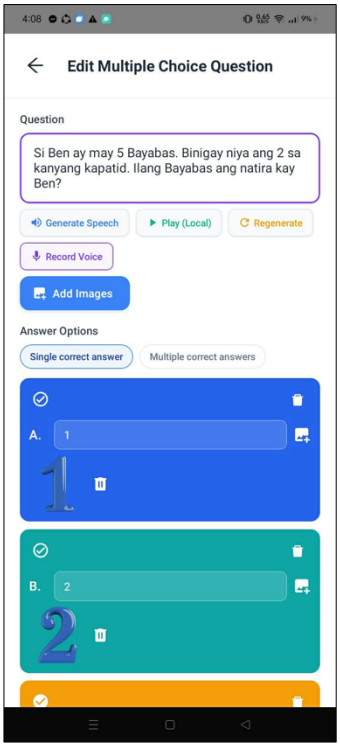

Welcome,
Lisa Manoban
Exercises Library
Create, manage, and assign exercises
Create
My Exercises
Public
Assigned
Assigned Exercises
Quarter 1
11 assignments
Patterns Fun Game
Section LSM
Open
11/21 students completed
52%
Due: 10/21/2025
Overdue
Peso Adventure!
Section LSM
Open
0/2 students completed
0%
Home
List
Class
Results
Exercises

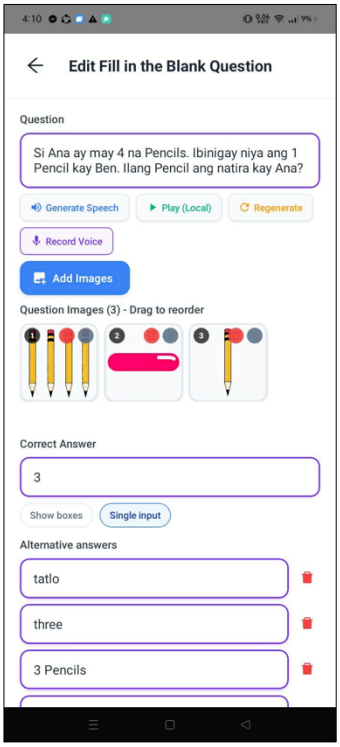

Create Exercise
Exercise Details
Title
Subtraction Exercise
Description
Let's learn to take away numbers! We'll practice subtraction with exciting pictures and solve easy problems together.
AI Improve
Category *
Subtraction
Time Limit Per Item
No Limit
Set Limit
Time per item (seconds)
300
5m 0s
Attempt Limit Per Item
Unlimited
Set Limit
Maximum attempts per item
1

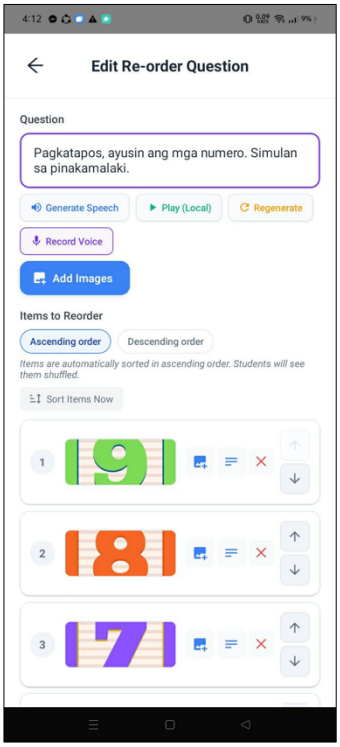

Attempt Limit Per Item
Unlimited
Set Limit
Maximum attempts per item
1
Students get only 1 attempt per question
Visibility
Private
Public
Private exercises are only visible to you
Attach Resource (optional)
Pick File
Questions (0)
Add Question
No questions added yet
Tap "Add Question" to get started
Create Exercise

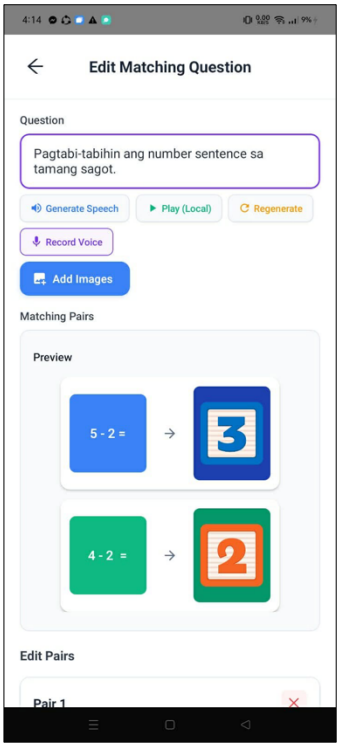

Attempt Limit Per Item
Unlimited
Set Limit
Maximum attempts per item
1
Students get only 1 attempt per question
Visibility
Private
Public
Select Question Type
Fill in the Blank
Students identify the correct answer
Multiple Choice
Students choose from given options
Matching
Students match items from two columns
Re-order
Students arrange items in correct order

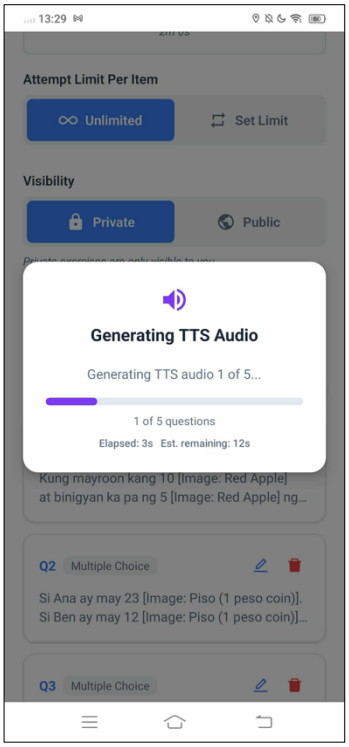

Attempt Limit Per Item
Unlimited
Set Limit
Maximum attempts per item
AI Questions Generator
Additional Requirements (Optional)
Halimbawa: 'Tungkol sa mga hayop sa bahay', 'Mga kulay at hugis', 'Bilang 1-10', 'Mga letra A-Z'
Generate AI TTS
Auto-generate speech for each question
Number of Questions
5
Question Type
Multiple Choice
Identification
Matching
Re-order
Generate Questions
Create Exercise

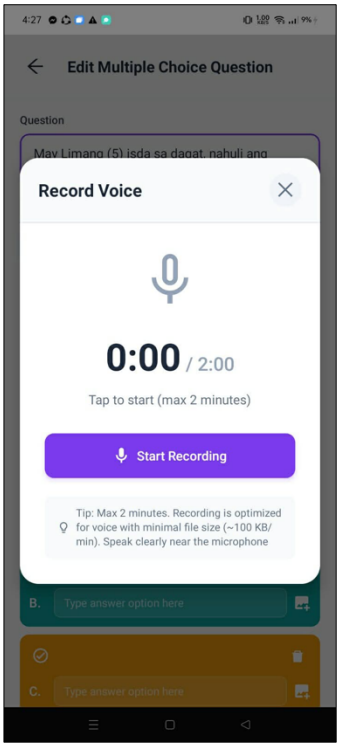

Edit Multiple Choice Question
Question
Si Ben ay may 5 Bayabas. Binigay niya ang 2 sa kanyang kapatid. Ilang Bayabas ang natira kay Ben?
Generate Speech
Play (Local)
Regenerate
Record Voice
Add Images
Answer Options
Single correct answer
Multiple correct answers
A. 1
B. 2

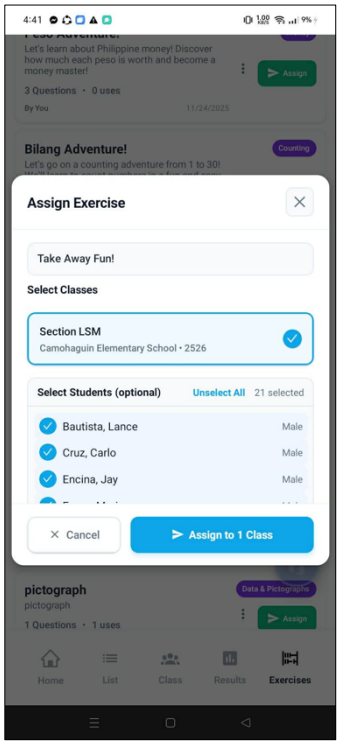

Edit Fill in the Blank Question
Question
Si Ana ay may 4 na Pencils. Ibinigay niya ang 1 Pencil kay Ben. Ilang Pencil ang natira kay Ana?
Generate Speech
Play (Local)
Regenerate
Record Voice
Add Images
Question Images (3) - Drag to reorder
Correct Answer
3
Show boxes
Single input
Alternative answers
tatlo
three
3 Pencils

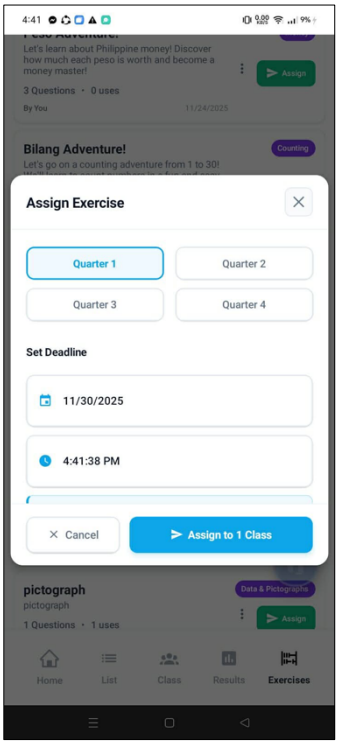

Edit Re-order Question
Question
Pagkatapos, ayusin ang mga numero. Simulan sa pinakamalaki.
Generate Speech
Play (Local)
Regenerate
Record Voice
Add Images
Items to Reorder
Ascending order
Descending order
Items are automatically sorted in ascending order. Students will see them shuffled.
Sort Items Now

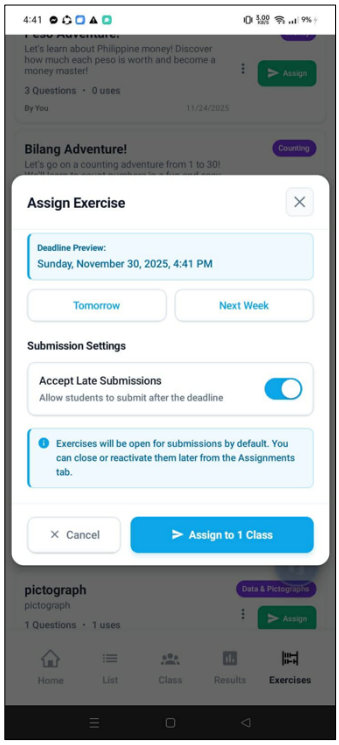

Edit Matching Question
Question
Pagtabi-tabihin ang number sentence sa tamang sagot.
Generate Speech
Play (Local)
Regenerate
Record Voice
Add Images
Matching Pairs
Preview
5 - 2 =
4 - 2 =
Edit Pairs

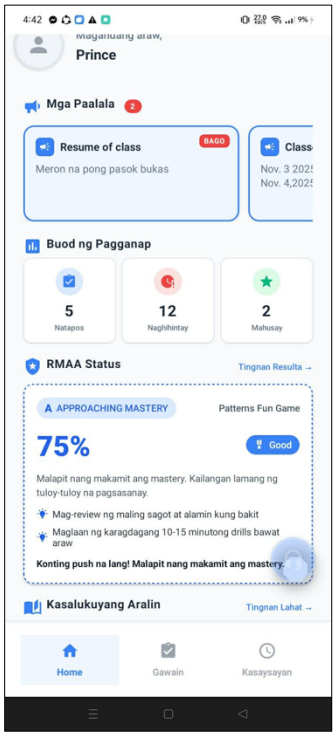

Attempt Limit Per Item
Unlimited
Set Limit
Visibility
Private
Public
Generating TTS Audio
Generating TTS audio 1 of 5...
1 of 5 questions
Elapsed: 3s Est. remaining: 12s
Q2 Multiple Choice
Q3 Multiple Choice

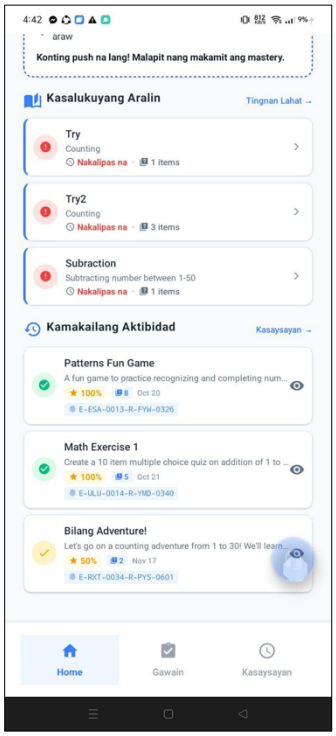

Edit Multiple Choice Question
Question
Record Voice
0:00 / 2:00
Tap to start (max 2 minutes)
Start Recording
Tip: Max 2 minutes. Recording is optimized for voice with minimal file size (~100 KB/min). Speak clearly near the microphone

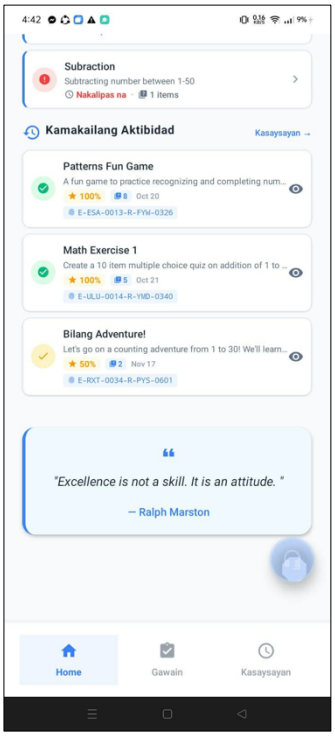

Assign Exercise
Take Away Fun!
Select Classes
Section LSM
Camohaguin Elementary School • 2526
Select Students (optional)
Unselect All
21 selected
Bautista, Lance
Male
Cruz, Carlo
Male
Encina, Jay
Male
Cancel
Assign to 1 Class

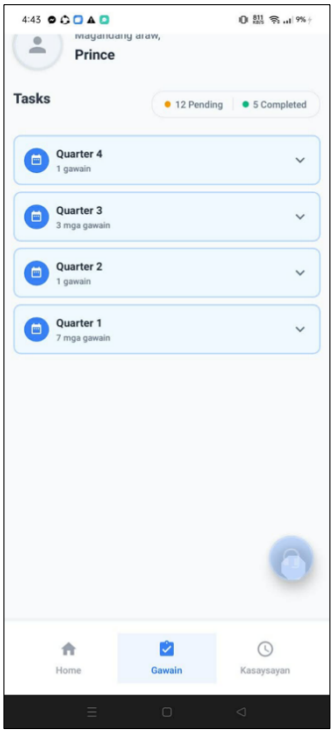

Assign Exercise
Quarter 1
Quarter 2
Quarter 3
Quarter 4
Set Deadline
11/30/2025
4:41:38 PM
Cancel
Assign to 1 Class

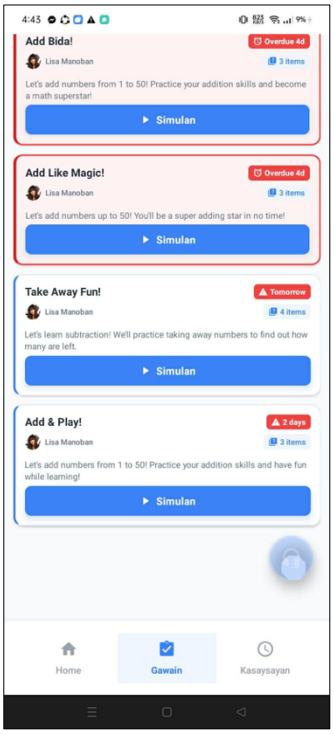

Assign Exercise
Deadline Preview:
Sunday, November 30, 2025, 4:41 PM
Tomorrow
Next Week
Submission Settings
Accept Late Submissions
Allow students to submit after the deadline
Exercises will be open for submissions by default. You can close or reactivate them later from the Assignments tab.
Cancel
Assign to 1 Class

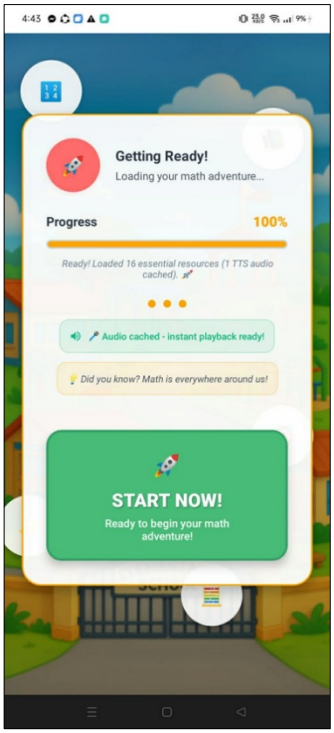

Prince
Mga Paalala
Resume of class
BAGO
Meron na pong pasok bukas
Buod ng Pagganap
5
Natapos
12
Naghihintay
2
Mahusay
RMAA Status
Tingnan Resulta
APPROACHING MASTERY
Patterns Fun Game
75%
Good
Malapit nang makamit ang mastery. Kailangan lamang ng tuloy-tuloy na pagsasanay.
Mag-review ng maling sagot at alamin kung bakit
Maglaan ng karagdagang 10-15 minutong drills bawat araw
Konting push na lang! Malapit nang makamit ang mastery.
Kasalukuyang Aralin
Tingnan Lahat
Home
Gawain
Kasaysayan

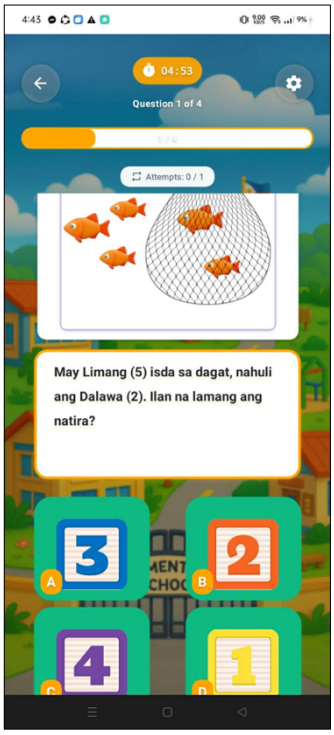

Konting push na lang! Malapit nang makamit ang mastery.
Kasalukuyang Aralin
Tingnan Lahat
Try
Counting
Nakalipas na
1 items
Try2
Counting
Nakalipas na
3 items
Subraction
Subtracting number between 1-50
Nakalipas na
1 items
Kamakailang Aktibidad
Kasaysayan
Patterns Fun Game
A fun game to practice recognizing and completing num...
100%
Oct 20
E-ESA-0013-R-FYW-0326
Math Exercise 1
Create a 10 item multiple choice quiz on addition of 1 to ...
100%
5
Oct 21
E-ULU-0014-R-YMD-0340
Bilang Adventure!
Let's go on a counting adventure from 1 to 30! We'll learn...
50%
2
Nov 17
E-RXT-0034-R-PYS-0601
Home
Gawain
Kasaysayan

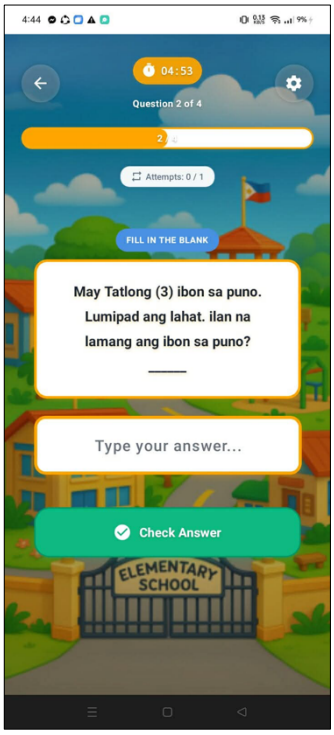

Subraction
Subtracting number between 1-50
Nakalipas na
1 items
Kamakailang Aktibidad
Kasaysayan
Patterns Fun Game
A fun game to practice recognizing and completing num...
100%
Oct 20
E-ESA-0013-R-FYW-0326
Math Exercise 1
Create a 10 item multiple choice quiz on addition of 1 to ...
100%
5
Oct 21
E-ULU-0014-R-YMD-0340
Bilang Adventure!
Let's go on a counting adventure from 1 to 30! We'll learn...
50%
2
Nov 17
E-RXT-0034-R-PYS-0601
"Excellence is not a skill. It is an attitude."
— Ralph Marston
Home
Gawain
Kasaysayan


Prince
Tasks
12 Pending
5 Completed
Quarter 4
1 gawain
Quarter 3
3 mga gawain
Quarter 2
1 gawain
Quarter 1
7 mga gawain
Home
Gawain
Kasaysayan


Add Bida!
Overdue 4d
Lisa Manoban
3 items
Let's add numbers from 1 to 50! Practice your addition skills and become a math superstar!
Simulan
Add Like Magic!
Overdue 4d
Lisa Manoban
3 items
Let's add numbers up to 50! You'll be a super adding star in no time!
Simulan
Take Away Fun!
Tomorrow
Lisa Manoban
4 items
Let's learn subtraction! We'll practice taking away numbers to find out how many are left.
Simulan
Add & Play!
2 days
Lisa Manoban
3 items
Let's add numbers from 1 to 50! Practice your addition skills and have fun while learning!
Simulan
Home
Gawain
Kasaysayan

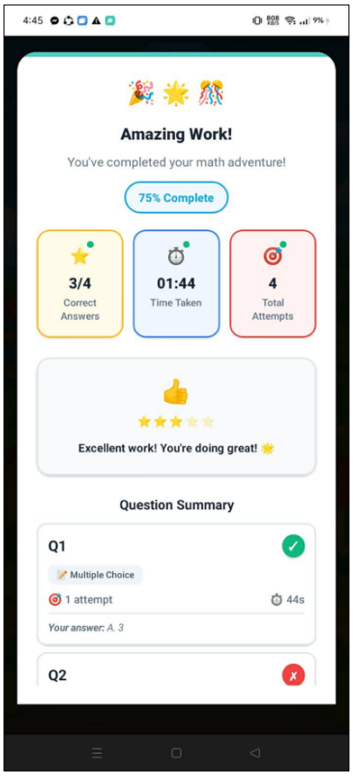

Getting Ready!
Loading your math adventure...
Progress
100%
Ready! Loaded 16 essential resources (1 TTS audio cached).
Audio cached - instant playback ready!
Did you know? Math is everywhere around us!
START NOW!
Ready to begin your math adventure!

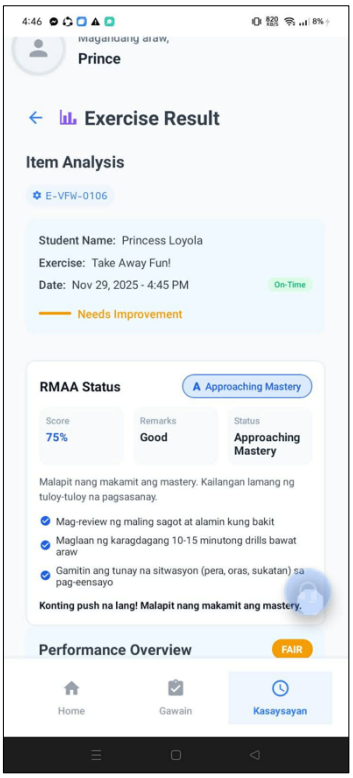

04:53
Question 1 of 4
Attempts: 0 / 1
May Limang (5) isda sa dagat, nahuli ang Dalawa (2). Ilan na lamang ang natira?
A
3
B
2
C
4
D
1

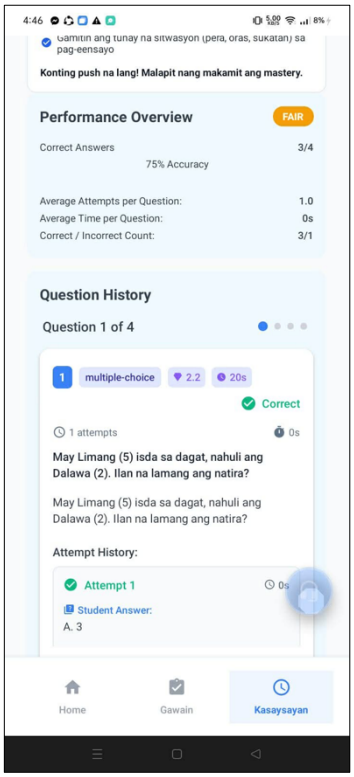

04:53
Question 2 of 4
Attempts: 0 / 1
FILL IN THE BLANK
May Tatlong (3) ibon sa puno. Lumipad ang lahat. ilan na lamang ang ibon sa puno?
Type your answer...
Check Answer
ELEMENTARY SCHOOL

04:44
Question 3 of 4
Attempts: 0 / 1
MATCHING
Pagtapatin ang Hanay A sa tamang bilang ng gamit sa Hanay B.
Stop
Column A
Column B
4
3
1

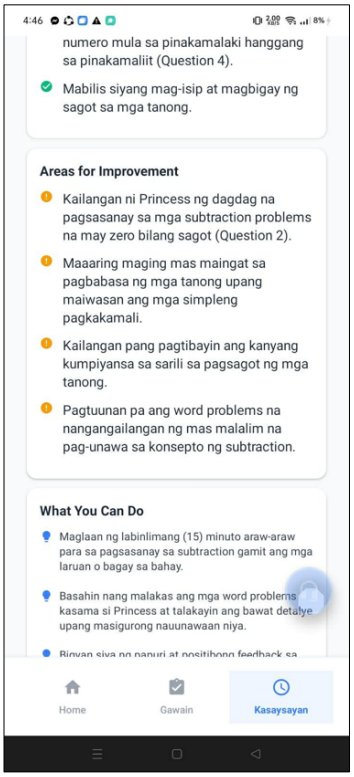
04:54
Question 4 of 4
4 / 4
Attempts: 0 / 1
RE-ORDER
Pagsunod sunodin mula sa pinaka malaking numero papunta sa pinaka maliit.
Reset
Drop here
Drop here
Drop here
Drop here
5
7
9
3

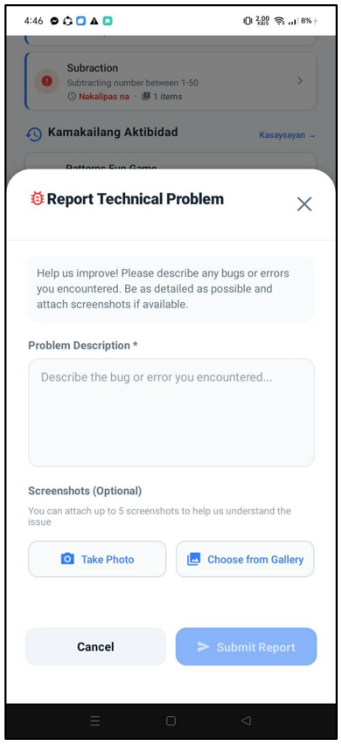

Amazing Work!
You've completed your math adventure!
75% Complete
3/4
Correct Answers
01:44
Time Taken
4
Total Attempts
Excellent work! You're doing great!
Question Summary
Q1
Multiple Choice
1 attempt
44s
Your answer: A. 3
Q2

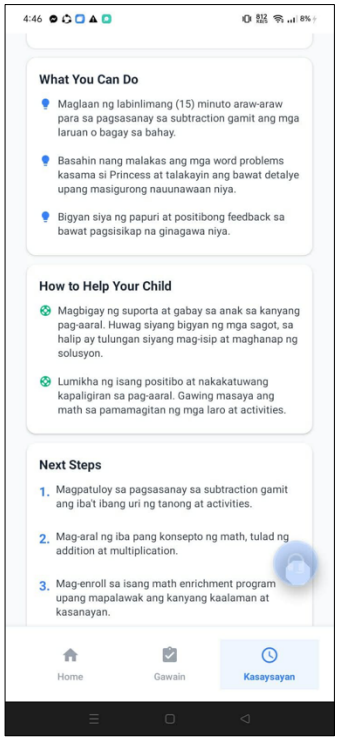

Prince
Exercise Result
Item Analysis
E-VFW-0106
Student Name: Princess Loyola
Exercise: Take Away Fun!
Date: Nov 29, 2025 - 4:45 PM
On-Time
Needs Improvement
RMAA Status
Approaching Mastery
Score
75%
Remarks
Good
Status
Approaching Mastery
Malapit nang makamit ang mastery. Kailangan lamang ng tuloy-tuloy na pagsasanay.
Mag-review ng maling sagot at alamin kung bakit
Maglaan ng karagdagang 10-15 minutong drills bawat araw
Gamitin ang tunay na sitwasyon (pera, oras, sukatan) sa pag-eensayo
Konting push na lang! Malapit nang makamit ang mastery.
Performance Overview
FAIR
Home
Gawain
Kasaysayan

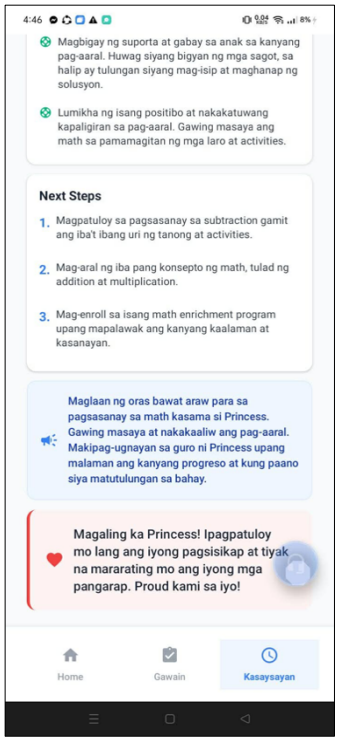

Gamitin ang tunay na sitwasyon (pera, oras, sukatan) sa pag-eensayo
Konting push na lang! Malapit nang makamit ang mastery.
Performance Overview
FAIR
Correct Answers
3/4
75% Accuracy
Average Attempts per Question:
1.0
Average Time per Question:
0s
Correct / Incorrect Count:
3/1
Question History
Question 1 of 4
1
multiple-choice
2.2
20s
Correct
1 attempts
0s
May Limang (5) isda sa dagat, nahuli ang Dalawa (2). Ilan na lamang ang natira?
May Limang (5) isda sa dagat, nahuli ang Dalawa (2). Ilan na lamang ang natira?
Attempt History:
Attempt 1
0s
Student Answer:
A. 3
Home
Gawain
Kasaysayan

3. 4
4. 1
Performance Analysis
MABUTI
B+
Magaling-galing si Princess sa pagsagot ng mga tanong sa Take Away Fun! Karamihan sa mga tanong ay nasagot niya nang tama. Kailangan lang ng konting dagdag na pagsasanay para mas maging bihasa.
Ang iskor na ito ay nagpapakita na mayroon siyang matatag na pag-unawa sa konsepto ng subtraction. Mayroon siyang ilang pagkakamali, ngunit ito ay normal lamang sa kanyang edad. Sa patuloy na pagsasanay, mas lalo pa siyang gagaling.
Strengths
Mahusay si Princess sa paglutas ng mga word problems gamit ang subtraction (Question 1).
Naiintindihan ni Princess ang konsepto ng pagtutugma ng mga bilang sa mga bagay (Question 3).
Kaya ni Princess na ayusin ang mga
Home
Gawain
Kasaysayan

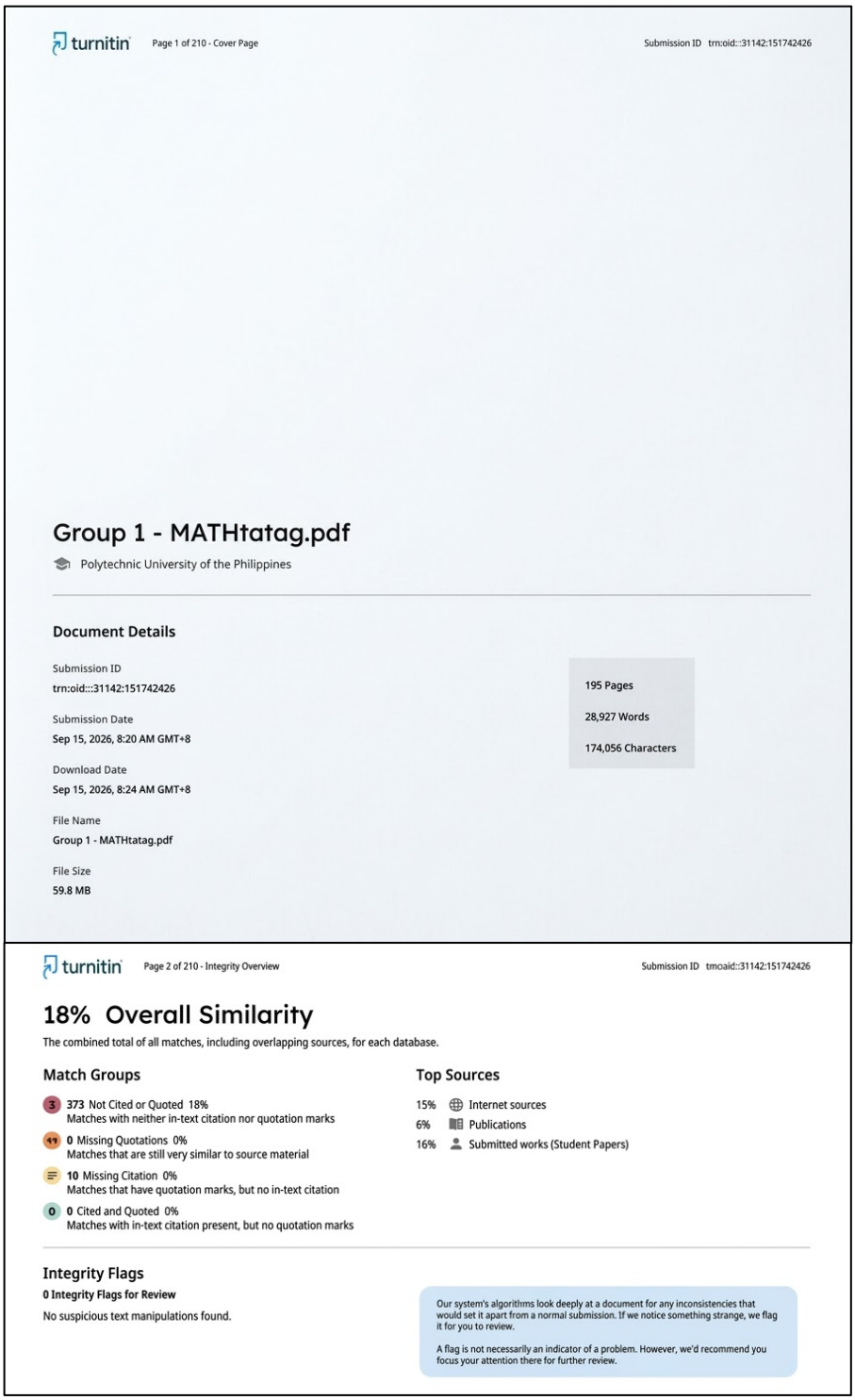

numero mula sa pinakamalaki hanggang sa pinakamaliit (Question 4).
Mabilis siyang mag-isip at magbigay ng sagot sa mga tanong.
Areas for Improvement
Kailangan ni Princess ng dagdag na pagsasanay sa mga subtraction problems na may zero bilang sagot (Question 2).
Maaaring maging mas maingat sa pagbabasa ng mga tanong upang maiwasan ang mga simpleng pagkakamali.
Kailangan pang pagtibayin ang kanyang kumpiyansa sa sarili sa pagsagot ng mga tanong.
Pagtuunan pa ang word problems na nangangailangan ng mas malalim na pag-unawa sa konsepto ng subtraction.
What You Can Do
Maglaan ng labinlimang (15) minuto araw-araw para sa pagsasanay sa subtraction gamit ang mga laruan o bagay sa bahay.
Basahin nang malakas ang mga word problems kasama si Princess at talakayin ang bawat detalye upang masigurong nauunawaan niya.
Home
Gawain
Kasaysayan


Subraction
Subtracting number between 1-50
Nakalipas na
1 items
Kamakailang Aktibidad
Kasaysayan
Report Technical Problem
Help us improve! Please describe any bugs or errors you encountered. Be as detailed as possible and attach screenshots if available.
Problem Description *
Describe the bug or error you encountered...
Screenshots (Optional)
You can attach up to 5 screenshots to help us understand the issue
Take Photo
Choose from Gallery
Cancel
Submit Report


What You Can Do
Maglaan ng labinlimang (15) minuto araw-araw para sa pagsasanay sa subtraction gamit ang mga laruan o bagay sa bahay.
Basahin nang malakas ang mga word problems kasama si Princess at talakayin ang bawat detalye upang masigurong nauunawaan niya.
Bigyan siya ng papuri at positibong feedback sa bawat pagsisikap na ginagawa niya.
How to Help Your Child
Magbigay ng suporta at gabay sa anak sa kanyang pag-aaral. Huwag siyang bigyan ng mga sagot, sa halip ay tulungan siyang mag-isip at maghanap ng solusyon.
Lumikha ng isang positibo at nakakatuwang kapaligiran sa pag-aaral. Gawing masaya ang math sa pamamagitan ng mga laro at activities.
Next Steps
1. Magpatuloy sa pagsasanay sa subtraction gamit ang iba't ibang uri ng tanong at activities.
2. Mag-aral ng iba pang konsepto ng math, tulad ng addition at multiplication.
3. Mag-enroll sa isang math enrichment program upang mapalawak ang kanyang kaalaman at kasanayan.
Home
Gawain
Kasaysayan


Magbigay ng suporta at gabay sa anak sa kanyang pag-aaral. Huwag siyang bigyan ng mga sagot, sa halip ay tulungan siyang mag-isip at maghanap ng solusyon.
Lumikha ng isang positibo at nakakatuwang kapaligiran sa pag-aaral. Gawing masaya ang math sa pamamagitan ng mga laro at activities.
Next Steps
1. Magpatuloy sa pagsasanay sa subtraction gamit ang iba't ibang uri ng tanong at activities.
2. Mag-aral ng iba pang konsepto ng math, tulad ng addition at multiplication.
3. Mag-enroll sa isang math enrichment program upang mapalawak ang kanyang kaalaman at kasanayan.
Maglaan ng oras bawat araw para sa pagsasanay sa math kasama si Princess. Gawing masaya at nakakaaliw ang pag-aaral. Makipag-ugnayan sa guro ni Princess upang malaman ang kanyang progreso at kung paano siya matutulungan sa bahay.
Magaling ka Princess! Ipagpatuloy mo lang ang iyong pagsisikap at tiyak na mararating mo ang iyong mga pangarap. Proud kami sa iyo!
Home
Gawain
Kasaysayan

APPENDIX 8
CERTIFICATION OF SIMILARITY CHECK/TURNITIN RESULTS



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APPENDIX 9

CERTIFICATION OF EDITING

POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

CERTIFICATE OF EDITING

This is to certify that the thesis with the title **MATHTATAG: A MOBILE APPLICATION FOR ONLINE ASSESSMENT AND PERFORMANCE MONITORING WITH FILIPINO TEXT-TO-SPEECH FOR GRADE 1 IN MATHEMATICS** prepared by **Ken P. Cas, Lloyd Jernell C. Loteriña, Cedick L. Artiaga,** and **Engelbert R. Morales** has been edited for language convention and has been aligned following the standard institutional format.

Issued this 24th day of June 2026 at PUP Lopez, Quezon Campus

Ms. JOSCELLE JOYCE L. RIVERA
Accredited Editor

APPENDIX 10
IMPLEMENTATION REPORT

The implementation of MATHtatag: A Mobile Application for Online Assessment and Performance Monitoring with Filipino Text-to-Speech for Grade 1 in Mathematics was conducted at Camohaguin Elementary School in Gumaca, Quezon. The implementation followed four phases: Initial Coordination, Deployment, Testing, and Evaluation. During the Initial Coordination Phase, the researchers met with the school principal on October 9 to discuss the project and its implementation. On October 13, they met with the Grade 1 teachers and conducted an interview regarding the implementation process. A partnership agreement was also signed on the same date to formalize the coordination between the researchers and the school.

During the Deployment Phase, seminar and training sessions were conducted for the teachers and school administrator on October 20, followed by training for the parents on October 21. These activities prepared the participants for the use of MATHtatag and the procedures of the study. The implementation schedule was then affected by the semestral break and Typhoon Uwan from October 22 to November 16.

The Testing Phase resumed on November 17 with the administration of the paper-based pretest to establish the learners' baseline mathematics performance. From November 17 to November 28, the learners completed teacher-assigned activities through the MATHtatag application. Teacher A assigned 14 activities, Teacher B assigned 11 activities, and Teacher C assigned 13 activities. On November 28, the learners completed the posttest through the application.

During the Evaluation Phase, surveys were conducted with the teachers and parents to assess the usability and acceptability of MATHtatag using the ISO/IEC 25010 Software Quality Model. The collected pretest, app-based activity, posttest, system testing, and evaluation data were then organized, calculated, and interpreted as part of the study. Overall, the implementation followed an organized sequence from school coordination and participant preparation to deployment, testing, and evaluation, allowing the researchers to gather the data needed to assess the application and the learner's performance.

APPENDIX 11
CAPSTONE PROJECT REVISION MATRICES

Panel 1 – Revision Matrix

No.	Recommendation	Solution/Revision
1.	The interactive assessments should not be limited to static activities because this may make the application less engaging and limit the number of learning activities that learners can answer.	The assessment component was improved to support different interactive question types, including matching, multiple choice, and re-ordering activities. This allows teachers to provide varied activities and gives learners more opportunities to interact with the mathematics content.
2.	Provide take-home assessments for the students.	The application was designed to support home-based learning. Teachers can assign activities to learners, while parents can monitor their children's tasks, scores, and progress through the parent dashboard.
3.	The application should have an innovation or unique feature that distinguishes it from other existing applications.	The unique features of MATHtatag were further emphasized, particularly its combination of MATATAG Curriculum-aligned Grade 1 Mathematics activities, Filipino Text-to-Speech, real-time teacher and parent monitoring, and home-school support in one platform. AI-supported features were also incorporated to support exercise generation and parent-facing feedback.
4.	The application should be usable throughout the school year and not be limited to a specific quarter.	The system was designed to support monitoring throughout the school year, with teacher and parent dashboards providing learner performance and activity records by quarter.

Panel 2 – Revision Matrix

No.	Recommendation	Solution/Revision
1.	Make sure that there are appropriate documents to support the contents of the study.	Supporting documents were reviewed and included in the capstone appendices, including the data-gathering instrument, client forms and reports, evaluation tools, test documents and results, user's manual, and project documentation. References and supporting sources were also reviewed and cited in the manuscript.
2.	The presenters should not simply read the content shown on the presentation slides.	simply read the content shown on the presentation slides. The presentation was revised to serve as a guide rather than a script. The presenters prepared explanations and supporting details for each topic and practiced presenting the capstone in a clear and conversational manner instead of simply reading the text displayed on the slides.

Panel 3 – Revision Matrix

No.	Recommendation	Solution/Revision
1.	Remove the socioeconomic profile feature.	The socioeconomic profile feature was removed from the application's intended user features and was no longer included as a system function. The study retained only the respondent information needed for the evaluation of the system.
2.	Find a unique feature for the application to make it unique from existing mobile applications.	The application was further distinguished by emphasizing the integration of Filipino Text-to-Speech, MATATAG Curriculum alignment, interactive Grade 1 Mathematics assessments, real-time teacher and parent dashboards, and home-based learning support within one mobile platform.

APPENDIX 12
ETHICS CLEARANCE AND TERMINAL REPORT

Republic of the Philippines
POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
OFFICE OF THE VICE PRESIDENT FOR CAMPUSES
LOPEZ, QUEZON CAMPUS

BAGONG PILIPINAS

Date: MAY 15, 2026

TO/FOR: **Ken P. Cas**
Lloyd Jernell C. Loterña
Cedick L. Artiaga
Engelbert R. Morales

Subject: Ethical Clearance

Recommendation From: *LESLEY ANN CHAN-MAGTIBAY, MSTHM*
Panel Chair

From (SGD) : *Aileen Camille B. Dimatatac, MC*
Chief, Research Ethics Section

This is to inform you that the documentary requirements you submitted for your research project titled **"MATHTATAG: A MOBILE APPLICATION FOR ONLINE ASSESSMENT AND PERFORMANCE MONITORING WITH FILIPINO TEXT-TO-SPEECH FOR GRADE 1 IN MATHEMATICS"** passed the evaluation of the Campus Research Ethics Committee (CMREC) under the requirements set by the Philippine Health Research Ethics Board (PHREB).

UREC Campus Code	CM-LOP-BSIT-2026-001
Type of Review	EXPIDITED
Approval Date	May 15, 2026
Expiry Date	May 14, 2027
Campus Review Ethics Committee Decision	**Approved**

The standard conditions of this approval are as follows:

1. Conduct the project following the submitted and approved research protocol and other documentary requirements.
2. If changes are to be made in the research project/study that will affect the research participants, an amendment of the research protocol must be submitted to the Campus Panel Chair Email Address before implementing such changes.
3. When ethical clearance is about to expire, the researcher must apply to resubmit the research protocol.
4. A final report/terminal report and co-authorship must be submitted when the research project/study is complete.
5. Researchers must advise the Campus Panel Chair Email Address in writing if the research project/study has been discontinued.
6. You may now commence your research project/study. Good luck.

APPENDIX 13
BIOGRAPHICAL STATEMENTS

Ken P. Cas is a 4th-year student of Polytechnic University of the Philippines – Lopez Branch, pursuing a Bachelor of Science in Information Technology. His journey in academic excellence and leadership began at Camohaguin Elementary School, where he was a consistent honor student and actively participated in student leadership as a representative or councilor.

He continued these achievements at Camohaguin National High School, where he remained an honor student and active student leader. He served in various leadership positions, with his highest position being Auditor of the Supreme Student Government. Beyond academics and leadership, he was also an active member of the school band and served as a Junior Fire Marshal, developing his skills in teamwork, discipline, and service.

For senior high school, he studied at Philtech Institute of Arts and Technology Inc. – Main Campus, where he graduated with High Honors and as Batch Valedictorian. He was also recognized with an Academic Excellence Award from the Mayor of Gumaca. During his senior high school OJT, he worked as a Graphics Design Intern at Purong Quezon Shirt, where he created shirt designs, graphic layouts, and other visual materials.

At PUP, he continued to pursue academic and leadership excellence, serving as Vice President for Communications of the Institute of Brilliant Information Technology Students organization. He also completed his college OJT at One CoreDev IT, Inc. as a Software Developer Intern, gaining hands-on experience in full-stack web development, system applications, and software development processes.

As he completes his final year at PUP, Ken continues to build upon his experiences in leadership, creativity, and technology. He carries with him a strong commitment to learning, innovation, and excellence as he prepares to begin his professional journey in the field of Information Technology.

Lloyd Jernell C. Loteriña was born on October 17, 2003, in Calauag, Quezon, and grew up in Tagkawayan, Quezon, where he currently resides. He completed his elementary education at Tagkawayan Central Elementary School and his secondary education at Our Lady of Lourdes Academy and Tagkawayan National High School, graduating With Honors in 2020 and 2022, respectively. He took the STEM strand in senior high school, where he participated in the 4th De La Salle University Senior High School Research Congress and earned 3rd Place in the 1st Quezon Division Science and Technology Fair. He is currently a fourth-year Bachelor of Science in Information Technology student at the Polytechnic University of the Philippines, Lopez, Quezon. In 2026, he completed a 500-hour On-the-Job Training as a Full-Stack Web Developer Intern at One CoreDev IT®, Inc. (CORE®). He was also part of the research team that earned 1st Place in the Southern Tagalog Consortium for Industry, Energy, and Emerging Technology Research and Development (STCIEERD) under the 2nd Undergraduate R&D Category for MATHtatag.

As one of the researchers of MATHtatag, he led the documentation and preparation of the capstone paper, including the development of research instruments and assessments, data gathering, and analysis. He also participated in school visits, coordinated with stakeholders, assisted in user training, and served as a quality assurance tester by identifying application errors and communicating user concerns to the development team. Through these

experiences, he developed skills in research, documentation, statistical analysis, software testing, communication, and educational technology. These experiences have strengthened his goal of using information technology to develop practical solutions that respond to real-world needs and create meaningful impact in the community.

Cedick L. Artiaga was born on June 20, 2004, in Gumaca, Quezon. He is the son of Mr. Alejandro Artiaga and Mrs. Airene Artiaga. He took his elementary education at Gumaca West Central School from the academic years 2010–2016. He completed his high school education at Gumaca National High School from the academic years 2016–2022, where he graduated with Honors. Currently, he is a fourth-year college student taking Bachelor of Science in Information Technology (BSIT) at the Polytechnic University of the Philippines – Lopez Campus. At present, he is a member of the Institute of Brilliant Information Technology Students (IBITS). He has developed an interest in programming, web development, and database management, which he continues to enhance as he prepares for a career in the Information Technology industry.

Engelbert R. Morales is a Bachelor of Science in Information Technology student at the Polytechnic University of the Philippines – Lopez, Quezon Campus. He began his education at Gumaca West Central School, where he completed elementary education with honor from 2010 to 2016. He continued his studies at Gumaca National High School, completing junior high school from 2016 to 2020 and senior high school from 2020 to 2022, earning honors in both levels. In 2026, he completed his On-the-Job Training at One Core Dev IT, where he gained practical experience in web development, system applications, frontend development, and UI/UX optimization. His work included developing interfaces for the Marketing Project Tracker, integrating frontend components with backend APIs, working with databases, resolving bugs, and improving responsive layouts. He also contributed to the Core Recruit UI Optimization project by enhancing the Departments & Jobs module based on Figma designs and using reusable components. As one of the researchers of MATHtatag, he contributed to its child-friendly user interface, conducted quality assurance testing, identified and reported application errors, and supported research documentation and curriculum-aligned content. These experiences strengthened his skills in UI/UX, frontend development, software quality assurance, and educational technology, supporting his goal of becoming an IT professional focused on practical and user-centered digital solutions.